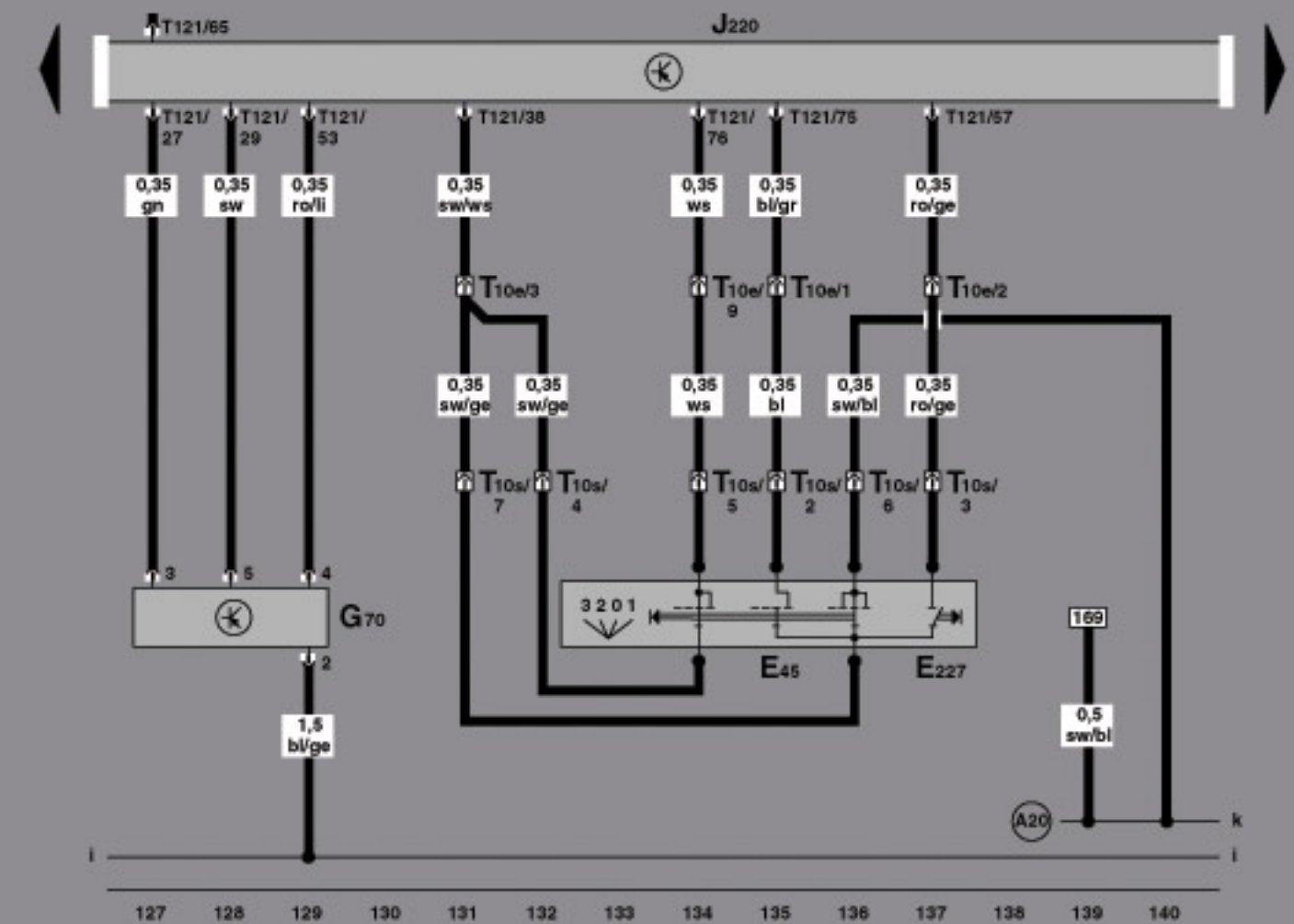




How to Read Wiring Diagrams

Symbols, Layout and Navigation



Volkswagen of America, Inc.
Service Training
Printed in U.S.A.
Printed 8/01
Course Number 873003

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Course Goals

Course goals

This course will enable you to:

- Follow current from its power source to the Ground point quickly and accurately.
- Understand the symbols of common components and circuit designations used in Volkswagen Wiring Diagrams.
- Practice Wiring Diagram navigation through practical exercises and hands on examples.

Introduction

As today's vehicles become increasingly complex, so does the job of the technician. Wiring diagram navigation skills are critical to diagnosing and repairing today's vehicle in a timely and accurate manner.

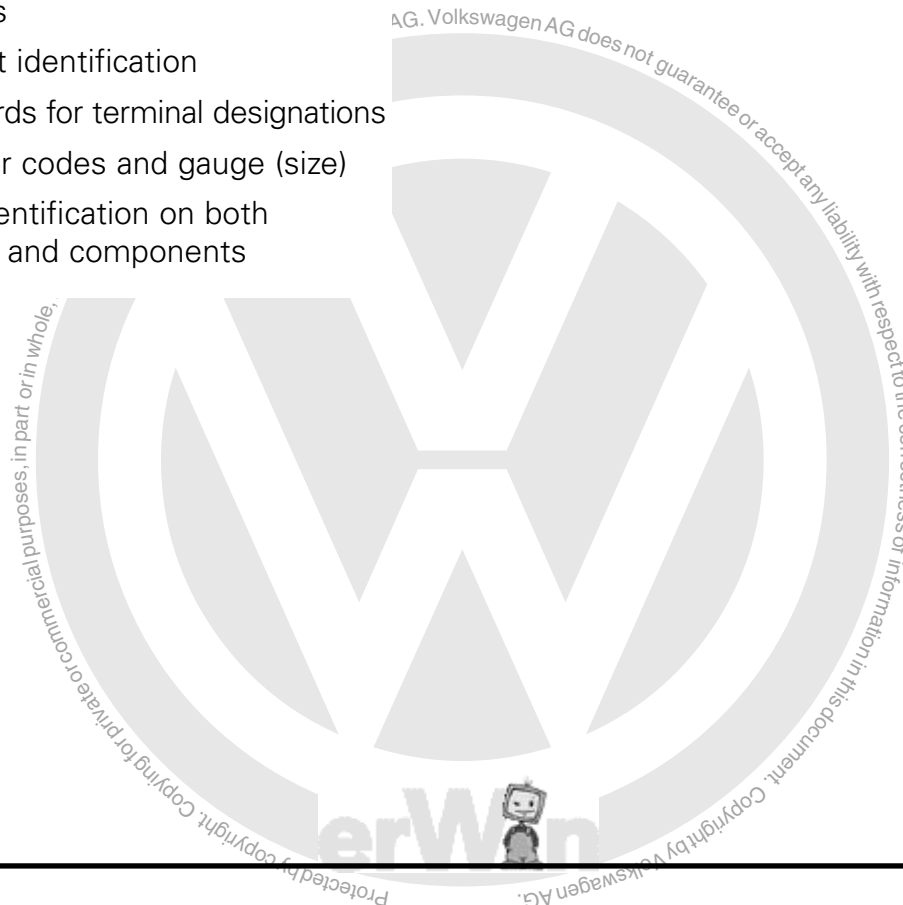
This Self Study Program is not intended to instruct the technician how the electrical system operates in a vehicle. Given an understanding of electrical operation, this Self Study Program will introduce you to the skills necessary to read Volkswagen wiring diagrams.

In this program you will be exposed to all aspects of wiring diagrams, including:

- Commonly used symbols and their meanings
- Current tracks, including how to follow circuits between pages or diagrams
- Component identification
- DIN standards for terminal designations
- Wiring color codes and gauge (size)
- Terminal identification on both connectors and components

This book will provide examples of various types of circuits. The Computer Based Training (CBT) CD included at the end of this self study will provide an opportunity to practice "hands-on" wiring diagram navigation.

Areas of text that give the technician an opportunity to practice the concepts in the CBT will be designated with an icon of a CD (below) at the beginning of that section.



Wiring Diagram Overview

Overview

Volkswagen wiring diagrams are a graphic representation of the actual vehicle wiring. They are developed from the engineering drawings that are used to produce the wiring harness. A consistent set of symbols are used to represent the actual components and conductors.

Volkswagen electrical systems and wiring diagrams follow the German DIN (Deutsche Industrie Norm/Deutsches Institut für Normung) standards. These standards are guidelines for manufacturing in Germany, similar to SAE (Society of Automotive Engineers) in the United States.

Layout

The layout of wiring diagrams is common to all Volkswagen vehicles. Called “Current Track” wiring diagrams, they show the power source at the top of the page and the Ground points at the bottom. Situated vertically between power and Ground are the current tracks, which contain electrical components and conductors.

This current track layout simplifies the wiring diagram. Conductor symbols crossing where they do not connect is kept to a minimum. Refer to the example on page 3 for the basic layout of the wiring diagram.

Central/relay panel

The central/relay panel is indicated in gray at the top of the wiring diagram page. The central/relay panel includes common power circuits, such as battery power (30), ignition switched power (15), load reduction (75/X), and Ground (31).

Ground Connections

Ground connections are represented as a line at the bottom of the wiring diagram page, directly above the current track numbers. All Ground connections, whether they occur as a splice in a harness, or the final Ground source, are numbered and identified in the wiring diagram.

Conductors and components

Between the central/relay panel and the vehicle ground at the bottom of the diagram are located the component symbols and conductors. Components are marked with a component code listed in the legend. Conductors are generally marked with wire color and size.

Current tracks

Individual current tracks are identified numerically along the base of the wiring diagram. These numbers are used to find the continuation of a conductor. Where the system or circuit layout is complex, this continuation may be on the same page, or on a different page.

For example, the number 191 inside of the small box on page 3 indicates that the wire is continued on current track 191. Following straight down on the diagram, we see that this wire is on current track 5. If we were to navigate to current track 191, we would see the same color and size wire with a small box containing the number 5.

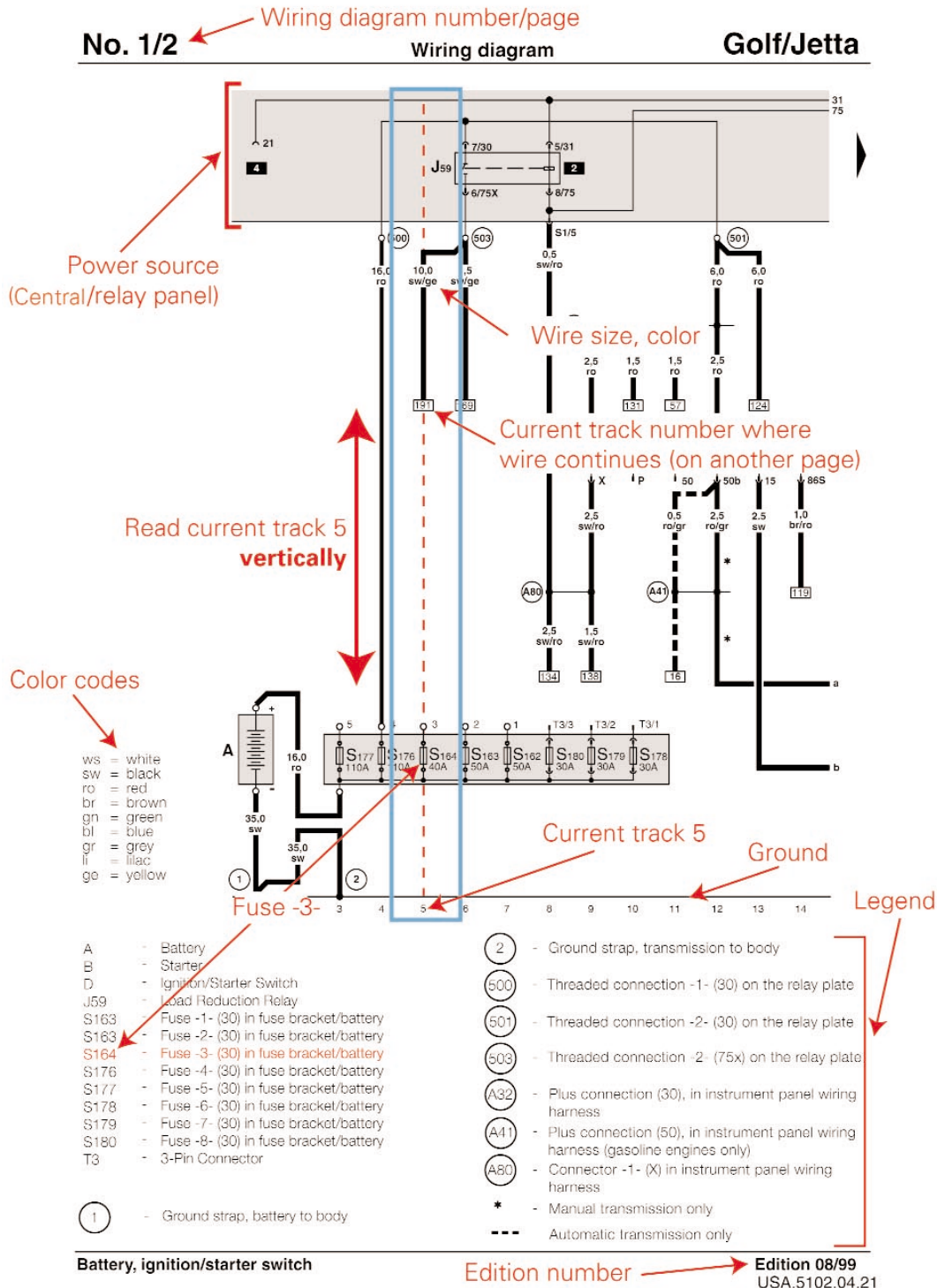
Legend

Below the current track numbers you'll find a legend of the components (by component code) found in the specific diagram. This will often detail the location of a given component or connection.

Wiring Diagram Overview

Navigation

Navigation in the wiring diagram is based around the use of the current track numbers. You will generally start with the affected component and then follow the associated circuit from there, tracing Power, Ground, and signals that affect the component's operation.



Elements of a Wiring Diagram

Elements of a wiring diagram

In this section, we will look at how various symbols are used in a wiring diagram to represent the actual components on the vehicle.

Every circuit needs a minimum of the following to operate:

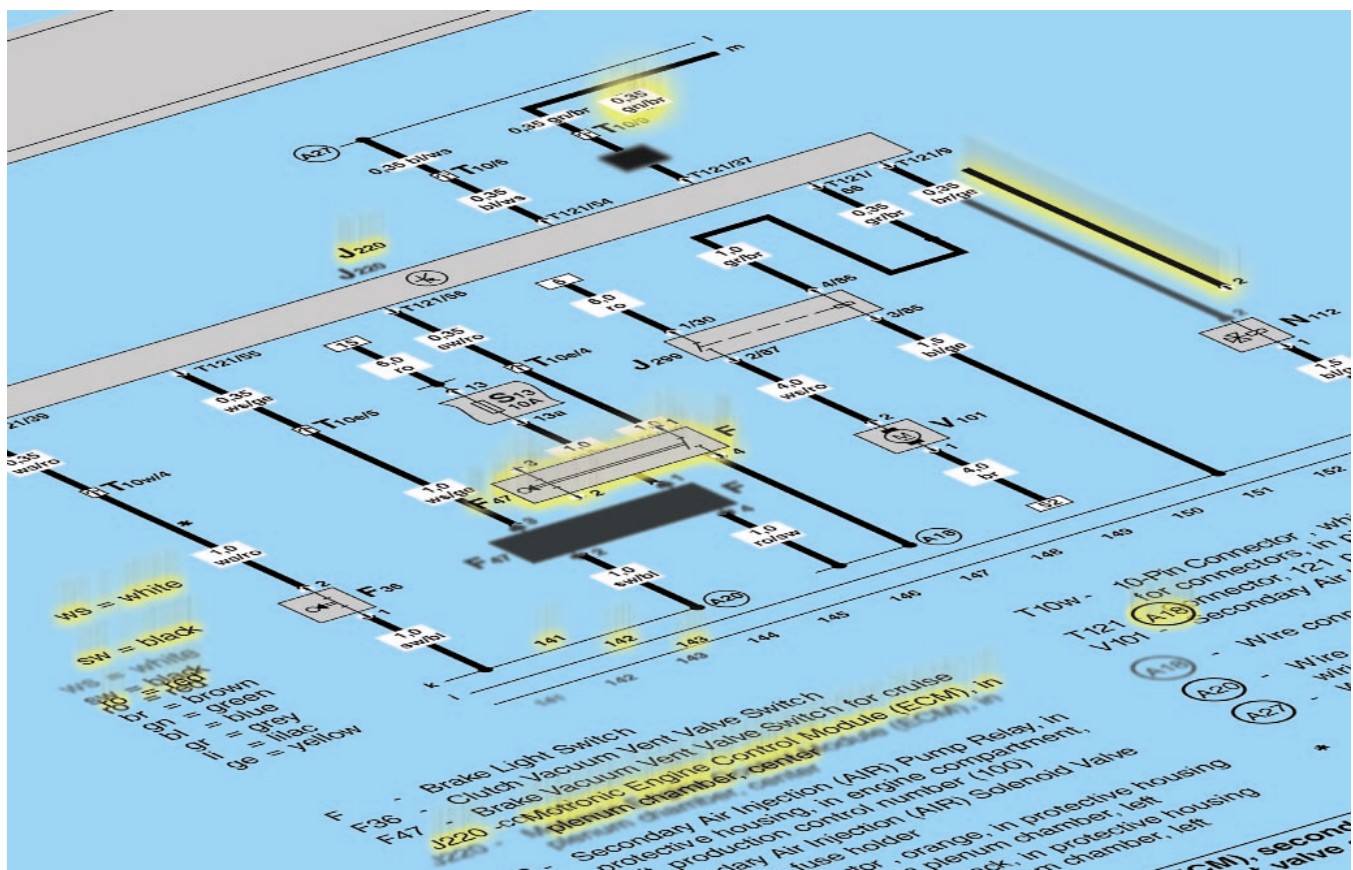
- Power supply
- Consumer (load)
- Ground
- Conductors (usually wire)

If any of these are missing, a complete circuit is broken and the consumer will not function. The ability to break down a circuit into its individual parts is the key to being able to diagnose failures in the circuit.

Wiring diagrams incorporate many symbols used to illustrate a complete circuit. These symbols can include:

- Current track numbers
- Components
- Terminal designations
- Conductors
- Connectors

Together these components make up a complete and accurate wiring diagram.



Symbols

Graphical representations called "symbols" are used to represent components and conductors in wiring diagrams. The key to reading wiring diagrams is understanding the symbols.

These symbols are standardized, allowing quick recognition of various components.

DIN Standard 72 552

This standard applies to the terminal designations for circuits. The purpose of the terminal designation system is to enable accurate connection identification from conductors (wires) to various components when diagnosis and repair is necessary. Examples of DIN standards for terminal designations are shown below:

1	Ignition coil primary
4	Ignition coil secondary
15	Ignition switched, on and start
30	Battery +
31	Ground
31b	Switched Ground
50	Starter control
75/X	Load Reduction/Ignition switched on only

The terminal designations do not identify the wires, but the type of circuit. For this reason, the designations are not placed on the wires in the diagram, but on the component. Refer to Appendix D on page 36 for a more complete list of terminal standards. Be aware that some abbreviations are used, and they may be abbreviations of German words. For example, "GRA" is the abbreviation for "cruise control," and VL is the abbreviation for "left front."



Elements of a Wiring Diagram

Introduction to Conductors

The wiring harness and related components may contain many different types of conductors, including wires, internal connections, threaded connections, welded connections, push-on connectors, multiple point connectors and Ground.

The complete list of wiring connections and symbols is shown in Appendix B.

Wires

Wires are conductors that carry current to components, and are usually indicated by a solid line. A wire shown as a dashed line in a wiring diagram indicates that the wire does not apply to all vehicles, and is noted in the wiring diagram legend.

Wire colors

Knowing the standards for wiring colors makes the job of reading and interpreting them easier. Some of the common standards include wiring color for specific circuits, as well as the terminal designation. For example:

Red	Battery +
Green.	Ignition (1)
Brown	Ground (31)
Yellow	Headlights (58)

Once the technician has an understanding of the color guides, the job of isolating systems becomes easier.

Wire colors are shown as abbreviations of the German word for the color.

The following list shows the German abbreviations to the English text for the most common colors. Refer to appendix C for other color code definitions.

bl.	Blue
br.	Brown
ge	Yellow
gn	Green
ro.	Red
sw	Black
li	Violet
ws	White

Wire sizes

Wiring diagrams also indicate the wire gauge used (shown in mm²), designating the cross sectional area of the wire. Because standards exist for the maximum permissible voltage drop across a circuit, wire gauge is critical. If the voltage drop across the wire is too high, one or more of the following may occur:

- The circuit may overheat
- The consumer may not operate properly (due to low voltage condition)
- Components may be damaged

If a wiring repair needs to be made and metric sized wire is not available, the technician may need to use American Wire Gauge (AWG) sized wire.

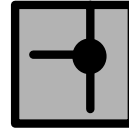
Note:

- *If the exact size wire is not available for a repair, use the next larger size.*
- *For more information on wiring, refer to the **Wiring Harness Inspection and Repair** SSP (course number 871003).*

Elements of a Wiring Diagram

Other conductor descriptions

Internal conductors exist inside components, acting as bridges between the wiring harness and the final consumer. In some components these conductors are labeled in the component. An example would be the 30 circuit (Battery +) in the central/relay panel. In other components the conductor is not labeled. Internal conductors are shown as thin, black lines.



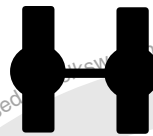
Physical contact

Some components, such as the starter or generator, may receive Ground where they are bolted to the engine or transmission. This is also shown as a thin black line.

Welded connections are used in wiring harnesses to join multiple, smaller gauge wires to a single larger wire which terminates at the fuse relay panel or chassis Ground.

Sometimes a welded connection is shown with the thin line not terminating at another wire. This symbolizes that this welded connection is used in other diagrams for the car. The technician may need to reference other diagrams to locate components or Grounds related to this connection.

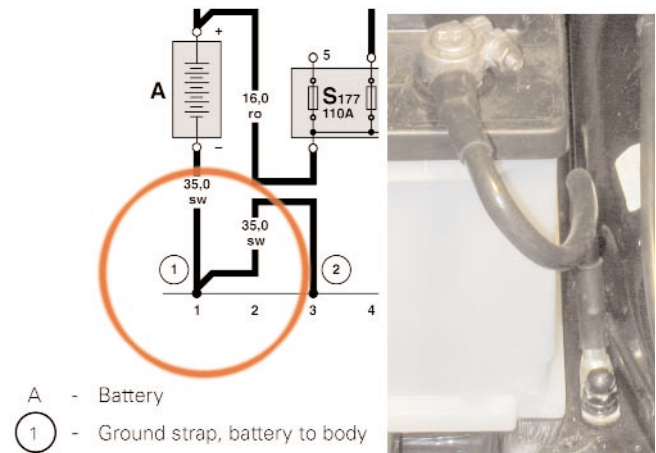
Threaded connectors are commonly used on the bottom of the fuse relay plate to distribute power and Ground to components. The common connectors include Battery power, Ground and load reduction (X).



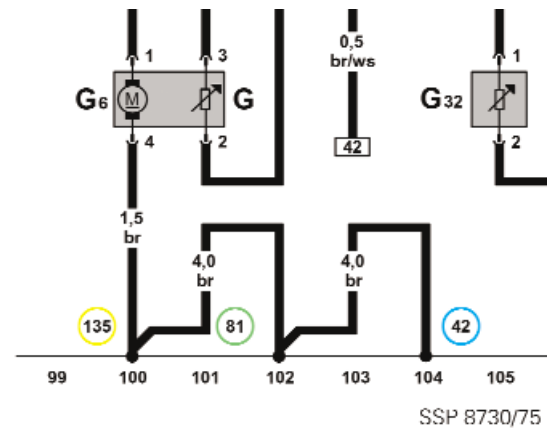
Elements of a Wiring Diagram

Ground

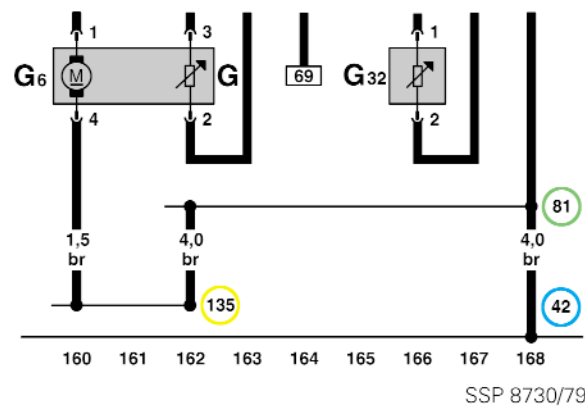
The vehicle chassis acts as a common Ground conductor throughout the vehicle, and is identified by the line at the bottom of the wiring diagram above the current track numbers.



Some wiring diagrams will show ground designations at the bottom of the wiring diagram that are actually welded ground connections in the harness (see illustration *SSP 8730/75*). In practice, you may need to search through the legend to find where a welded ground connection finally attaches to the vehicle chassis.



Other wiring diagrams will show welded Ground connections in the wiring harness, as shown in illustration *SSP 8730/79*.



- 42 - Ground connection, beside steering column
- 81 - Ground connection -1-, in instrument panel wiring harness
- 135 - Ground connection -2-, in instrument panel wiring harness

Elements of a Wiring Diagram

Connector pin assignments

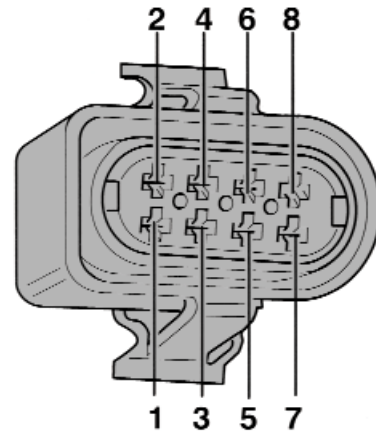
Wiring diagrams tell the user at which pin numbers the wires terminate, simplifying diagnosis.

There are 4 main types of terminal designations:

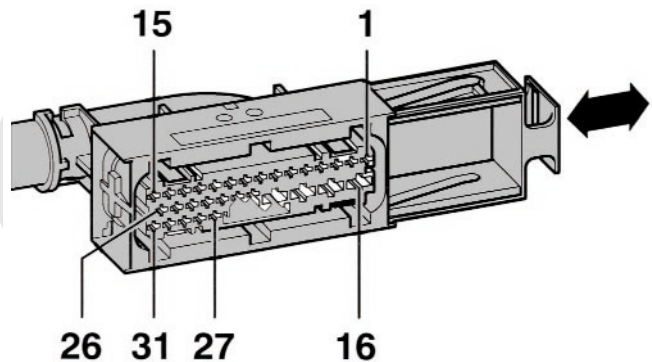
- Push-on/multi-point connections
- Component/multi-point
- Central/relay panel
- Relay

Push-on/multi-pin connectors use the "T" designation, and are identified in the legend. For example, T8a/5 designates an 8-pin connector, with the specified wire located in terminal 5 of the connector housing. The legend will give additional information about this connector. For example: "8-Pin connector, brown, in engine compartment, in wiring duct, left" (see Wiring Diagram example 41 beginning on page 61 for specific examples).

Generally, pin assignments are labeled on the plastic hard-shell connector housing and/or the corresponding component. On larger connectors, pin assignments are labeled at either end of a row. For example, the Engine Control Module (ECM) plug often has 2 or 3 rows of 12 or more terminals. Each row will be marked on each end to facilitate easier diagnosis.



SSP 8/30/33



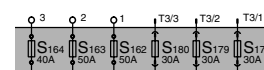
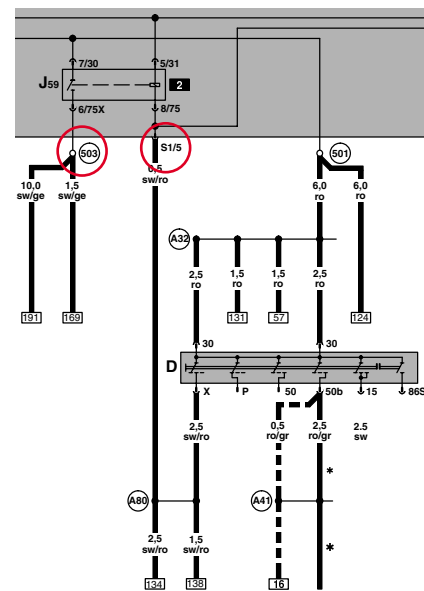
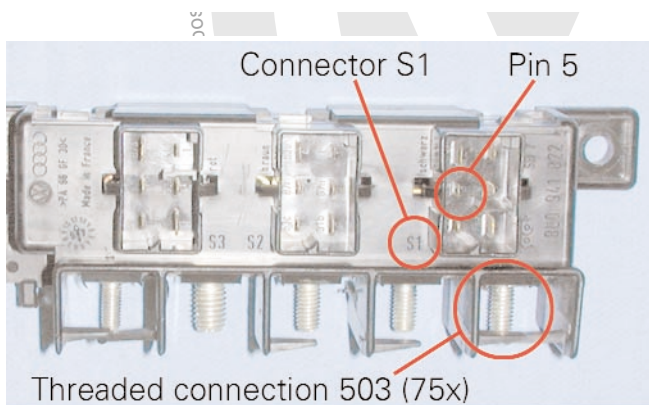
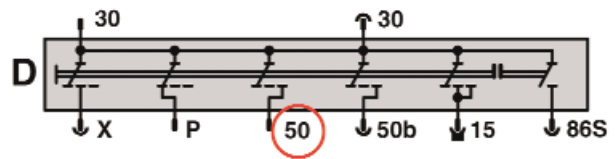
SSP 8730/48

Elements of a Wiring Diagram

Component/multi-point pin assignments may or may not use the "T" designation. Some are numbered sequentially. Others may use DIN circuit designations, or a combination of these methods. See example at right, and find two examples in wiring diagram 29 (Appendix E).

Central/relay panel connections enter or exit as either threaded connections, or multi-point connectors. Threaded connectors are identified in the legend, and may be identified on the component with the DIN circuit designation.

Multi-point connectors on the back of the panel are not identified in the legend. They will have a letter and possibly a number to identify location, followed by the pin number. For example S1/5 would be connector S1, pin 5.



SSP 8730/55

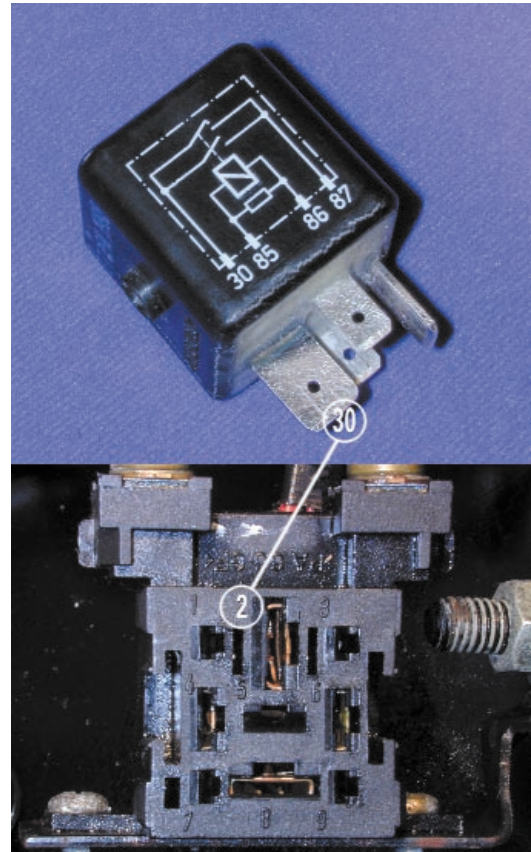
Elements of a Wiring Diagram

Relay connections list the terminal cavity number on the relay socket, followed by the pin identification on the relay. Illustration SSP 8730/65 shows that cavity 2 of the harness connector is associated with the DIN terminal 30 of the relay. Both may be used during diagnosis of the circuit.

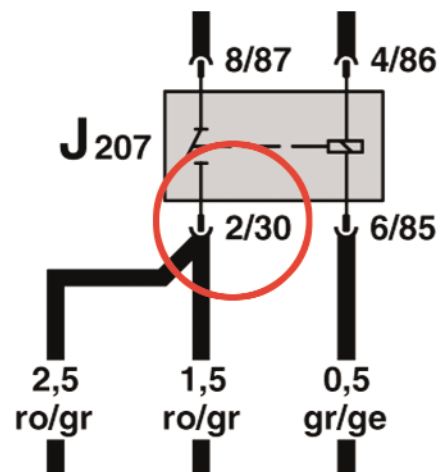
Note:

When diagnosing electrical concerns using the VAG 1598 pin-out box, the pin numbers on the control module and multi pin connector usually match the pin numbers on the tool.

When using the VAG 1466 pin-out box, the numbers on the relay or control module and socket usually **do not** match those on the tool. These must be noted before beginning diagnosis.



SSP 8730/65



J207 - Starting Interlock Relay

SSP 8730/76

Elements of a Wiring Diagram

Components

Components in wiring diagrams are given an alphanumeric designation for identification. The first portion of the code separates the component into basic groups. An *F* for example, designates a switch, while a *Z* would be used for a heating element. A complete list of these designations are shown in Appendix C on page 43. The second part (numeric) designates which component is covered inside of these main groups.

Example: Evaporative Emissions (EVAP) Canister Purge Regulator Valve N80. N designates a solenoid valve, 80 clarifies which solenoid is being dealt with.



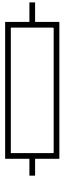
SSP 8730/57



SSP 8730/56



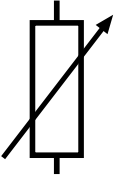
SSP 8730/21



Resistor

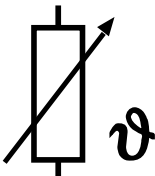
A commonly used symbol is the **resistor**, used to create a voltage drop in a circuit. Below are three types of resistors:

A **standard resistor** has a fixed resistance.



Variable Resistor (Rheostat)

A **rheostat**, or **potentiometer**, varies its resistance based on mechanical input. An example of this is the Throttle Position Sensor (TPS) on a Motronic equipped car (SSP 8730/57).



Temperature Dependent Resistor

A **temperature dependent resistor** varies its resistance based on temperature. This type of resistor is used as the Engine Coolant Temperature (ECT) sensor on an engine management system, or the Interior Temperature sensor on models with climate control (SSP 8730/56).

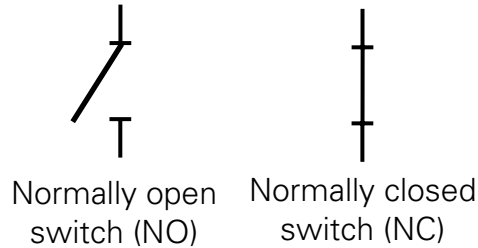
When a resistor is used as a sensor, it generally carries a component designator "G."

Note:

Any time the temperature symbol (left) is attached to another symbol, it signifies that the operation of that component will vary with temperature.

Elements of a Wiring Diagram

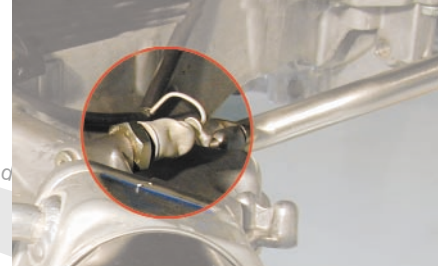
The **switch** is another component used to control current flow. The basic symbols for an open and closed switch are shown in illustration SSP 8730/22. An example of this simple two-position switch would be a glove compartment light switch. A simple switch uses the component designation "F".



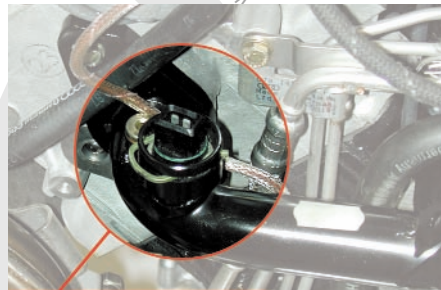
SSP 8730/22

There are many types of designs, including mechanically actuated, pressure actuated, temperature actuated and momentary. Examples include:

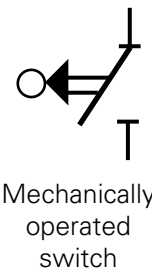
- Oil pressure switch (pressure actuated)
- Cooling fan thermo-switch (temperature actuated)
- Brake Switch -F- (mechanically operated)



SSP 8730/60



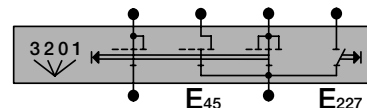
SSP 8730/59a



SSP 8730/58

More complex circuits may require a switch with multiple sets of contacts. An example of this would be the cruise control switch.

As you can see in illustration SSP 8730/25, there are multiple sets of contacts within the assembly. Depending on the position of the switch, various sets of contacts are open or closed. Careful study of the symbols allows the technician to follow the circuit through the switch under any condition.



SSP 8730/25

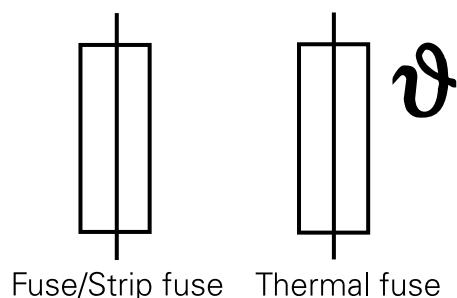
A complex switch uses the component designation "E".

Note:

All switches and relays are shown in a non-operated state.

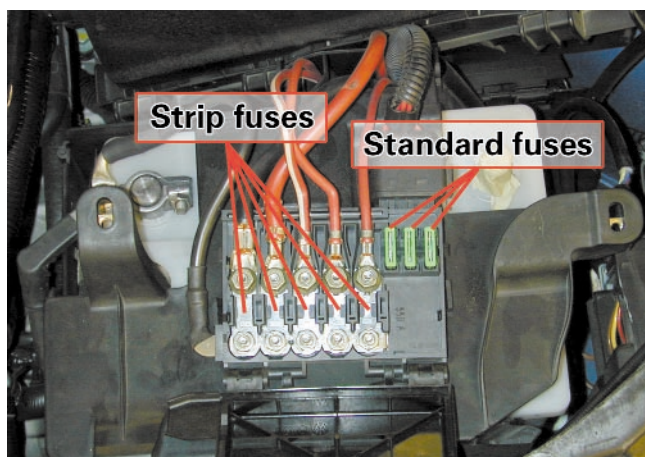
Elements of a Wiring Diagram

Fuses are used to prevent excessive current from damaging other components in a circuit. There are various types, including standard fuses, thermal fuses (circuit breakers), and strip fuses. Fuses use the component designation "S"; their symbols are shown below:



SSP 8730/23

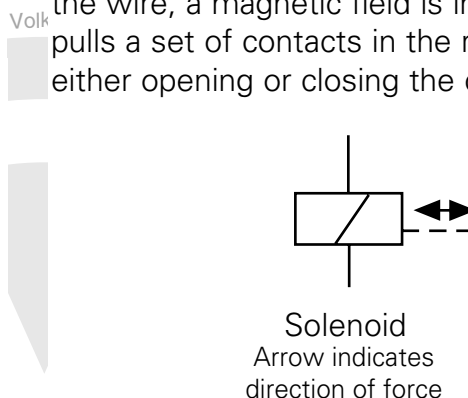
The example below shows strip fuses and standard fuses as seen in a current model vehicle.



SSP 8730/45

Solenoids are used to actuate many different components, including fuel injectors and relays.

A solenoid is a coil of wire wrapped around an iron core. When current is passed through the wire, a magnetic field is induced. This pulls a set of contacts in the relay closed, either opening or closing the circuit.



SSP 8730/24

Complex Symbols

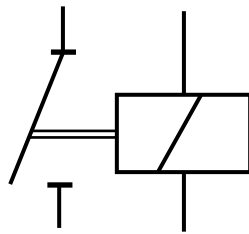
Often the internal schematic of the component is shown to allow the technician to follow current flow through the component.

These internal symbols are a combination of several basic symbols. This allows the technician to take a more complex symbol and break it down into its smaller components. Even the most complex components are nothing more than a combination of smaller basic symbols.

More complex components may contain complex control circuitry. This will be indicated with the symbol of a transistor in the component symbol (see control module).

Elements of a Wiring Diagram

A **relay** is an example of a combination of symbols in a single component.

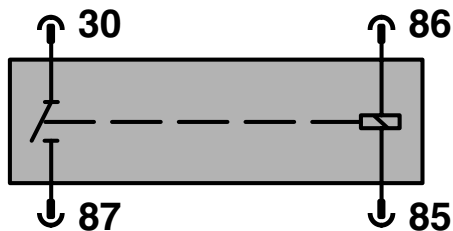


Relay

SSP 8730/26

Relays require a signal from an outside source to activate. Volkswagen vehicles use common Bosch®-type 4-pin relays on many circuits, to remove electrical load from the switch. Relays share the component designator "J" with control units.

The basic 4-pin relay (below) contains two separate components: a switch and a solenoid.



SSP 8730/49

The coil in the solenoid is energized with low current, creating a magnetic pull that closes or opens the switch.

Note:

All switches and relays are shown in a non-operated state.

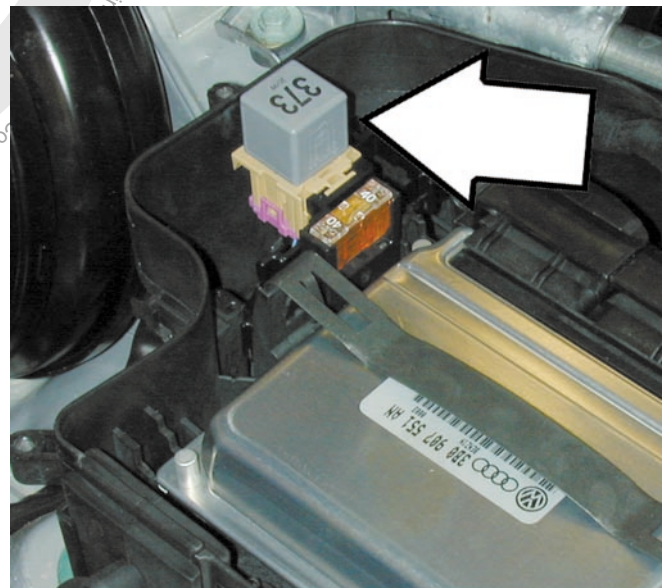
The 4 pins of a standard relay are generally (but not always) numbered as follows:

- 30** Receives Battery power (switched to consumer)
- 87** High load to the electrical consumer
- 86** Ignition switch, Battery+, load reduction (X)
- 85** Receives a switched ground to activate the solenoid winding in the relay

Note:

A production number may appear on top of the relay (see illustration SSP 8730/61). This number may be referred to in the wiring diagram for diagnostic purposes. See page 45 for examples.

However, do not rely solely on this information when diagnosing a circuit, as this number may change in production. Always refer to the parts information system for the current replacement part numbers.



SSP 8730/61

Elements of a Wiring Diagram

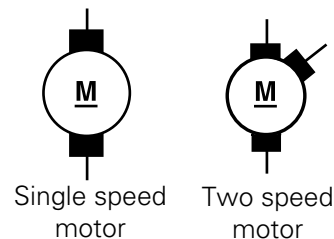
The last of the more common symbols we will look at are the **Electric motor** and the **Electronic Control Module**.

Electric motors are used throughout the vehicle for numerous applications, including the Windshield Wiper Motor and the Fuel Pump.

At the top and bottom of the motor symbol there are dark squares, representing the brushes in the electric motor.



SSP 8730/62



SSP 8730/28

If there are only two brushes, it is a single speed motor. The rear window wiper motor shown in illustration SSP 8730/62 is an example of a single speed motor.

If there are three brushes, the motor has two speeds. In the case of the two speed motor, the upper set of brushes are the low and high speed brushes.

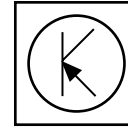
The windshield wiper motor shown in illustration SSP 8730/66 is an example of a two speed motor.



SSP 8730/66

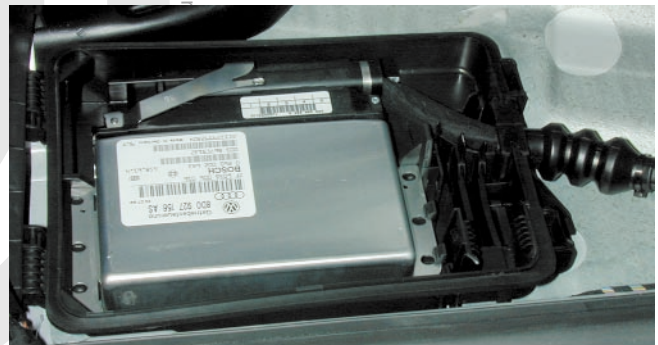
Elements of a Wiring Diagram

Control Modules can make the task of reading and interpreting wiring diagrams more difficult because multiple signals enter and/or exit, but the internal schematic is not always shown.



Control Module
SSP 8730/44

The Transmission Control Module (TCM) J217 seen in illustration SSP 8730/63 is an example of a common control module.

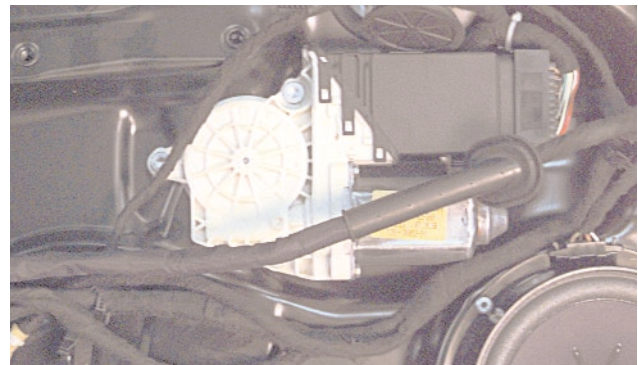
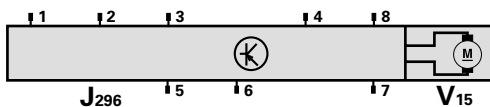


SSP 8730/63

Some control modules may be integrated with other components.

Take, for example, the window motor with control module. Both the motor and the control module are combined to make a single component (SSP 8730/74).

The symbol below represents the combined components.



SSP 8730/74

Elements of a Wiring Diagram

Layout

- 1. Relay location number**
Indicates location on relay panel. See page 1 of individual wiring diagram for details.
- 2. Arrow**
Indicates wiring circuit is continued on the previous and/or next page.
- 3. Connector designation - relay/control module on relay panel**
Shows relay panel terminals with corresponding relay terminal.
For example: 17/30 equals terminal 30 of relay connects to terminal 17 of central relay panel.
- 4. Threaded pin on relay panel**
White circle shows a threaded removable connection.
- 5. Fuse designation**
For example: S228 equals Fuse 28, 15 amps, in fuse holder.
- 6. Reference of wire continuation (current track number)**
Number in frame indicates current track where wire is continued (see page 3 for example).
- 7. Wire connection designation in wiring harness**
Location of wire connections are indicated in the legend.
- 8a. Terminal designation on a multipoint connector**
8b. Terminal designation on a component
Designation which appears on the actual component and/or terminal number of a multi-point connector.
- 9. Ground connection designation in wire harness**
Locations of ground connections are indicated in legend.
- 10. Component designation**
Use legend at bottom of page to identify the component code.
- 11. Component symbols**
A graphical representation of a component type. See Appendix A, page 35.
- 12. Wire cross section size (in mm²) and wire colors**
Abbreviations are explained in the color chart beside the wiring diagram.
- 13. Component symbol with open drawing side**
Indicated component is continued on another wiring diagram. The number of the corresponding wiring diagram can be found in the table of contents.
- 14. Internal connections (thin lines)**
These connections are not wires. Internal connections are current carrying and are listed to allow tracing of current flow inside components and wiring harness.
 - a. Internal Harness Splice (Welded Connection)**
 - b. Physical Contact (Mounted to engine)**
- 15. Reference of continuation of wire to component (inset)**
For example: Control module for anti-theft immobilizer J362 on 6-pin connector terminal 2.
- 16. Central Relay panel connectors**
Shows wiring of multi-point or single connectors on central relay panel.
For example: S3/3 equals Multi-point connector S3, terminal 3.
- 17. Reference of internal connection continuation**
Letters indicate where connection continues on previous and/or next page.
- 18. Central Relay Panel**
- 19. Ground Path**
ex.: from welded harness connection 135 to welded harness connection 81 to welded harness connection 42.

Wiring Diagram Number and Page



Oil pressure switch, fuel pump (FP), fuel pump (FP) relay, engine coolant level sensor, speedometer vehicle speed sensor (VSS)

Working With Wiring Diagrams

Working with wiring diagrams

When working to diagnose electrical concerns on vehicles, it is important that you have the correct wiring diagram. Not only are there different models and year, wiring harnesses will often change in the middle of a model year. There may be differences between models of the same type and year, but built in different factories.

Within models, there are also different levels of equipment. For example, the wiring for the fresh air blower of a vehicle with Climatronic will be different from one without Climatronic. Different engines, transmissions, even trim levels will mean differences. Always check that the wiring diagram is right for the vehicle in question.

It may be necessary to check more than one wiring diagram for the model you are servicing. A vehicle's wiring harness is often split into several different wiring diagrams. Engine, transmission and power accessory wiring diagrams will only show wiring for those specific systems.

When choosing a wiring diagram for diagnosis of an electrical concern, you should first confirm the vehicle model, model year, and production date, where applicable. Next, identify if the affected system is part of vehicle "standard equipment" or "additional equipment". Locate the appropriate wiring diagram by using the index.

A standard equipment diagram will show the vehicle with its base level wiring. Base level wiring is defined as the most basic possible rolling chassis, and this may be different than what is normally considered "standard equipment".

For example, all 2001 m.y. Golfs and Jettas sold in the US and Canada come with Daytime Running Lamps as "standard equipment". However, the Daytime Running Lamps are shown in a separate wiring diagram.

Wiring diagrams are automatically shipped to your dealer. Periodic updates can include both new wiring diagrams and updated pages for existing diagrams.

Wiring diagrams also exist as part of the Volkswagen Electronic Service Information System (VESIS). The advantages of electronic wiring diagrams include easy updating, pages that don't get dirty or lost, and search capabilities that can make searching for a component much easier.

Practical Examples

The preceding sections of this book gave examples of the symbols and layout of our current track wiring diagrams. In the following section we will examine current flow using navigation exercises. In these examples, we will look at:

- how battery power is provided to a component
- how consumers are actuated using relays
- how to split up a circuit to simplify diagnosis
- the importance of looking in multiple wiring diagrams

Appendix E, starting on page 39, contains the complete wiring diagrams number 29 and 41. Within the navigation exercises, magnified views of specific sections of these Wiring Diagrams may be shown. Note that, for clarification purposes, certain components found in the Wiring Diagrams may not be shown in these magnifications.



Navigation 1



Navigation 1

In this example we will look at the circuit for the load reduction relay. The following page contains the complete diagram for this example.

The Load Reduction circuit supplies power to consumers such as the windshield wiper motor and fresh air blower motor when the ignition switch is in the "ON" position. With the ignition switch in the "START" position, the circuit is de-energized. This lowers the load on the electrical system when starting the vehicle.

Locate the Load Reduction Relay J59 on the wiring diagram on page 23. The relay consists of two separate circuits: the solenoid circuit, and the switch circuit. In order to supply power to the consumers in the load reduction circuit, the solenoid in the relay must be energized. To do this, terminal 8/75 must be supplied with power. The ground side of the solenoid receives an internal ground from the central relay panel at pin 5/31.

Starting at the Positive (+) terminal of the Battery (A), follow battery power (30) to the fuse box through a 16.0 ro wire.

Power then passes through S176, a 110A fuse, and exits the fuse box at terminal 4. From terminal 4, follow the 16.0 ro wire to threaded connection 500 on the central relay panel.

Power passes through the central relay panel and exits at threaded connection 501. From the relay plate, follow the 6.0 ro wire to welded connection A32 in the instrument panel wiring harness.

Note that the wire that supplies the power to the welded connection is larger than other wires that leave the connection. This fact can be useful in diagnosis. When tracing out a weld with many wire connections, the largest wire will usually be the one that goes directly to power or ground.

From the welded connection, a 2.5 ro wire goes to terminal 30 of the Ignition/Starter switch D. When the Ignition switch is moved to the "ON" position, power continues through the switch to the X contact. Power exits the Ignition switch through the 2.5 sw/ro wire on current track 9.

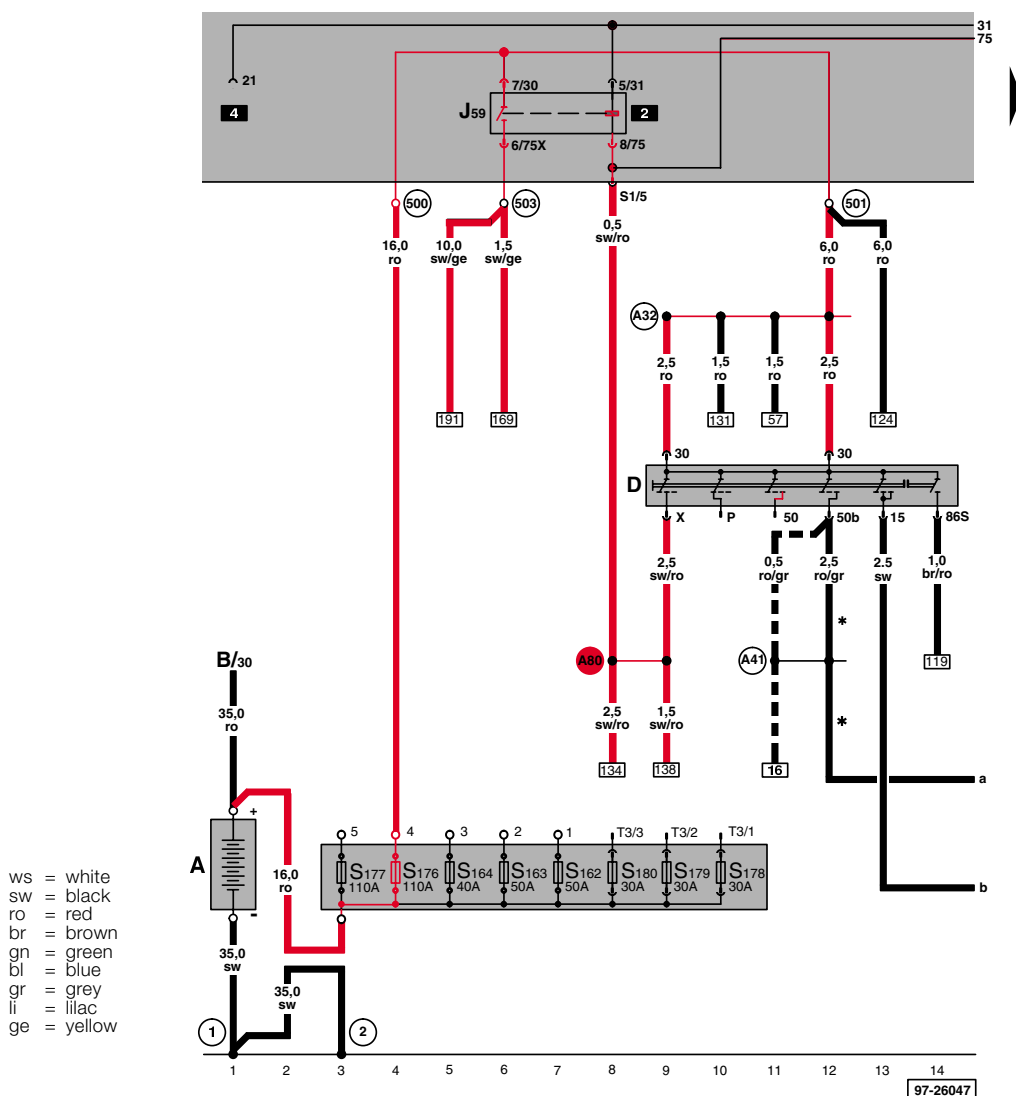
Follow the wire to welded connection A80. Another similar wire leaves this welded connection on current track 8, which connects to the central relay panel at terminal S1/5. Power flows from this terminal through the internal connection to terminal 8/75 of the relay.

Because relay terminal 5/31 is already grounded at the central relay panel, the solenoid is energized. This closes the switch and supplies power from terminal 500 (30), to terminal 503 (75/X) on the relay panel. From terminal 503, power can flow to the consumers on the circuit as needed.

No. 1/2

Wiring diagram

Golf/Jetta



- | | | | |
|------|---|-------|--|
| A | - Battery | (2) | - Ground strap, transmission to body |
| B | - Starter | (500) | - Threaded connection -1- (30) on the relay plate |
| D | - Ignition/Starter Switch | (501) | - Threaded connection -2- (30) on the relay plate |
| J59 | - Load Reduction Relay | (503) | - Threaded connection -2- (75x) on the relay plate |
| S163 | - Fuse -1- (30) in fuse bracket/battery | (A32) | - Plus connection (30), in instrument panel wiring harness |
| S163 | - Fuse -2- (30) in fuse bracket/battery | (A41) | - Plus connection (50), in instrument panel wiring harness (gasoline engines only) |
| S164 | - Fuse -3- (30) in fuse bracket/battery | (A80) | - Connector -1- (X) in instrument panel wiring harness |
| S176 | - Fuse -4- (30) in fuse bracket/battery | * | - Manual transmission only |
| S177 | - Fuse -5- (30) in fuse bracket/battery | --- | - Automatic transmission only |
| S178 | - Fuse -6- (30) in fuse bracket/battery | | |
| S179 | - Fuse -7- (30) in fuse bracket/battery | | |
| S180 | - Fuse -8- (30) in fuse bracket/battery | | |
| T3 | - 3-Pin Connector | | |
| (1) | - Ground strap, battery to body | | |

Battery, ignition/starter switch

Edition 08/99
USA.5102.04.21



Navigation 2

This example will follow the flow of power from its source, through a switch, and to a consumer. This example will build on our understanding of the Load Reduction circuit as seen in Navigation example 1.

The example we will use is a typical horn circuit. The horn circuit on all Volkswagen vehicles is similar in design. The main components of the system are as follows.

- Mechanically operated switch (horn button)
- Relay
- Horns (high and low tone)
- Connecting wires

In order to follow these types of circuits, it is best to split the circuit into two sections. We will first look at the solenoid side of the relay circuit, then the switch side.

Note:

Refer to wiring diagram 29 in appendix E for this example.

From the table of contents for wiring diagram 29 (page 39), we see that the horn circuit is shown on page 29/15.

Locate the horn relay (J4) on page 29/15. Starting at the solenoid side of the symbol, follow the internal connection from terminal 4/75 across the central relay panel until it terminates at an internal connection that runs between terminal S1/5 on the back of the central relay panel, and terminal 8/75 of relay J59 on wiring diagram page 29/2. Looking at the legend, we see that J59 is the Load reduction relay.

Looking at the wiring for J59 we see that the internal connection we are concerned with here is the power that activates the load reduction relay. Follow the 0.5 sw/ro wire down current track 8 to plus connection A80, across to current track 9 and up to terminal X of component D.

We can see that power will flow from the battery, through fuse S176, and through the ignition switch to the horn relay when the ignition switch is in the on position. Now, back on page 29/15, let's take a look at the ground side of the Horn relay solenoid.

From J4, follow the ground side of the circuit from terminal 1/71 of the relay to S1/6. A 0.35 br/bl wire runs from the central relay panel to T5b/3. The legend identifies T5b as a 5 pin connector on the airbag spiral spring (F138).

From the spiral spring, the circuit continues through another 5 pin connector, T5j, through manually operated push button switch H (the horn contact), and then back through the spiral spring to T5b/2. From this terminal a 0.35 br wire runs to welded ground connection 135, in the instrument panel wiring harness.

Looking at 135, we see no less than 9 wires in the weld! In practice, you may need to trace out every wire in the weld to find a problem. Let's use the trick we discussed in the last example to narrow down the field. Looking at the left side of connection 135, we see a 4.0 br wire connecting to another internal connection, marked with the continuation "o."

Follow "o" in both directions. It is identified as welded ground connection 81, in the instrument wiring harness. It is connected to ground at two points; ground 49 at current track 182, and ground 42 at current track 201. We now have the complete circuit for the solenoid side of the relay.

Let's look at the horns and the switch circuit of the relay. Follow the internal connection "m" from terminal 3/30 at the relay to the fuse box at current track 153. Continue to follow the source of the power through the fuse and the current track continuation back to page two of the wiring diagram.

As seen in Navigation example 1, the power comes from the battery, through fuse S176, to threaded connection 500. From there it runs through the central relay panel to threaded connection 501, to the fuse box at fuse 40 (S240), and to the relay through the central relay panel. When the relay is activated by pushing the horn contact, power will flow to the horns through S2/2, through welded connection A90, to terminal 2 on each horn.

Tracing the ground side of the horns, we see that they jump to current tracks 74 and 75 from tracks 183 and 185, respectively. Going to current tracks 74 and 75, follow the brown wires to internal connection marked "d". We follow that along to find that it is welded connection 179, which goes to Ground through a 4.0 br wire at ground connection 12, in engine compartment left, shown on current track 61. We now have the complete circuit for the horns.

Navigation 3



Navigation 3

In this exercise, we will again examine the activation of an electrical consumer through a simple 4-pin relay. Let's look at the navigation from the point of view of diagnosing an inoperative motor. In this circuit we will look at the Secondary Air Injection (AIR) pump motor V101.

To diagnose this circuit, we need to understand that the secondary air system is activated by the Motronic ECM at certain times when the engine is running, and that we can trigger this function using the VAS 5051 scan tool. Assume that we have already attempted this, and the pump does not run. Also assume that the only DTC in the systems memory is the Secondary Air DTC.

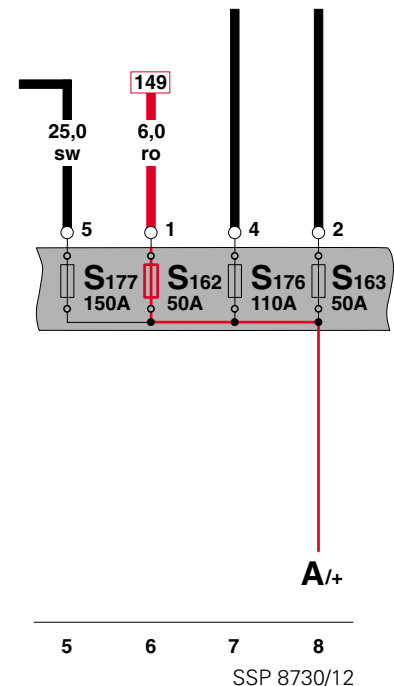
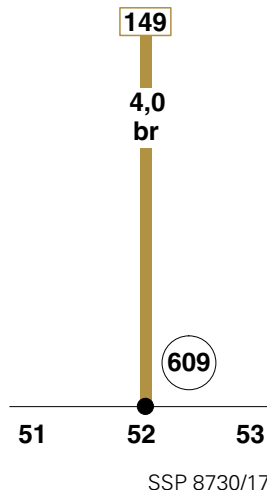
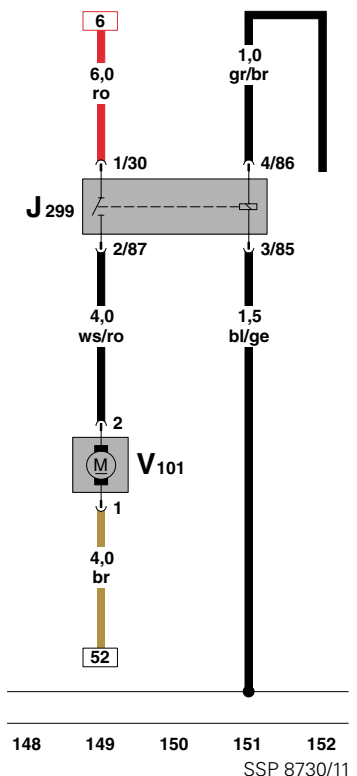
Note:

Refer to the illustrations on pages 26-28 and wiring diagram 41 in appendix E for this example.

From the table of contents for wiring diagram 41 (page 40), we see that the Secondary Air circuit is shown on page 41/12.

Locate the Secondary Air Injection (AIR) pump relay J299 on wiring diagram page 41/12. We will once again split the circuit into two parts to simplify diagnosis. Let's start with the switch side of the relay.

From motor V101, follow the 4.0 br wire down to the current track continuation (illustration SSP8730/11), and over to page 41/5. The wire goes directly to Ground 609 in the right side of the plenum chamber. Now let's trace the power side.



Returning to current track 149, we see that the 4.0 ws/ro wire runs to the switch portion of the relay (illustration SSP8730/11).

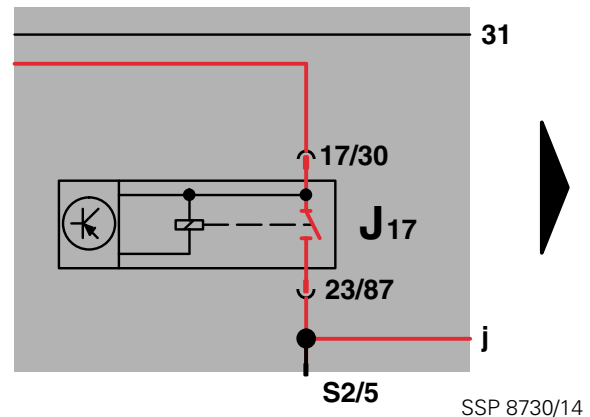
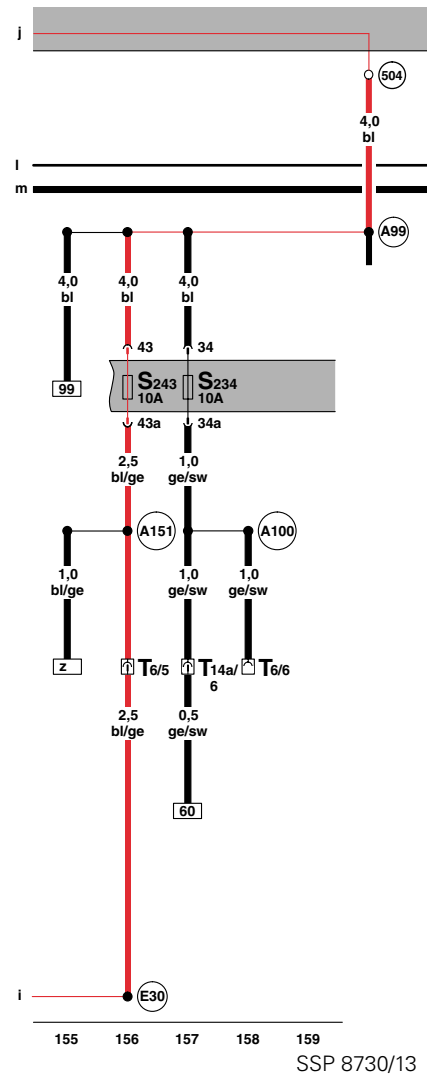
Following the 6.0 ro wire from where it leaves terminal 1/30 on the relay, through its current track continuation, we see that the motor receives direct battery power (30) from fuse S162. Checking the fuse on the vehicle shows it to be good.

Locate relay J299, and pull it from its socket. Using a wire from the VAS 1978 wiring repair kit of the proper size, and with the proper size terminals, we can jump the relay socket from terminal 1/30 to 2/87. The Secondary Air Injection (AIR) pump motor V101 runs, verifying that this section of the circuit is functioning correctly.

Let's trace out the power side of the solenoid in the relay. Terminal 85 on a standard 4-pin relay is usually the power side. Following the 1.5 bl/ge wire down, we see that it leads to welded connection continuation "i".

In both directions along "i", we see that this is welded connection E30 in the engine wiring harness. It supplies power to many of the actuators in the engine management system. Since there are no other faults in the Engine Management System, we can assume that fuse 43 (S243) on current track 156 is good. To test this circuit correctly, let's trace that power back to its source.

Following the circuit from Fuse S243 up to the central relay panel and over internal connection "j", we see that this power comes from the switch side of J17, the fuel pump relay. This is important for testing. We know that we will only have power at terminal 3/85 of the Secondary Air Injection (AIR) pump relay J299 when the fuel pump relay is energized.

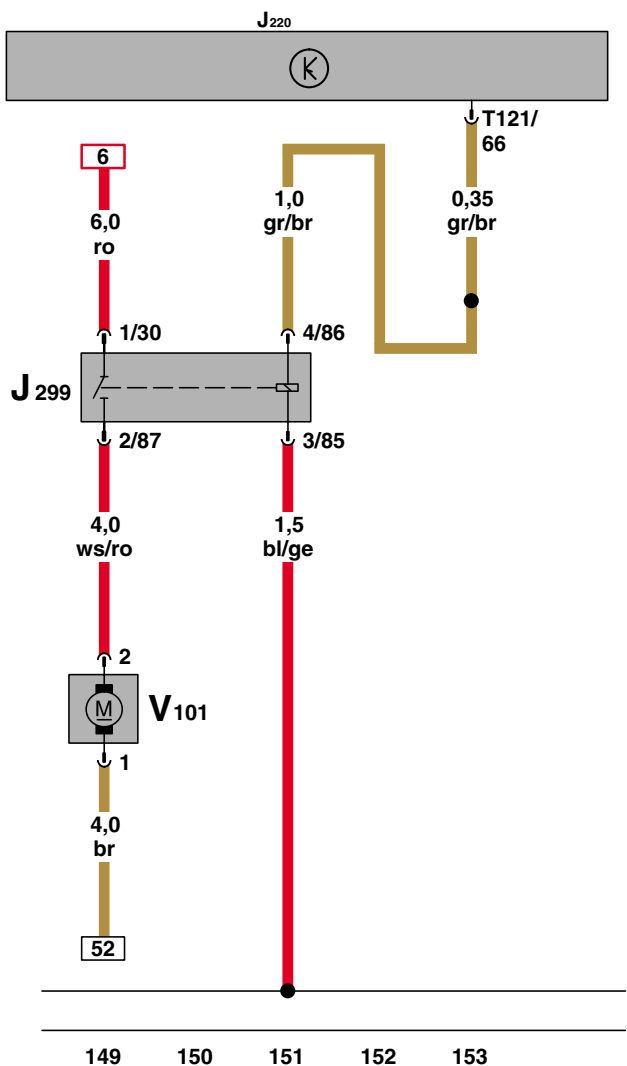


Navigation 3

Go to the Secondary Air pump relay J299 on page 41/12 (or see below). Follow the 1.0 gr/br wire from terminal 4/86 to terminal T121/66 on the Motronic ECM J220. Note that somewhere in the harness it appears that the wire enters an unidentified connection, and changes size. This wire carries the ground signal that energizes the relay.

Now let's test the solenoid side of the circuit. Connect the appropriate test equipment, such as an LED test light, multimeter, or the Digital Storage Oscilloscope of the VAS 5051, between terminals 4/86 and 3/85 of the relay socket. We then activate the circuit (using the VAS 5051) and see that the proper signal is going to the relay socket.

Since the circuit appears to be complete, the next step would be to replace the relay and retest. Doing this, we find that the circuit operates correctly. We can now check, and if necessary, erase any DTCs, quality check the vehicle, and return it to the customer.



SSP 8730/16



Navigation 4

Often, when we are diagnosing an electrical problem on a vehicle, we will need to look at more than one wiring diagram. In this final example, we will look at a simple problem that shows the importance of considering all of the applicable wiring diagrams for the vehicle that we are diagnosing.

Let's consider a customer concern of an inoperative speedometer. The vehicle is a 2000 Jetta GLS, equipped with a 1.8T engine and manual transmission.

A road test verifies that neither the speedometer nor the odometer is operating. There is also a DTC stored in the engine ECM memory, showing no signal from the vehicle speed sensor (VSS).

Note:

Refer to wiring diagrams 29 and 41 in appendix E for this example.

From the table of contents for wiring diagram 41 (page 40), we see that the vehicle speed sensor circuit is shown on page 41/14.

Looking at wiring diagram page 41/14, we locate the VSS (G22), and trace the power back to fuse 7 (S7). We find that the fuse is blown. After locating the 14 pin connector in the engine compartment wiring duct (T14a), and some testing with a multimeter, we find there is a short to ground between terminal 7a on the fuse box, and terminal T14a/2.

Looking at welded connection B163 shows it to be open ended. This often indicates that the weld is continued on another wiring diagram. Since we located the VSS in the 1.8T engine wiring diagram, one place to look would be the Standard Equipment wiring diagram for this vehicle, number 29.

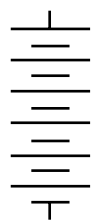
Looking through the legend at the bottom of each page in wiring diagram 29, we find weld connection B163 on page 29/3.

Following the current track continuation from B163 to current track 122, we find that the 1.0 sw/gn wire runs to the backup light switch. In the engine compartment, we find that this wire has been pinched in the engine compartment wiring harness near the starter. We repair the wire using the VAS 1978 wiring repair kit, replace the fuse. Check, and if necessary, erase any DTCs, quality check the vehicle, and return it to the customer.

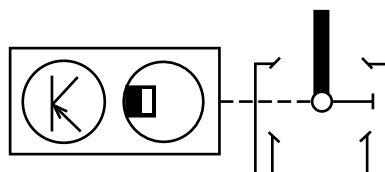
One thing to note: in vehicles equipped with an automatic transmission, the circuit for fuse S7 would be different. We would need to look in the wiring diagram for that system as well.



Appendix A: Component Symbols



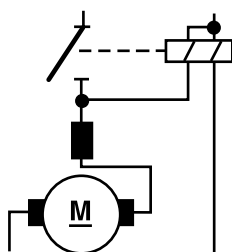
Battery



Distributor
(electrical)



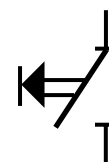
Thermally
operated
switch



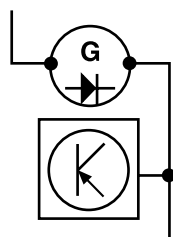
Starter



Spark plug
connector
and plug



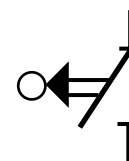
Push-button switch
(manually operated)



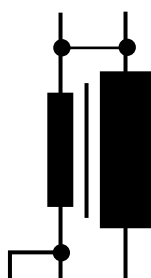
Generator



Glow plug
heater element



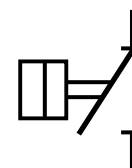
Mechanically
operated
switch



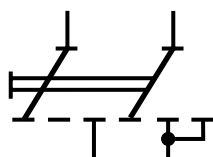
Ignition coil



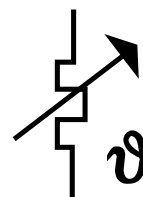
Manually
operated
switch



Pressure
operated
switch



Multiple switch
(manually operated)



Heater element
(temperature dependent)

SSP 8730/02

Appendix A: Component Symbols



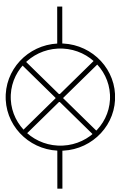
Diode



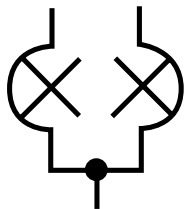
Zener diode



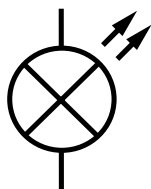
Diode
(light sensitive)



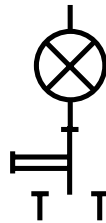
Light bulb



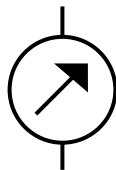
Light bulb
(dual filament)



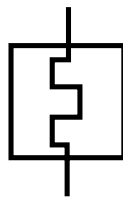
LED



Interior light



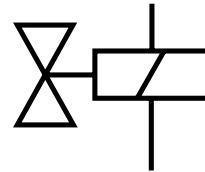
Instrument
(gauge)



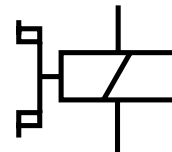
Rear window defogger
heat element



Cigarette
lighter



Solenoid valve



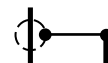
Magnetic clutch



Wire connection
in wiring harness



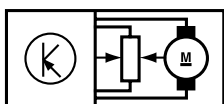
Resistance wire



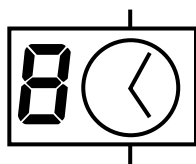
Shield wire

SSP 8730/03

Appendix A: Component Symbols



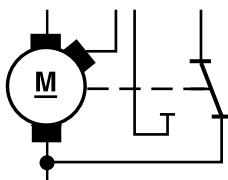
Control motor,
headlight range
adjustment



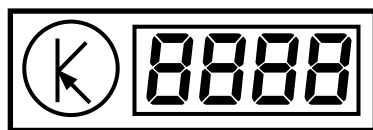
Digital clock



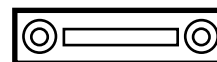
Antenna with
electronic antenna
amplifier



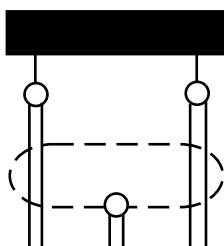
Wiper motor
(2-speed)



Multifunction
indicator



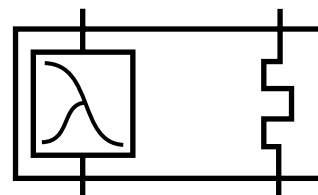
Radio



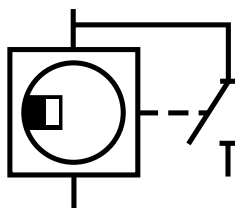
Crankshaft
position sensor



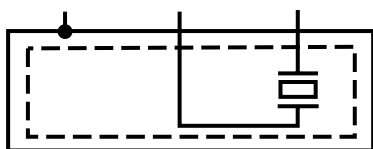
Airbag spiral
spring



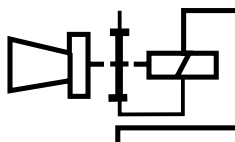
Heated oxygen
sensor



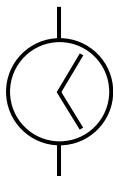
Speed sensor



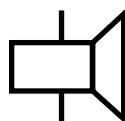
Camshaft
position sensor



Horn



Analog clock



Speaker

SSP 8730/04

Appendix B: Wiring Connections



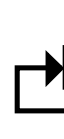
Wiring
junction



Internal connection
in a component



Disconnected
wire terminal



Slip contact



Multipoint connector
or component



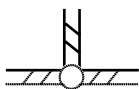
Push-on
connector



Wires
connected



Connected
wire terminal



Wires connected
(detachable junction)



Wires not
connected

SSP 8730/30

Appendix C: Component Codes and Wiring

Component Code Prefixes

- A** Battery
- B** Starter
- C** Alternator/Generator
- D** Ignition/Starter switch
- E** Switches - these are usually more complex switches: A/C control head, sunroof regulator
- F** Switches - door, hood, brake, clutch, trunk, multifunction, etc.
- G** Gauges and sensors
- H** Horn
- J** Control modules, Relays, Electronic Relays
- K** Indicator/Warning lights
- L** Lights
- M** Lights
- N** Solenoids/Inductors/Ignition - Fuel injectors, Purge, Ignition coil, Ignition module, etc.
- P** Spark plug connector
- Q** Spark Plugs
- R** Radio, CD, Telephone, Navigation
- S** Fuse, circuit breaker, protection device
- T** Wire connector
- V** Motor - Window motors, vacuum pumps, etc.
- W** Lights - Interior, Trunk
- Z** Heating elements - O₂ sensor heater, heated seats, heated mirrors, etc.

Color Codes

German Abbreviation to English

bl	Blue
br	Brown
ge	Yellow
gn	Green
ro	Red
sw	Black
li	Violet
ws	White

Occasionally in a complex circuit, other colors may be used. These are listed below.

el	Cream
nf	Neutral
og	Orange
rs	Pink
hbl	Light Blue
hgn	Light Green
rbr	Maroon
x	Braided cable
y	High tension
z	Non-cable

American Wire Gauge (AWG) Conversion to Metric

The conversion from AWG to Metric (mm²) is shown below.

AWG	mm ²
22	0.35
20	0.50
18	0.75
16	1.00
14	1.50
12	2.50
10	4.00
8	6.00
4	16.0
2	25.0
2	35.0

Appendix D: DIN Standards

DIN standards for terminal designations

The following are examples of the common Volkswagen DIN list for terminal designations:

- 1 Ignition coil, ignition distributor - Low voltage
- 1a To contact breaker I (Ignition distributor with two separate circuits)
- 1b To contact breaker II (Ignition distributor with two separate circuits)
- 4 Ignition coil, ignition distributor - high voltage
- 4a From ignition coil I (Ignition distributor with two separate circuits)
- 4b From ignition coil II (Ignition distributor with two separate circuits)
- 15 Switched (+) downstream of battery (output of ignition/driving switch)
- 15a Output at ballast resistor to ignition coil and starter
- 17 Glow plug and starter switch - Start
- 19 Glow plug and starter switch - Preheat
- 30 Input from battery (+) term., direct 12/24 V series-parallel battery switch
- 30a Input from (+) terminal of battery II
- 31 Battery negative terminal, or ground, direct
- 31b Return line to battery negative terminal, or ground via switch or relay (switched negative)
- 31a Return line to battery II, negative (12/24 V series-parallel battery switch)
- 31c Return line to battery I, negative (12/24 V series-parallel battery switch)

Electric motors

- 32 Return line
- 33 Main terminal connection
- 33a Self-parking switch-off
- 33b Shunt field
- 33f For second lower-speed range
- 33g For third lower-speed range
- 33h For fourth lower-speed range
- 33L Counterclockwise rotation
- 33R Clockwise rotation

Starters

- 45 Separate starter relay, output; starter, input (main current)
Two-starter parallel operation - Starting relay for engagement current
- 45a Output, starter I, Input, starters I and II
- 45b Output, starter II
- 48 Terminal on starter & on start-repeating relay for monitoring starting

Turn-signal flashers (pulse generators)

- 49 Input
- 49a Output
- 49b Output, second circuit
- 49c Output, third circuit

Starter control

- 50 Starter control (direct)
- 50a Series-parallel battery switch - Output for starter control
- 50b with parallel operation of two starters with sequential control
Starting relay for sequential control of the engagement current during parallel operation of two starters
- 50c Input at starting relay for starter I
- 50d Input at starting relay for starter II
- 50e Start-locking relay Input
- 50f Start-locking relay Output
- 50g Start-repeating relay Input
- 50h Start-repeating relay Output

Wiper motors

- 53 Wiper motor, input (+)
- 53a Wiper (+), self-parking switch-off
- 53b Wiper (shunt winding)
- 53c Electric windshield-washer pump
- 53e Wiper (brake winding)
- 53l Wiper motor with permanent magnet and third brush (for higher speed)

Lighting

- 55 Fog lamp
- 56 Headlamp
- 56a High beam, high-beam indicator lamp
- 56b Low beam
- 56d Headlamp-flasher contact
- 57a Parking lamp
- 57L Parking lamp, left
- 57R Parking lamp, right
- 58 Side-marker, tail, license plate, and instrument panel lamps
- 58b Dimmer
- 58d Dimmer
- 58L License-plate lamp, left
- 58R License-plate lamp, right

Alternators and voltage regulators

- 61 Alternator charge-indicator lamp
- B+ Battery positive
- B- Battery negative
- D+ Dynamo positive
- D- Dynamo negative
- DF Dynamo field
- DF1 Dynamo field 1
- DF2 Dynamo field 2
- U,V,W Alternator terminals
- 75 Radio, cigarette lighter
- 76 Speakers

Switches

Break contact (NC) and changeover switches

- 81 Input
- 81a Output 1, NC side
- 81b Output 2, NC side

Make contact (NO) switches

- 82 Input
- 82a Output 1
- 82b Output 2
- 82z Input 1
- 82y Input 2

Multiple-position switches

- 83 Input
- 83a Output, position 1
- 83b Output, position 2
- 83L Output, left-hand position
- 83R Output, right-hand position

Appendix D: DIN Standards

Relays/Current relays

- 84 Input, actuator and relay contact
- 84a Output, actuator
- 84a Output, relay contact

Switching relays

- 85 Output, actuator (end of winding to ground or negative)
- 86 Input, actuator (start of winding)
- 86a Start of winding or 1st winding
- 86b Winding tap or 2nd winding

Relay contact for break (NC) and changeover contacts

- 87 Input
- 87a Output 1 (NC side)
- 87b Output 2
- 87c Output 3
- 87z Input 1
- 87y Input 2
- 87x Input 3

Relay contact for make (NO) contact and changeover contacts

- 88 Input
- 88a Output 1
- 88b Output 2
- 88c Output 3
- 88z Input 1
- 88y Input 2
- 88x Input 3

Directional signals (turn signal flashers)

- C Indicator lamp 1
- C2 Indicator lamp 2
- C0 Main terminal connection for separate indicator circuits actuated by the turn signal switch
- C3 Indicator lamp 3 (e.g., when towing two trailers)
- L Turn-signal lamps, left
- R Turn-signal lamps, right

Appendix E: Wiring Diagrams

Golf/Jetta - Standard Equipment, from May 1999

No./Page

Relay locations	29/1
Battery, ignition/starter switch	29/2
Instrument cluster, clutch pedal position (CPP) switch, starting interlock relay	29/3
Instrument cluster, tachometer, speedometer, odometer, display antitheft immobilizer, warning light for anti-theft immobilizer sensor . . .	29/4
Instrument cluster, generator (GEN) warning light, digital clock, low fuel level warning light	29/5
Instrument cluster, left front brake pad wear sensor, brake pad wear indicator light, headlight high beam indicator light	9/6
Instrument cluster, parking brake warning light switch, brake fluid level warning switch, low windshield washer fluid level indication, left and right turn signal indicator lights	29/7
Turn signal switch, headlight dimmer/flasher switch, left front turn signal light, light for side marker front left, left headlight	29/8
Emergency flasher switch, turn signal relay, right front turn signal light, light for side marker front right, right headlight	29/9
Back-up light switch, left rear turn signal light, left back-up light, left brake/tail light, high-mount brake light (Golf only)	29/10
Brake light switch, right rear turn signal light, right back-up light, right brake/tail light, high mount brake light (Jetta only)	29/11
Data Link Connector (DLC) radio connection, cigarette lighter	29/12
Light switch, rear fog light switch, instrument panel vent illumination	29/13
License plate light	29/14
Dual tone horn, fuel tank door remote	29/15
Instrument panel light dimmer switch, glove compartment light, rear window, defogger switch, heated rear window	29/16
Fresh air blower, fresh air/recirculating flap switch, fresh air blower	29/17
Motor for rear windshield wiper, windshield and rear window washer pump	29/18
Windshield wiper/washer switch, windshield wiper/intermittent regulator, windshield wiper motor, wiper/washer intermittent relay	29/19

Appendix E: Wiring Diagrams

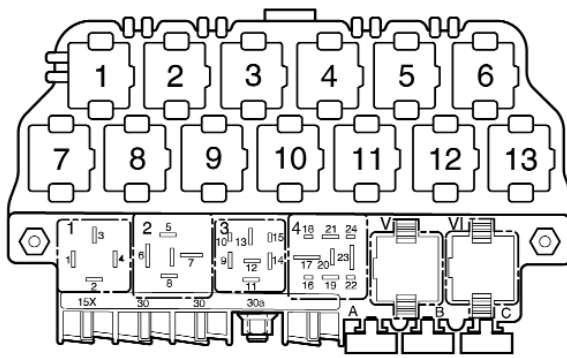
1.8L - Engine - Motronic Multiport Fuel Injection (MFI)/110kW, code AWD, from November 1999

	No./Page
Generator (GEN), starter	41/2
Motronic engine control module (ECM) power supply relay	41/3
Motronic engine control module (ECM) ignition system	41/4
Motronic engine control module (ECM)	41/5
Motronic engine control module (ECM), (ECT) sensor, charge air pressure sensor, camshaft position (CMP) sensor 2, wastegate bypass regulator valve, recirculating valve for turbocharger	41/6
Motronic engine control module (ECM), angle sensor for throttle drive (power accelerator actuation), intake air temperature (IAT) sensor, knock sensor (KS) 1, leak detection pump (LDP)	41/7
Motronic engine control module (ECM), pressure switch/power steering, engine speed (RPM) sensor, knock sensor (KS) 2	41/8
Motronic engine control module (ECM), heated oxygen sensor (HO2S), injectors	41/9
Motronic engine control module (ECM), throttle position (TP) sensor, oxygen sensor (O2S) behind three way catalytic converter (TWC), evaporative emission (EVAP) canister purge	41/10
Motronic engine control module (ECM), fuel pump (FP) relay, cruise control switch, mass air flow (MAF) sensor	41/11
Motronic engine control module (ECM), secondary air injection (AIR) pump system, brake light switch, clutch vacuum vent valve switch, brake vacuum vent valve switch	41/12
Fuel pump (FP), fuel level sensor, engine coolant level (ECL) sensor	41/13
Instrument cluster, oil pressure switch, speedometer vehicle speed sensor (VSS), oil pressure warning light	41/14
Instrument cluster, engine coolant temperature (ECT) gauge, tachometer, speedometer, generator (GEN) warning light	41/15
Instrument cluster, multi-function indicator (MFI), outside air temperature sensor, electronic power control (EPC) warning lamp	41/16

Golf /Jetta - Standard Equipment,

from May1999

Deviate relay location and fuseplacements as well as the locations of multiple connectors see section "component locations".



97-14163

Relay location on the thirteenfold auxiliary relay panel, above relay panel:

3 Starting Interlock Relay (53)

Relay panel:

1 Dual Horn Relay (53)

2 Load Reduction Relay (18)

V

Wiper/Washer Intermittent Relay (377)

VI

Note: Number in parentheses indicates production control number stamped on relay housing.

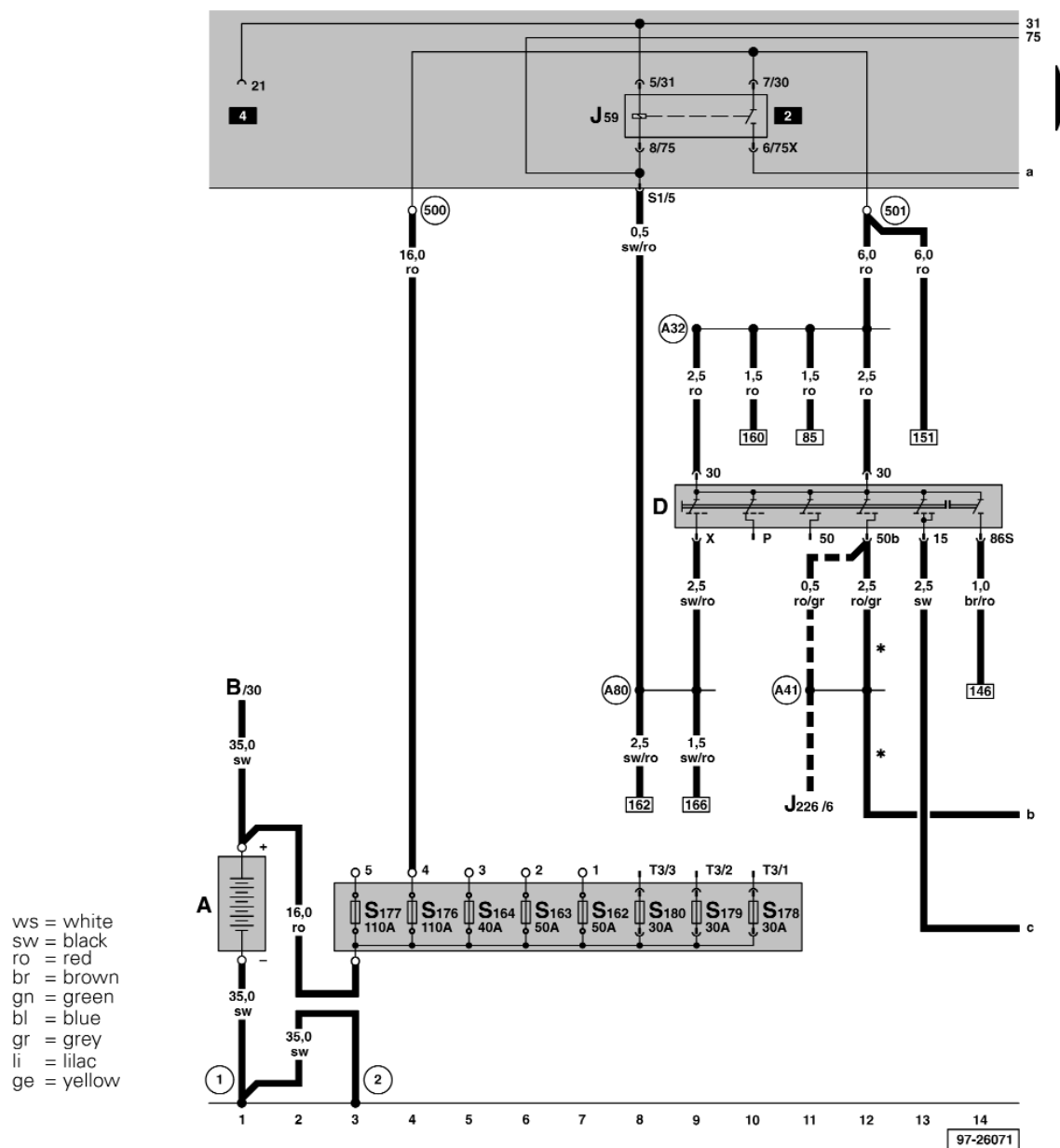
Fuse colors

30 A - green
25 A - white
20 A - yellow
15 A - blue
10 A - red
7,5 A - brown
5 A - beige
3 A - violet

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...with respect to the correctness of information...

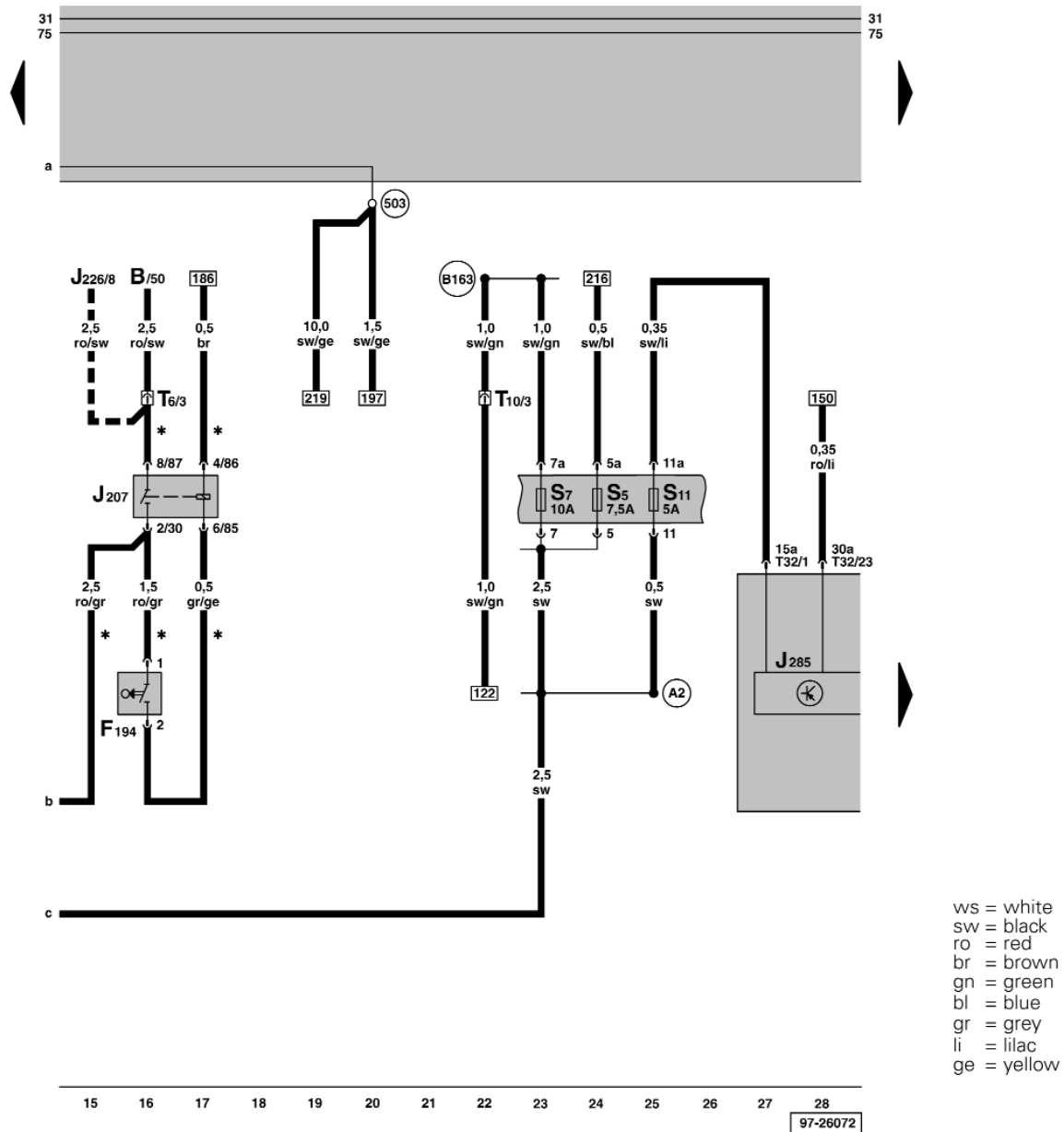


- A - Battery
- B - Starter
- D - Ignition/Starter Switch
- J59 - Load Reduction Relay
- J226 - Park/Neutral Position (PNP) Relay
- S162 - Fuse -1- (30) in fuse bracket/battery
- S163 - Fuse -2- (30) in fuse bracket/battery
- S164 - Fuse -3- (30) in fuse bracket/battery
- S176 - Fuse -4- (30) in fuse bracket/battery
- S177 - Fuse -5- (30) in fuse bracket/battery
- S178 - Fuse -6- (30) in fuse bracket/battery
- S179 - Fuse -7- (30) in fuse bracket/battery
- S180 - Fuse -8- (30) in fuse bracket/battery
- T3 - 3-Pin Connector

1 - Ground strap, battery to body

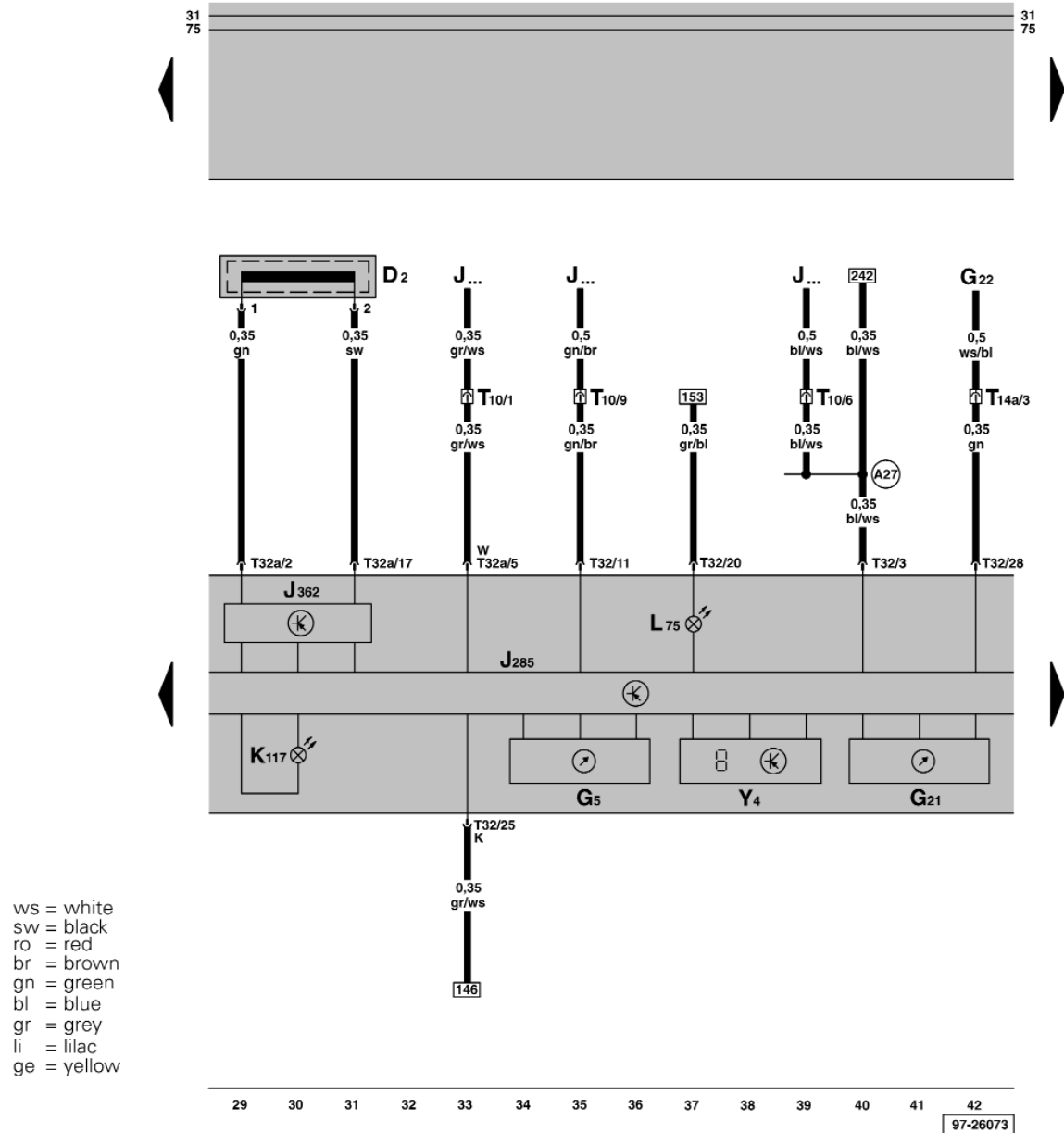
Battery, ignition/starter switch

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- B - Starter
- F194 - Clutch Pedal Position (CPP) Switch*
- J207 - Starting Interlock Relay*
- J226 - Park/Neutral Position (PNP) Relay
- J285 - Control module with indicator unit in instrument panel insert
- S5 - Fuse 5 in fuse holder
- S7 - Fuse 7 in fuse holder
- S11 - Fuse 11 in fuse holder
- T6 - 6-Pin Connector, brown, in protective housing for connectors, in plenum chamber, left
- T10 - 10-Pin Connector, orange, in protective housing for connectors, in plenum chamber, left
- T32 - 32-Pin Connector, blue

- (A2) - Plus connection (15), in instrument panel wiring harness
- (B163) - Plus connector -1- (15) in wiring harness interior
- (503) - Threaded connection -2- (75x) on the relay plate
- * - Manual transmission only
- Automatic transmission only



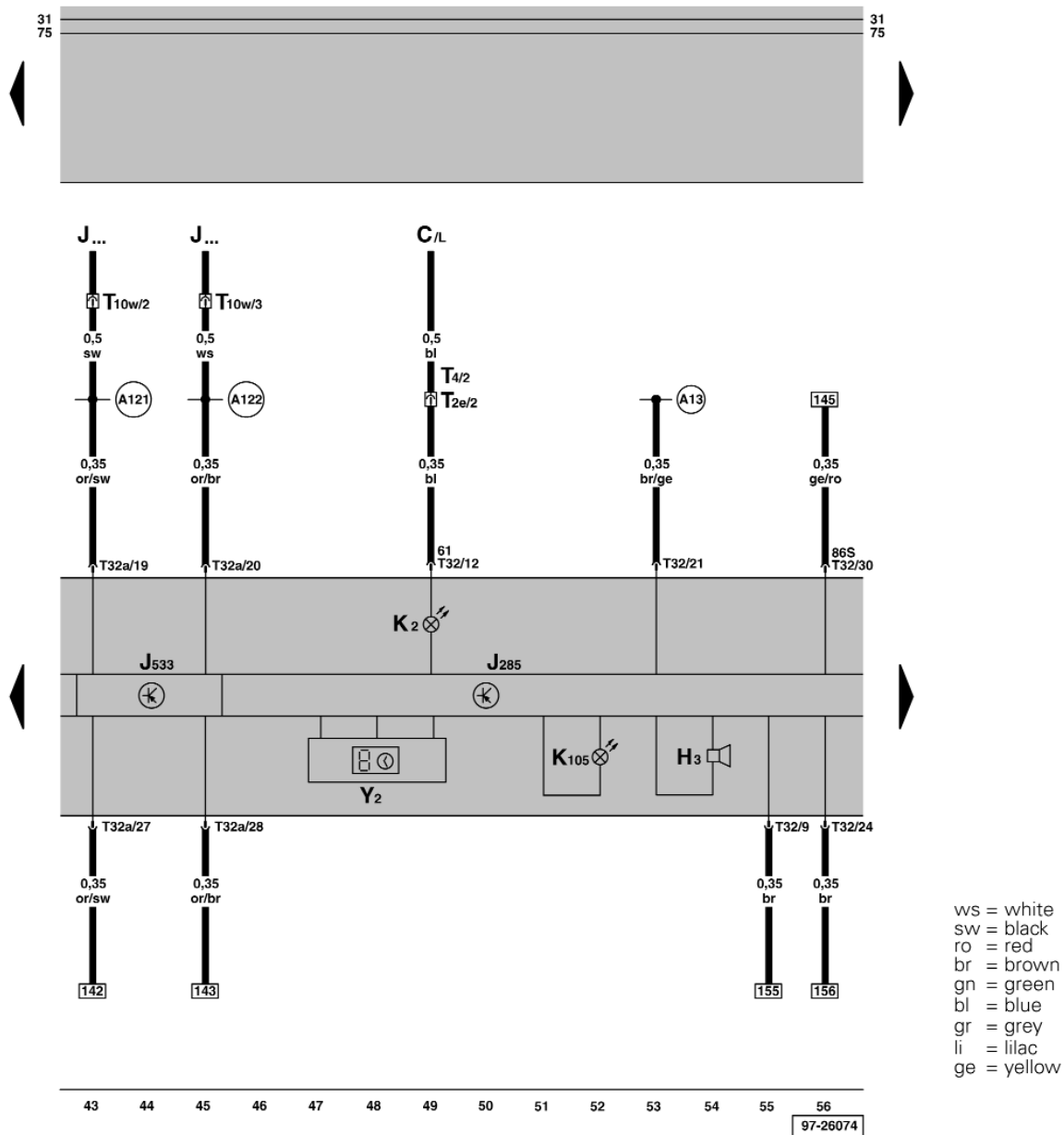
- D2 - Induction coil of anti-theft immobilizer
G5 - Tachometer
G21 - Speedometer
G22 - Speedometer Vehicle Speed Sensor
J285 - Control module with indicator unit in instrument panel insert
J362 - Control module for anti-theft immobilizer
J... - Engine Control Module (ECM)
K117 - Warning light for anti-theft immobilizer sensor
L75 - Digital Display Light
T10 - 10-Pin Connector, orange, in protective housing for connectors, in plenum chamber, left
T14a - 14-Pin Connector, in engine compartment, in wiring duct, left
T32 - 32-Pin Connector, blue

- T32a - 32-Pin Connector, green
Y4 - Odometer Display

- (A27) - Wire connection (vehicle speed signal), in instrument panel wiring harness

Instrument cluster, tachometer, speedometer, odometer display anti-theft immobilizer, warning light for anti-theft immobilizer sensor

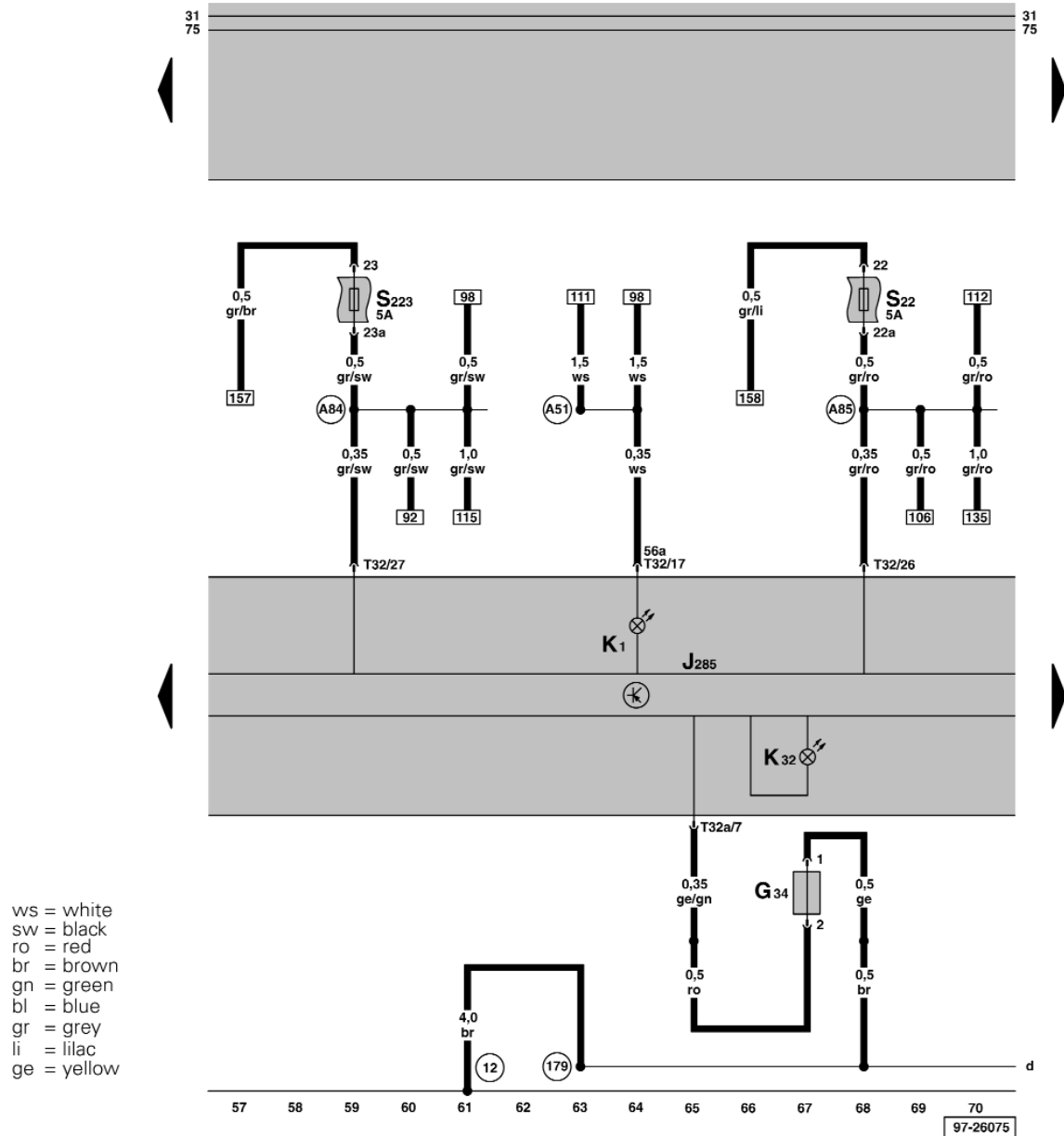
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- C - Generator (GEN)
- H3 - Warning buzzer
- J285 - Control module with indicator unit in instrument panel insert
- J533 - Data Bus On Board Diagnostic Interface
- J... - Engine Control Module (ECM)
- K2 - Generator (GEN) Warning Light
- K105 - Low Fuel Level Warning Light
- T2e - Double Connector, near starter (in vehicles without air conditioning)
- T4 - 4-Pin Connector, near starter (in vehicles with air conditioning)
- T10w- - 10-Pin Connector, white, in protective housing for connectors, in plenum chamber, left
- T32 - 32-Pin Connector, blue
- T32a - 32-Pin Connector, green

Y2 - Digital Clock

- (A13) - Wire connection (door contact switch) in instrument panel wiring harness
- (A121) - Connection (high bus), in instrument panel wiring harness
- (A122) - Connection (low bus), in instrument panel wiring harness



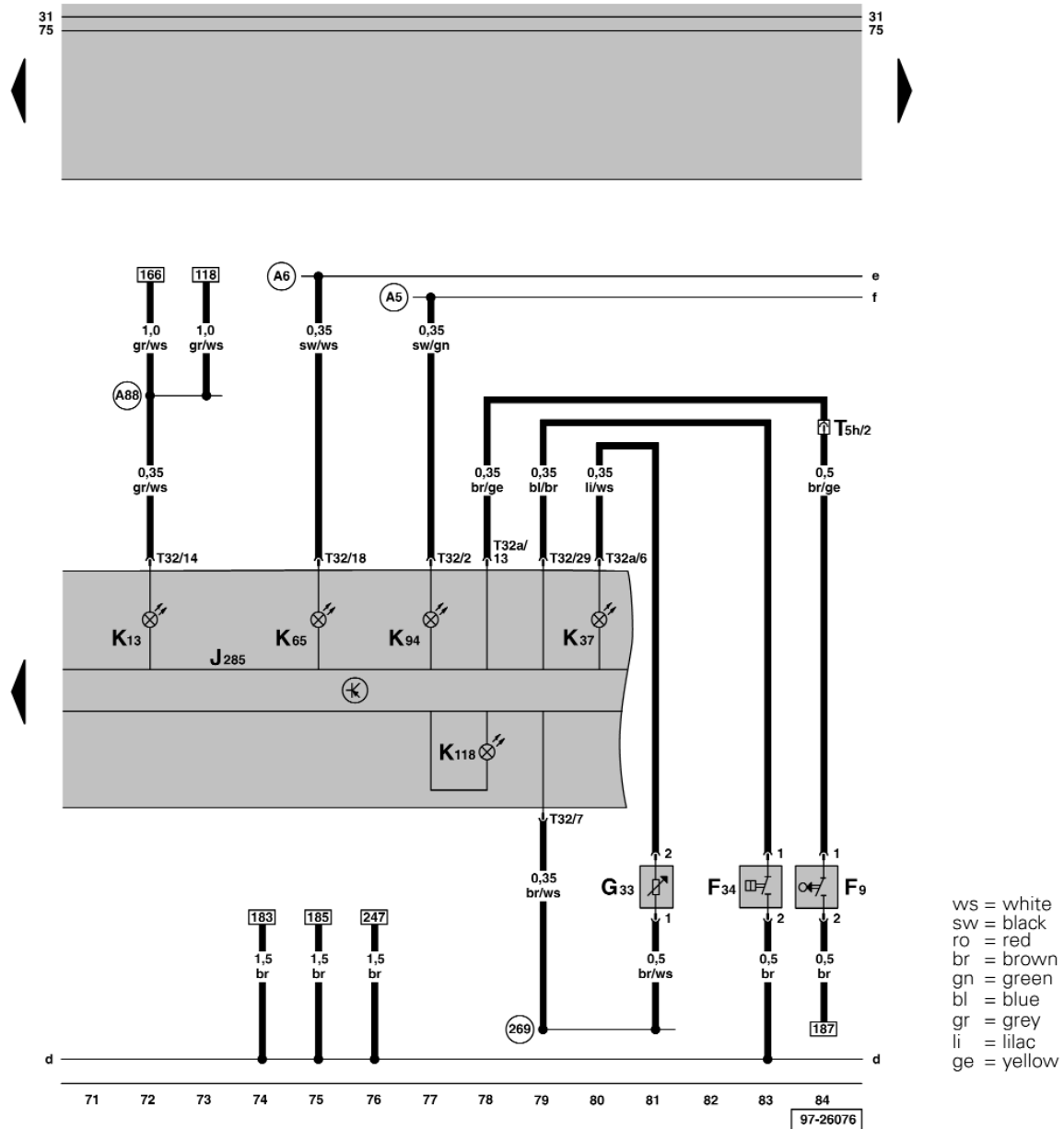
G34 - Left Front Brake Pad Wear Sensor
J285 - Control module with indicator unit in instrument panel insert
K1 - Headlight High Beam Indicator Light
K32 - Brake Pad Wear Indicator Light
S22 - Fuse 22 in fuse holder
S223 - Fuse 23 in fuse holder
T32 - 32-Pin Connector, blue
T32a - 32-Pin Connector, green

A84 - Connector (58L) in instrument panel wiring harness
A85 - Connector (58R) in instrument panel wiring harness

12 - Ground connection, in engine compartment, left
179 - Ground connection, in left headlight wiring harness
A51 - Wire connection (56), in instrument panel wiring harness

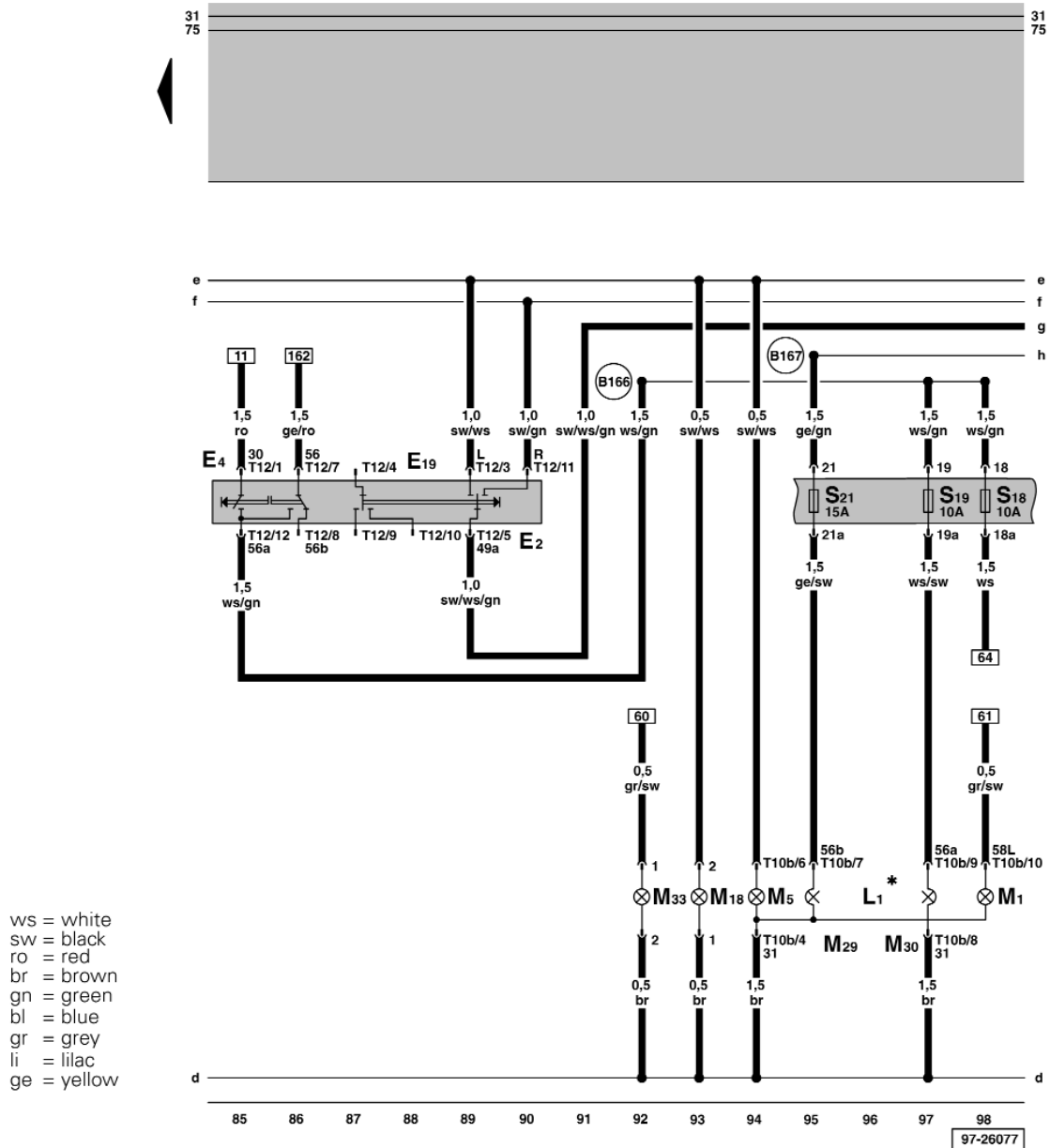
Instrument cluster, left front brake pad wear sensor, brake pad wear indicator light, headlight high beam indicator light

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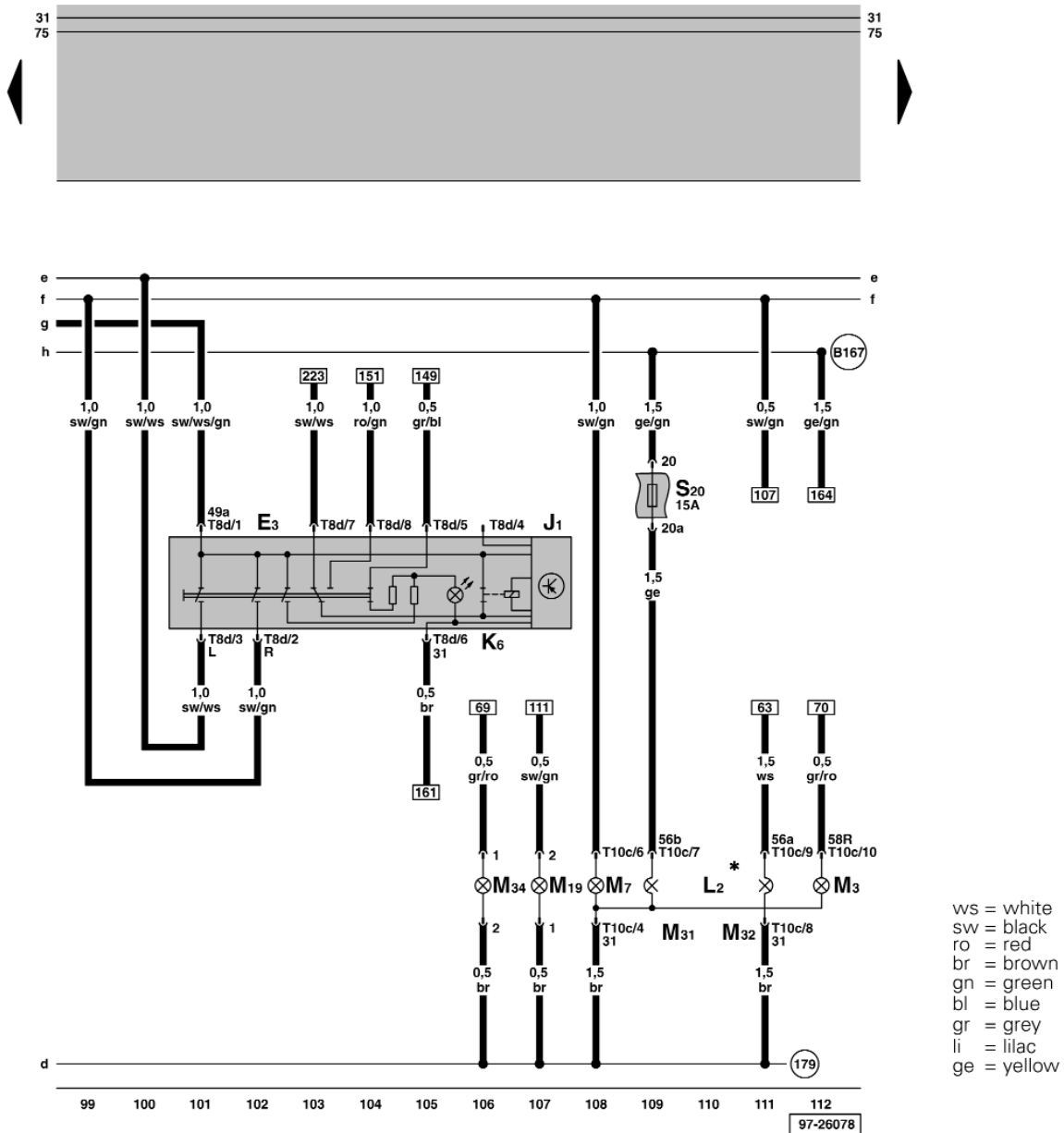
- F9 - Parking Brake Warning Light Switch
- F34 - Brake Fluid Level Warning Switch
- G33 - Windshield Washer Fluid Level Sensor
- J285 - Control module with indicator unit in instrument panel insert
- K13 - Rear Fog Light Indicator Light
- K37 - Low Windshield Washer Fluid Level Indicator Light
- K65 - Left Turn Signal Indicator Light
- K94 - Right Turn Signal Indicator Light
- K118 - Warning light for brake system
- T5h - 5-Pin Connector, near left A-pillar, lower part, in harness
- T32 - 32-Pin Connector, blue
- T32a - 32-Pin Connector, green

- (269) - Ground connector (sensor ground) -1-, in instrument panel wiring harness
- A5 - Plus connection (right turn signal), in instrument panel wiring harness
- A6 - Plus connection (left turn signal), in instrument panel wiring harness
- A88 - Connector fog light, in instrument panel wiring harness



Turn signal switch, headlight dimmer/flasher switch, left front turn signal light, light for side marker front left, left headlight

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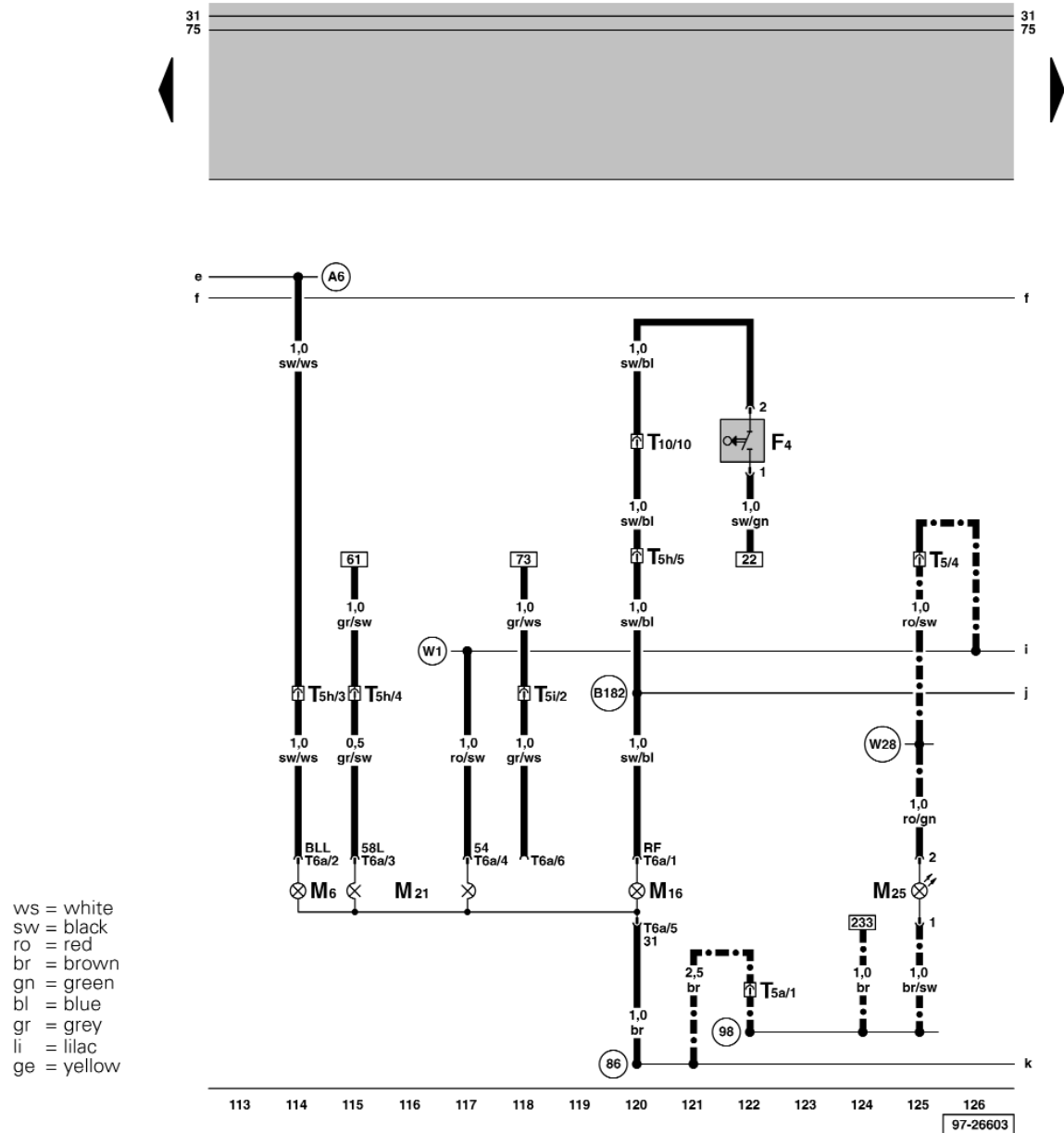


- E3 - Emergency Flasher Switch
- J1 - Turn Signal Relay
- K6 - Indicator light for emergency flasher system
- L2 - Right Headlight*
- M3 - Right Parking Light
- M7 - Right Front Turn Signal Light
- M19 - Right, Side Turn Signal Light
- M31 - Right Low Beam Headlight (Golf only)
- M32 - Right High Beam Headlight (Golf only)
- M34 - Light for side marker front right
- S20 - Fuse 20 in fuse holder
- T8d - 8-Pin Connector
- T10c - 10-Pin Connector

- (179) - Ground connection, in left headlight wiring harness

- (B167) - Connection (56b), in passenger compartment wiring harness

- * - Jetta only



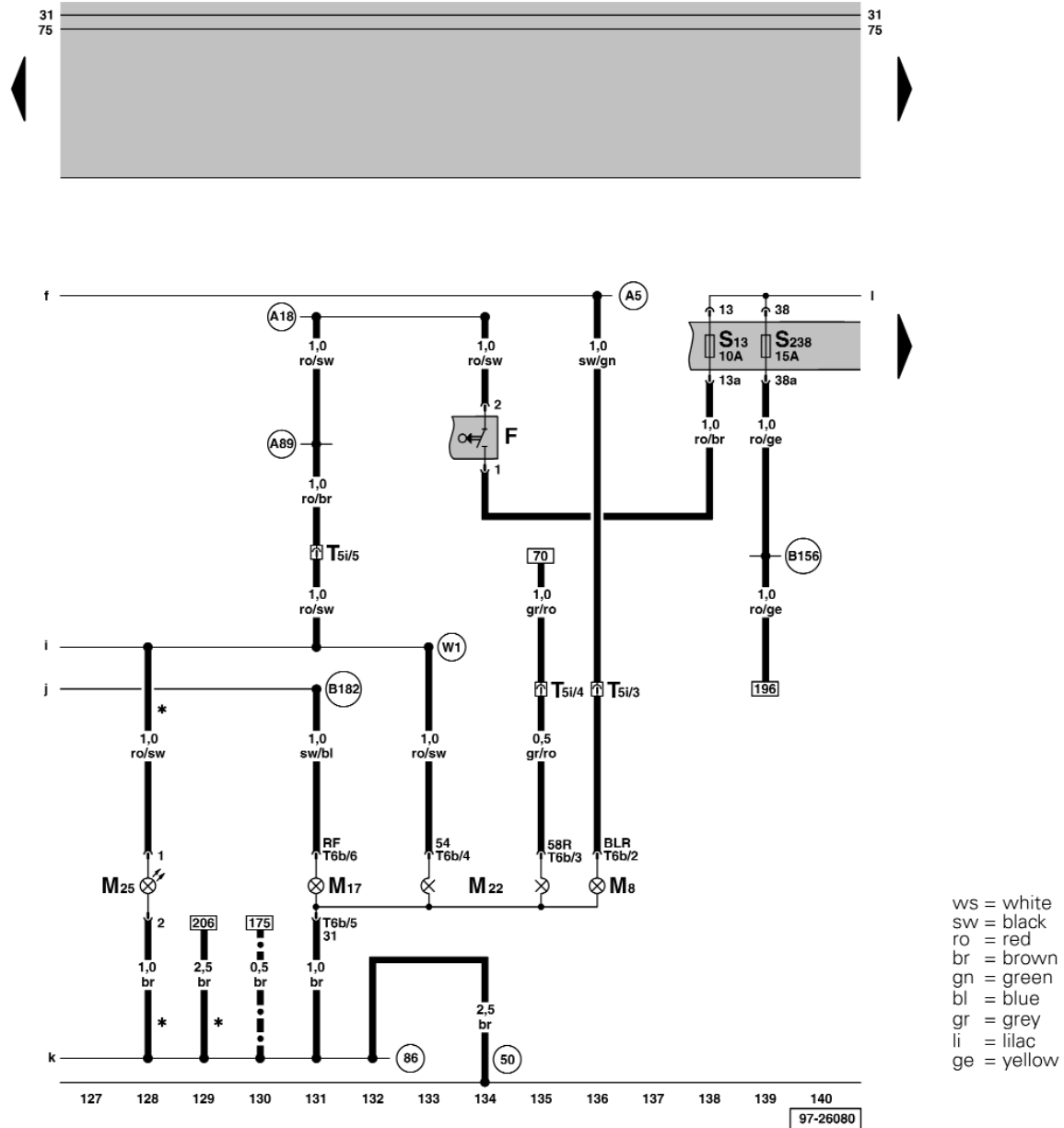
- F4 - Back-Up Light Switch
- M6 - Left Rear Turn Signal Light
- M16 - Left Back-Up Light
- M21 - Left Brake/Tail Light
- M25 - High-mount Brake Light (32 light emitting diodes)
- T5 - 5-Pin Connector, black, connector station C-pillar, left
- T5a - 5-Pin Connector, pink, connector station C-pillar, left (Golf only)
- T5h - 5-Pin Connector, near left A-pillar, lower part, in harness
- T5i - 5-Pin Connector, near left A-pillar, lower part, in harness
- T6a - 6-Pin Connector

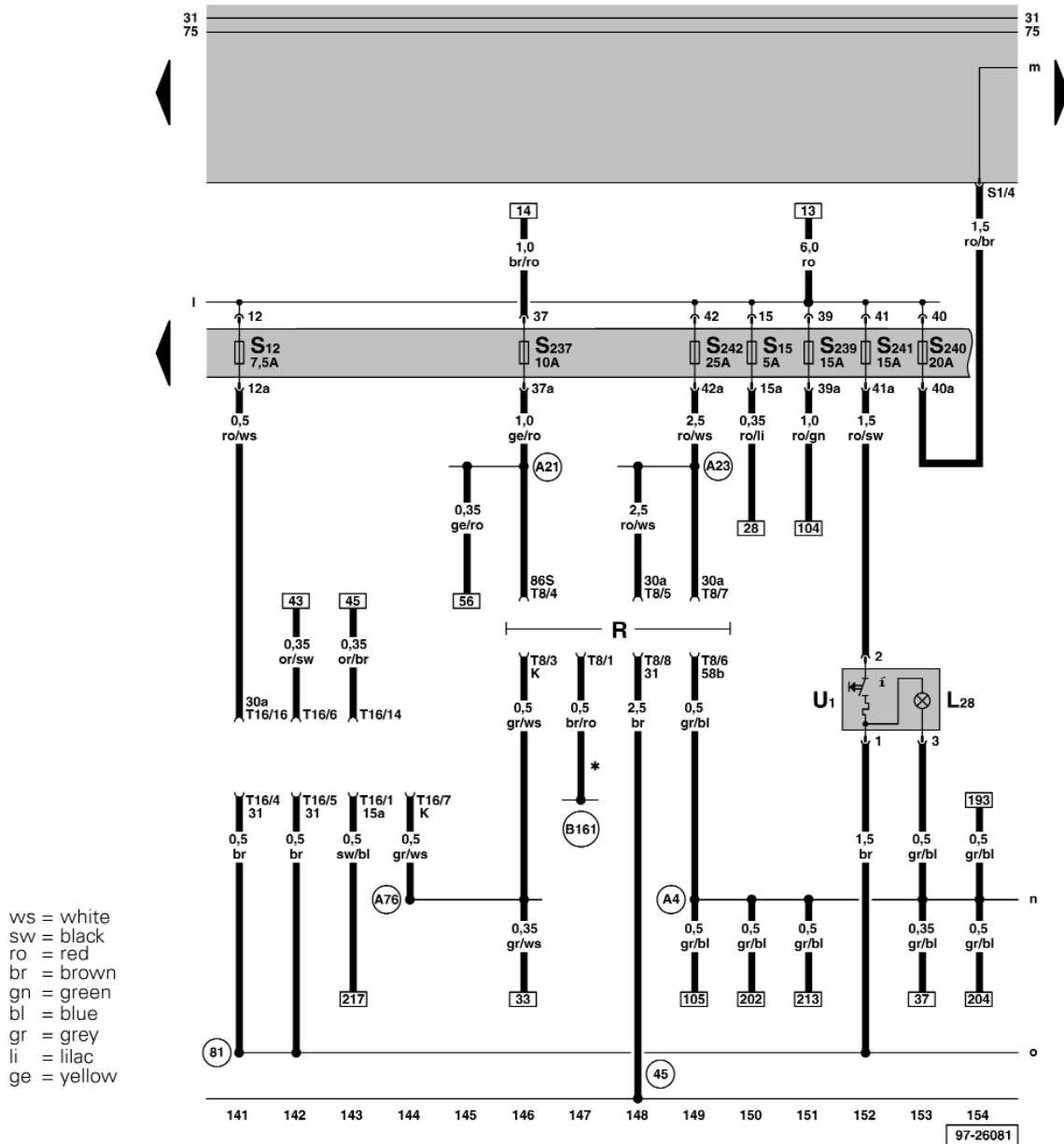
- T10 - 10-Pin Connector, orange, in protective housing for connectors, in plenum chamber, left

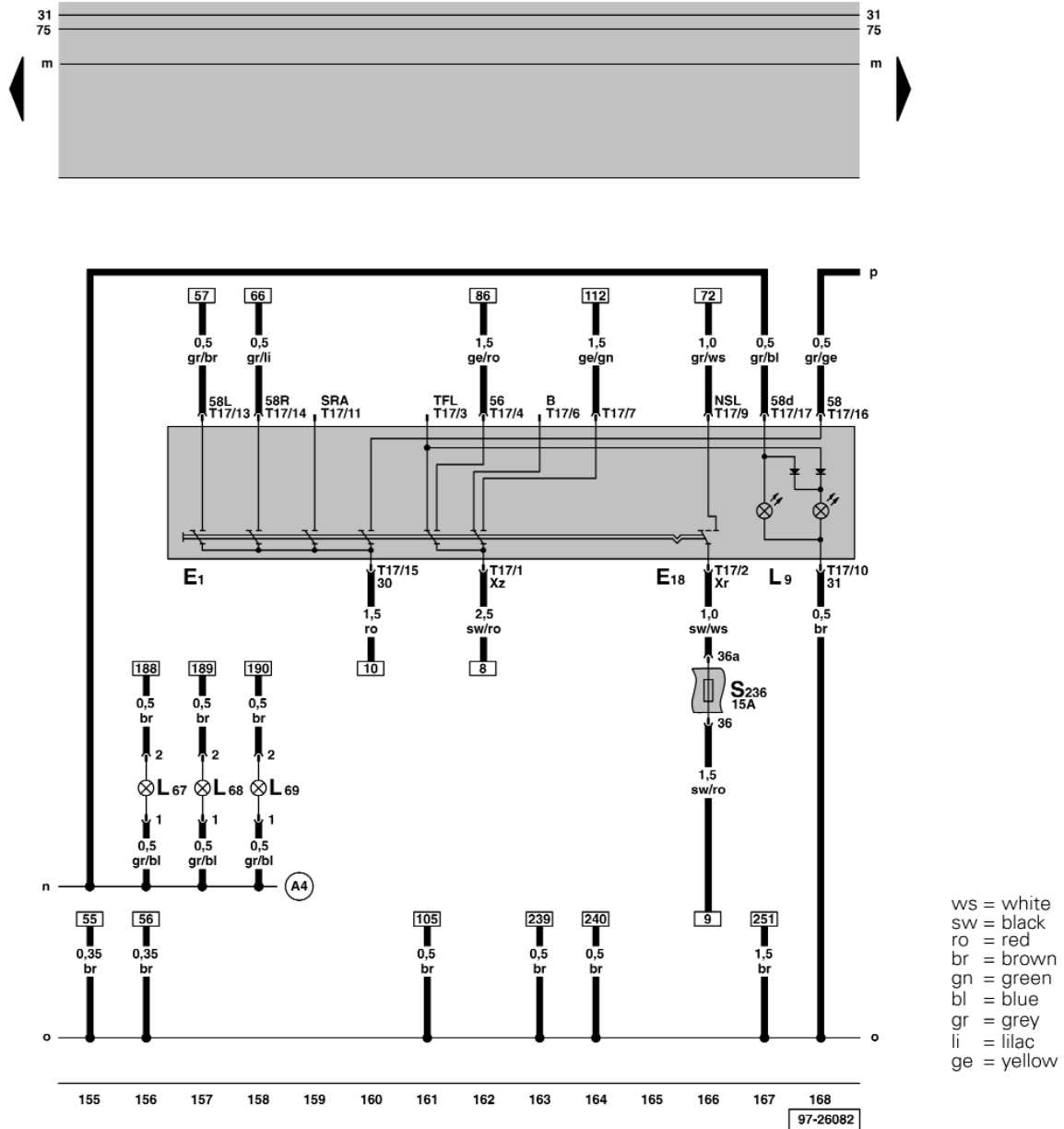
- (86) - Ground connection -1-, in rear wiring harness
- (98) - Ground connection, in rear lid wiring harness
- (A6) - Plus connection (left turn signal), in instrument panel wiring harness
- (B182) - Connection (RF), in passenger compartment wiring harness
- (W1) - Plus connection (54), in rear wiring harness
- (W28) - Plus connector -2- (54), in wiring harness taillight assembly
- Golf only

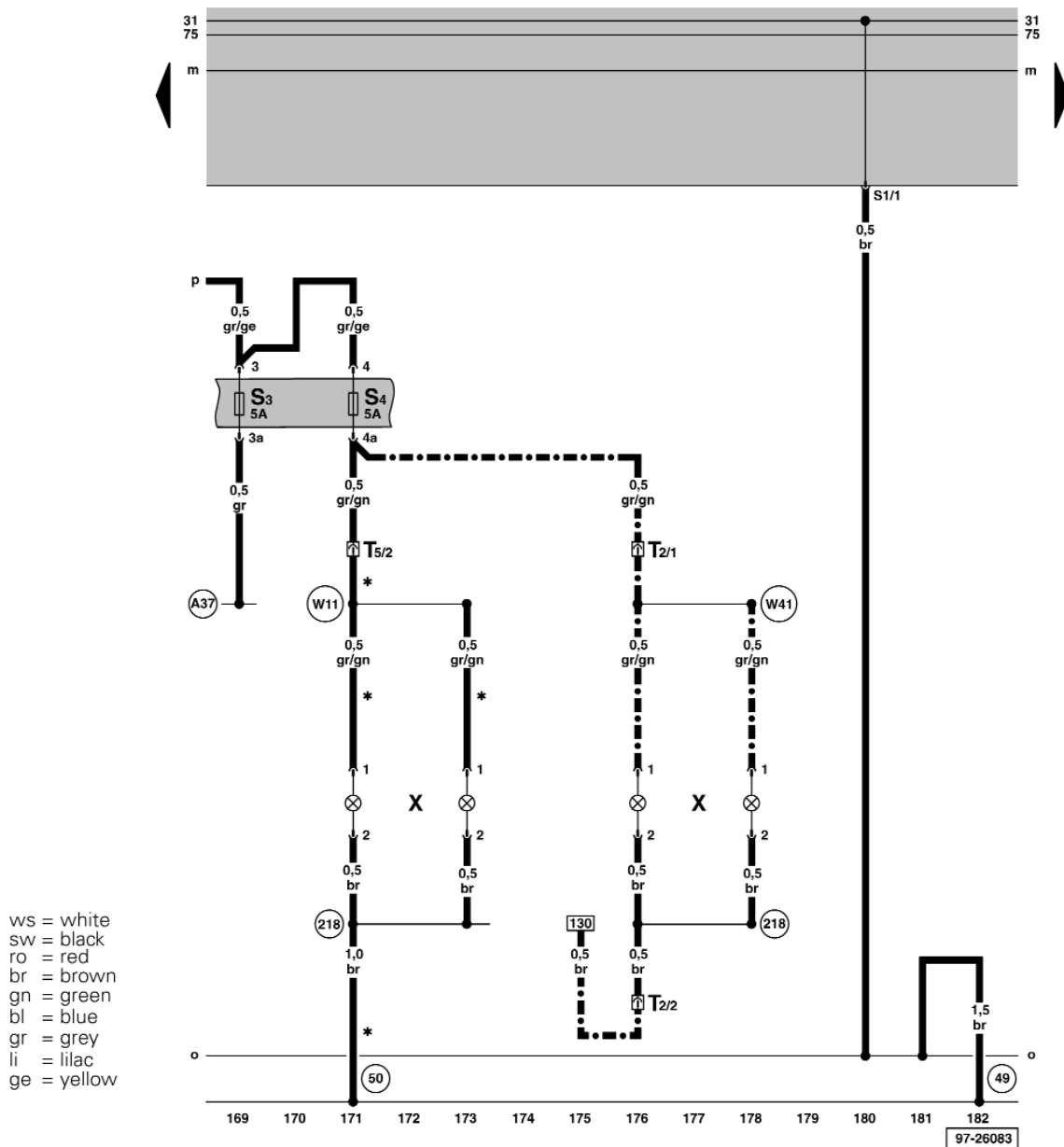
Back-up light switch, left rear turn signal light, left back-up light, left brake/tail light, high-mount brake light (Golf only)

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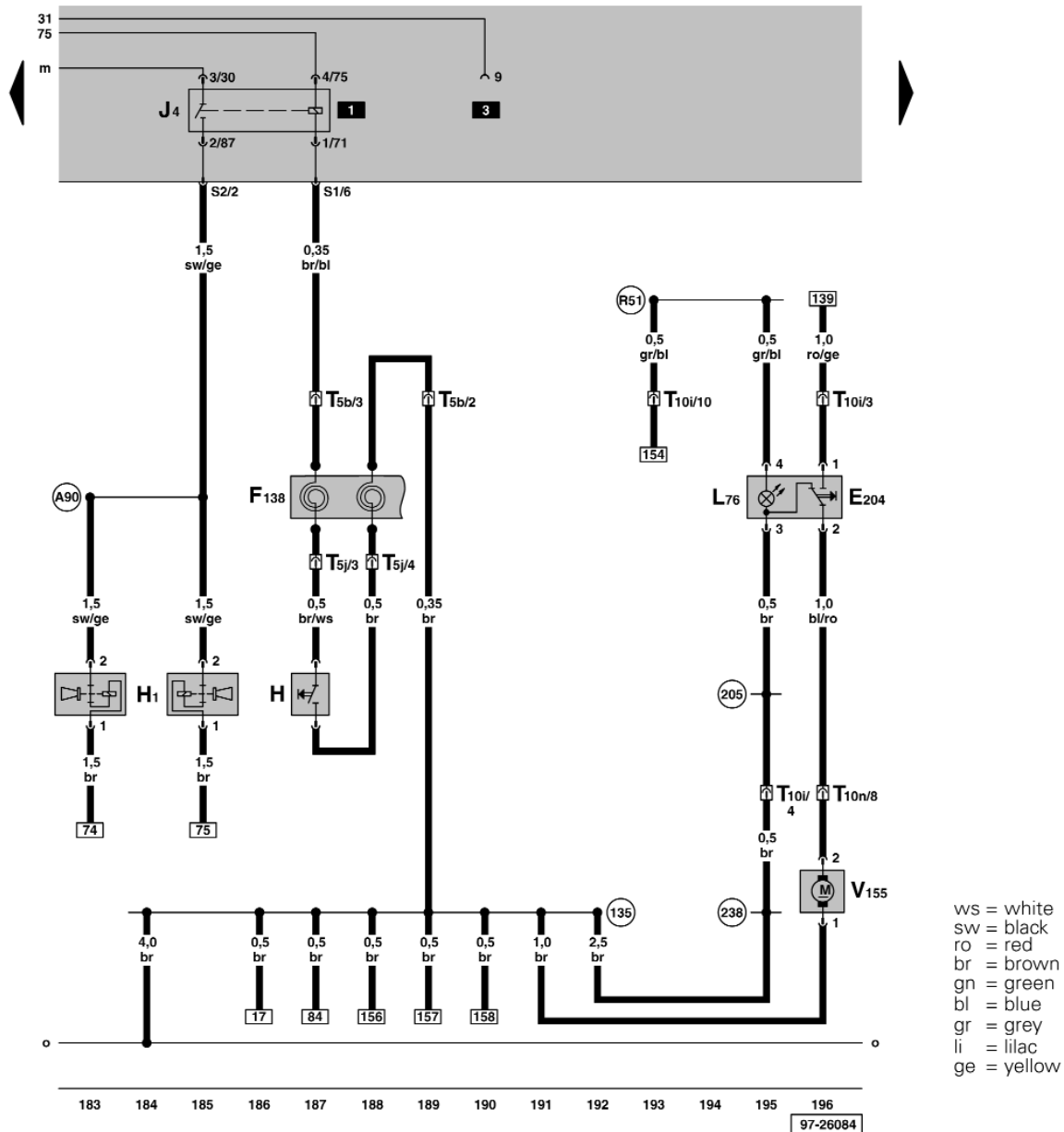






License plate light

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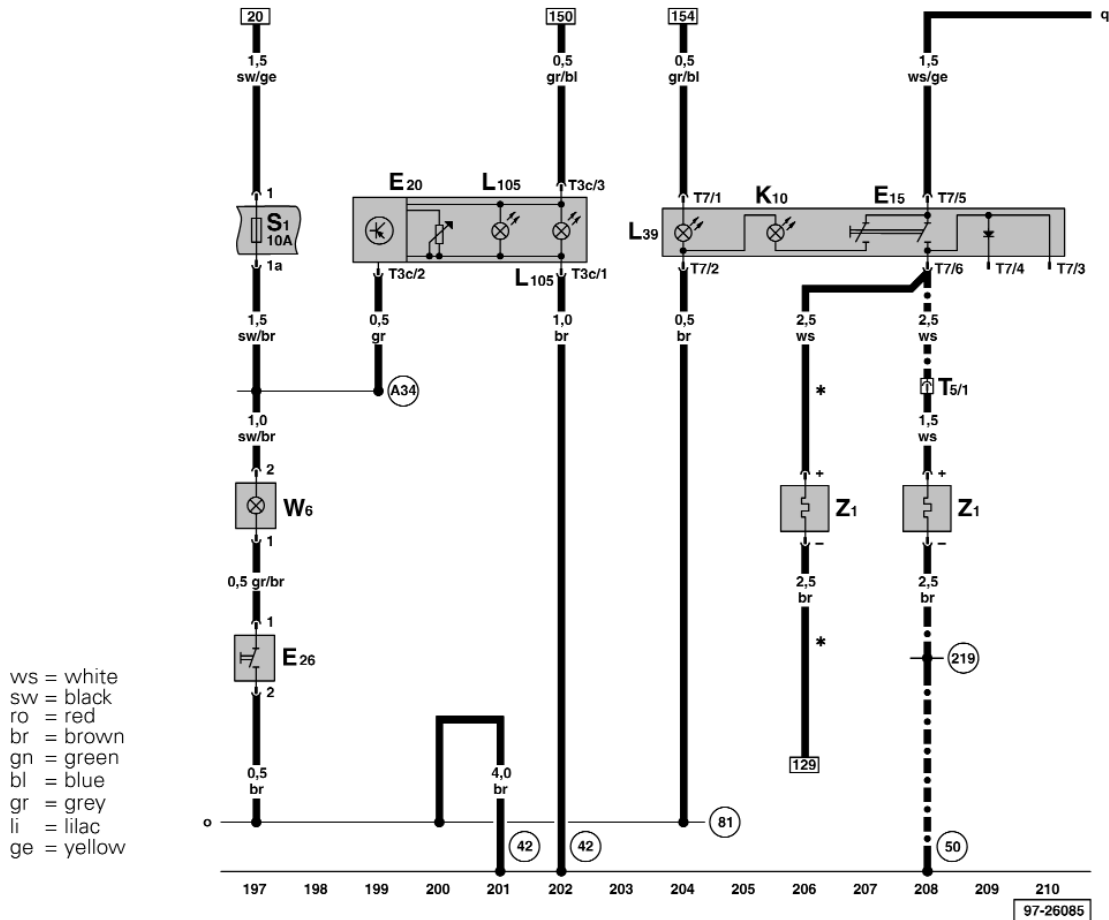


- E204 - Switch for remote/fuel tank door (also see Comfort System with power windows wiring diagram)
- F138 - Airbag Spiral Spring/Return Spring with Slip Ring
- H - Signal horn activation
- H1 - Dual tone horn
- J4 - Dual Horn Relay
- L76 - Push Button Light
- T5b - 5-Pin Connector, on airbag spiral spring
- T5j - 5-Pin Connector, behind driver's airbag
- T10i - 10-Pin Connector, black, connector station A-pillar, left
- T10n - 10-Pin Connector, brown, connector station A-pillar, left
- V155 - Motor for fuel tank lid unlock

- 135 - Ground connection -2-, in instrument panel wiring harness
- 205 - Ground connection, in driver's door wiring harness
- 238 - Ground connection -1-, in wiring harness interior
- A90 - Connector (dual tone horn), in instrument panel wiring harness
- R51 - Connection (58b) in door wiring harness, driver side

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Dual tone horn, fuel tank door remote

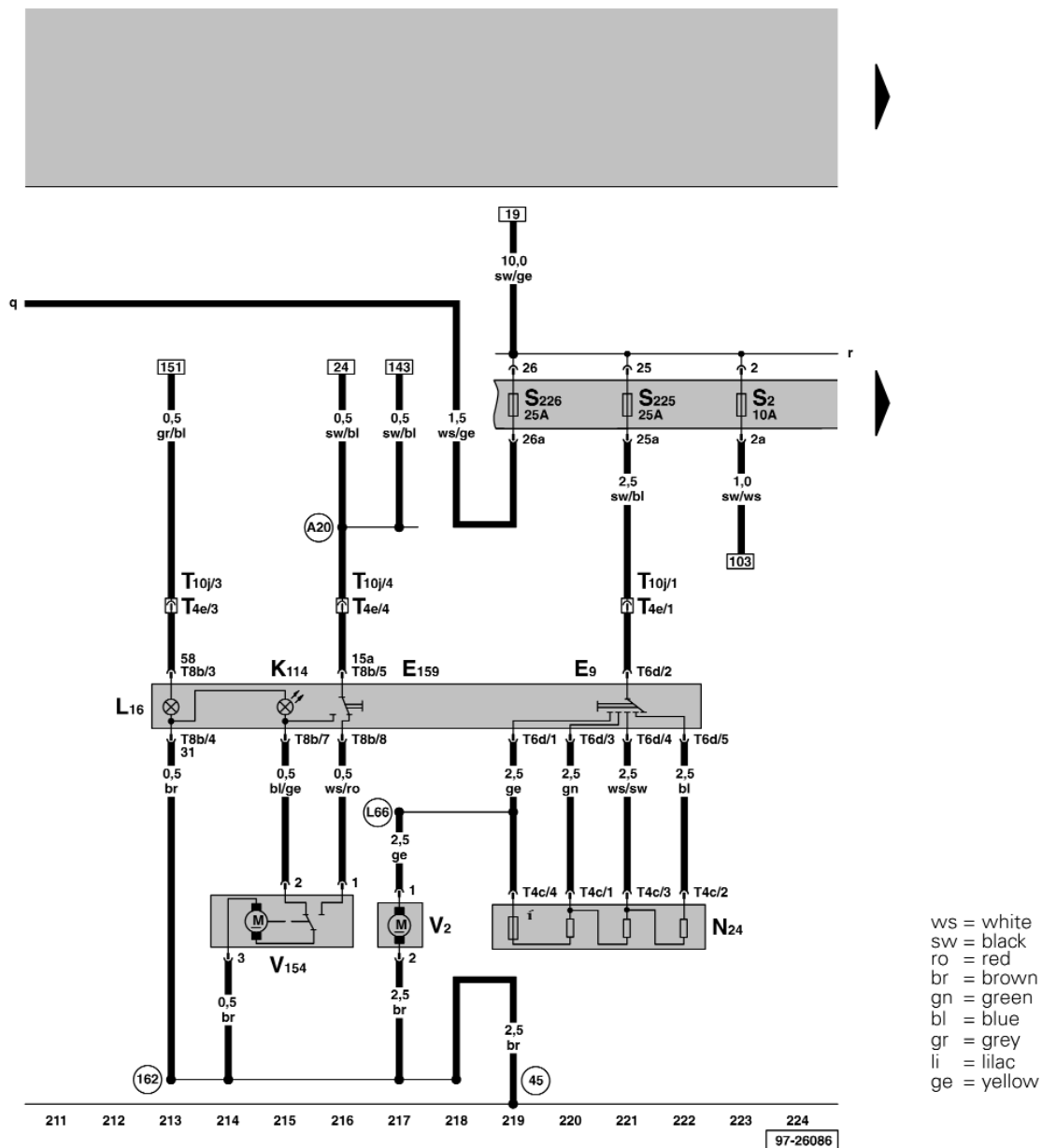


- E15 - Rear window defogger switch
- E20 - Instrument Panel Light Dimmer Switch
- E26 - Glove compartment light switch
- K10 - Indicator light for heated rear windshield
- L39 - Rear Window Defogger Switch Light
- L105 - Illumination for lighting controller
- S1 - Fuse 1 in fuse holder
- T3c - 3-Pin Connector
- T5 - 5-Pin Connector, black, connector station C-pillar, left (Golf only)
- T7 - 7-Pin Connector
- W6 - Glove Compartment Light
- Z1 - Heated rear window

- (42) - Ground connection, beside steering column

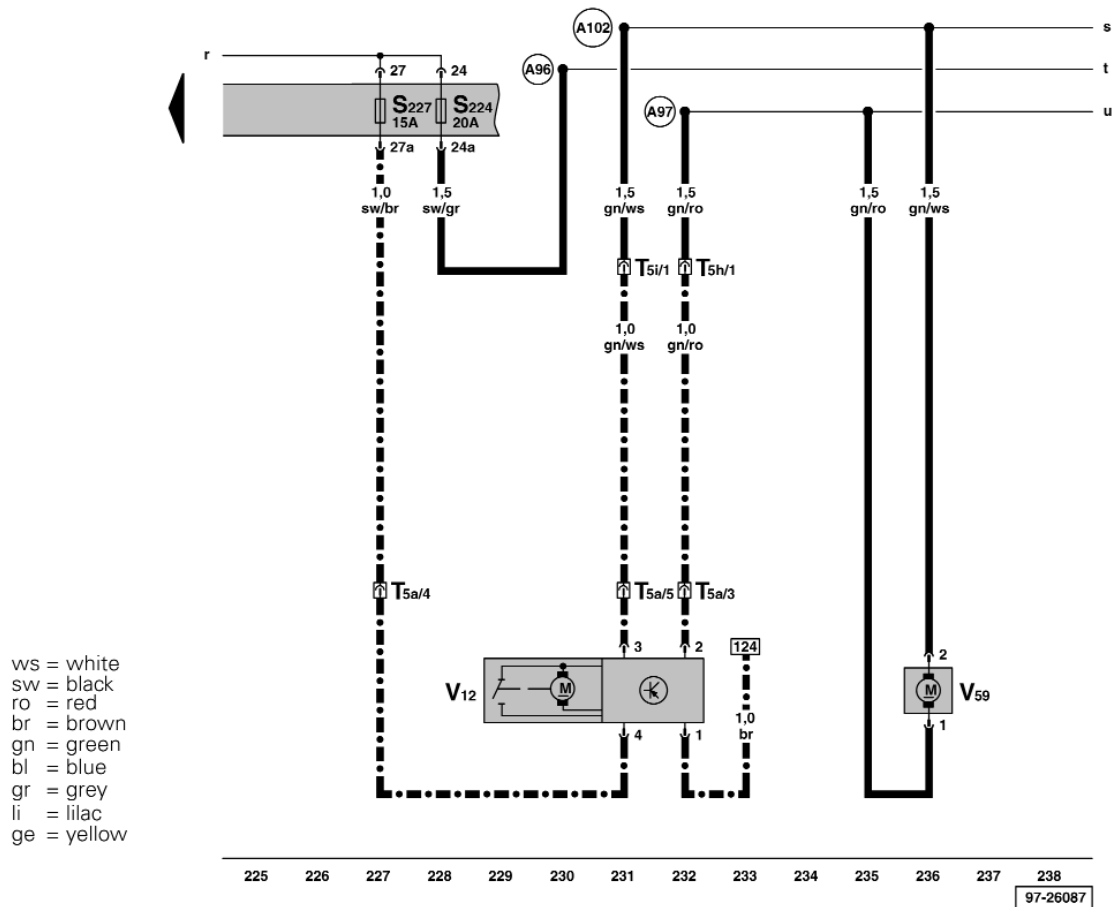
- (50) - Ground connection, in luggage compartment, left
- (81) - Ground connection -1-, in instrument panel wiring harness
- (219) - Ground connection -2-, in rear lid wiring harness
- (A34) - Wire connection (75x), in instrument panel wiring harness

- * - Jetta only
- - Golf only



- E9 - Fresh Air Blower Switch
- E159 - Fresh Air/Recirculating Flap Switch
- K114 - Fresh Air and Recirculating Air Mode Indicator Light
- L16 - Fresh Air Control Lever Light
- N24 - Fresh Air Blower Series Resistance With Fuse
- S2 - Fuse 2 in fuse holder
- S225 - Fuse 25 in fuse holder
- S226 - Fuse 26 in fuse holder
- T4c - 4-Pin Connector
- T4e - 4-Pin Connector (in vehicles without air conditioning)
- T6d - 6-Pin Connector
- T8b - 8-Pin Connector
- T10j - 10-Pin Connector, behind instrument panel, center (in vehicles with air conditioning)

- V2 - Fresh Air Blower
- V154 - Servo motor for fresh/recirculating air door
- (45) - Ground connection, behind instrument panel, center
- (162) - Ground connection, in blower motor wiring harness
- (A20) - Wire connection (15a), in instrument panel wiring harness
- (L66) - Connector, in wiring harness heater blower



S224 - Fuse 24 in fuse holder

S227 - Fuse 27 in fuse holder

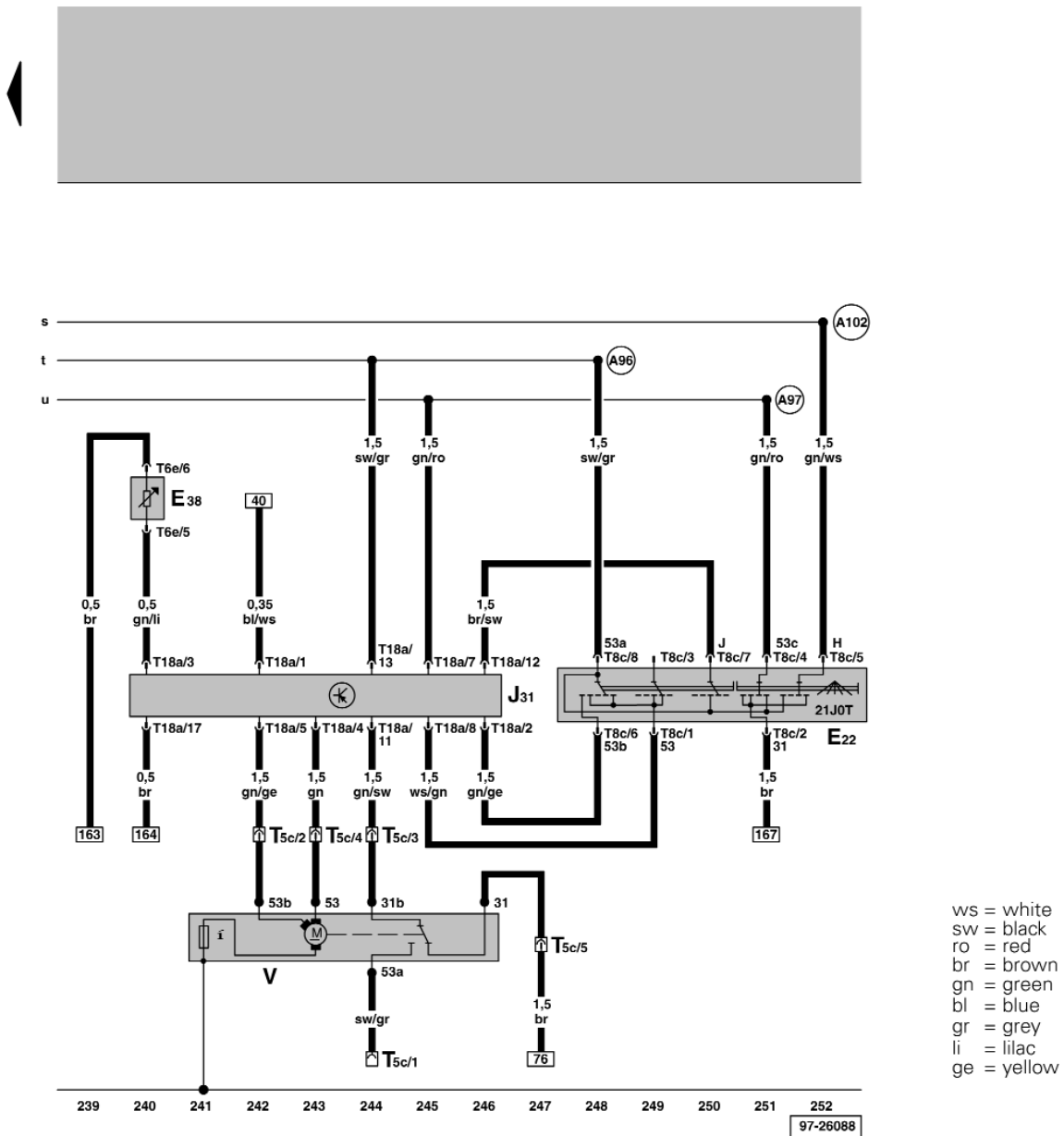
T5a - 5-Pin Connector, pink, connector station
C-pillar, left (Golf only)T5h - 5-Pin Connector, near left A-pillar, lower part, in
harnessT5i - 5-Pin Connector, near left A-pillar, lower part, in
harness

V12 - Motor for rear windshield wiper

V59 - Windshield and Rear Window Washer Pump

(A96) - Connector (53a), in instrument panel wiring
harness(A97) - Connector (53), in instrument panel wiring
harness(A102) - Connector (windshield wiper), in instrument
panel wiring harness

- - - - - Golf only



- E22 - Windshield Wiper/Washer Switch
- E38 - Windshield Wiper Intermittent Regulator
- J31 - Wiper/Washer Intermittent Relay, on relays panel
- T5c - 5-Pin Connector, on windshield wiper motor
- T6e - 6-Pin Connector
- T8c - 8-Pin Connector
- T18a - 18-Pin Connector
- V - Windshield Wiper Motor

- (A96) - Connector (53a), in instrument panel wiring harness
- (A97) - Connector (53), in instrument panel wiring harness
- (A102) - Connector (windshield wiper), in instrument panel wiring harness

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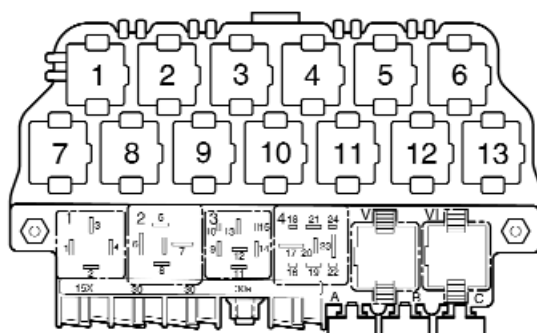
Windshield wiper/washer switch, windshield wiper intermittent regulator,
windshield wiper motor, wiper/washer intermittent relay



1.8L - Engine - Motronic Multiport Fuel Injection (MFI)/110 kW, code AWD,

from November 1999

Deviate relay location and fuse placements as well as the locations of multiple connectors see section "component locations".



97-14163

Relay location on the thirteenfold auxiliary relay panel, above relay panel:

Relay panel:

- 2** Load Reduction Relay (100)
- 4** Fuel Pump (FP) Relay (409)

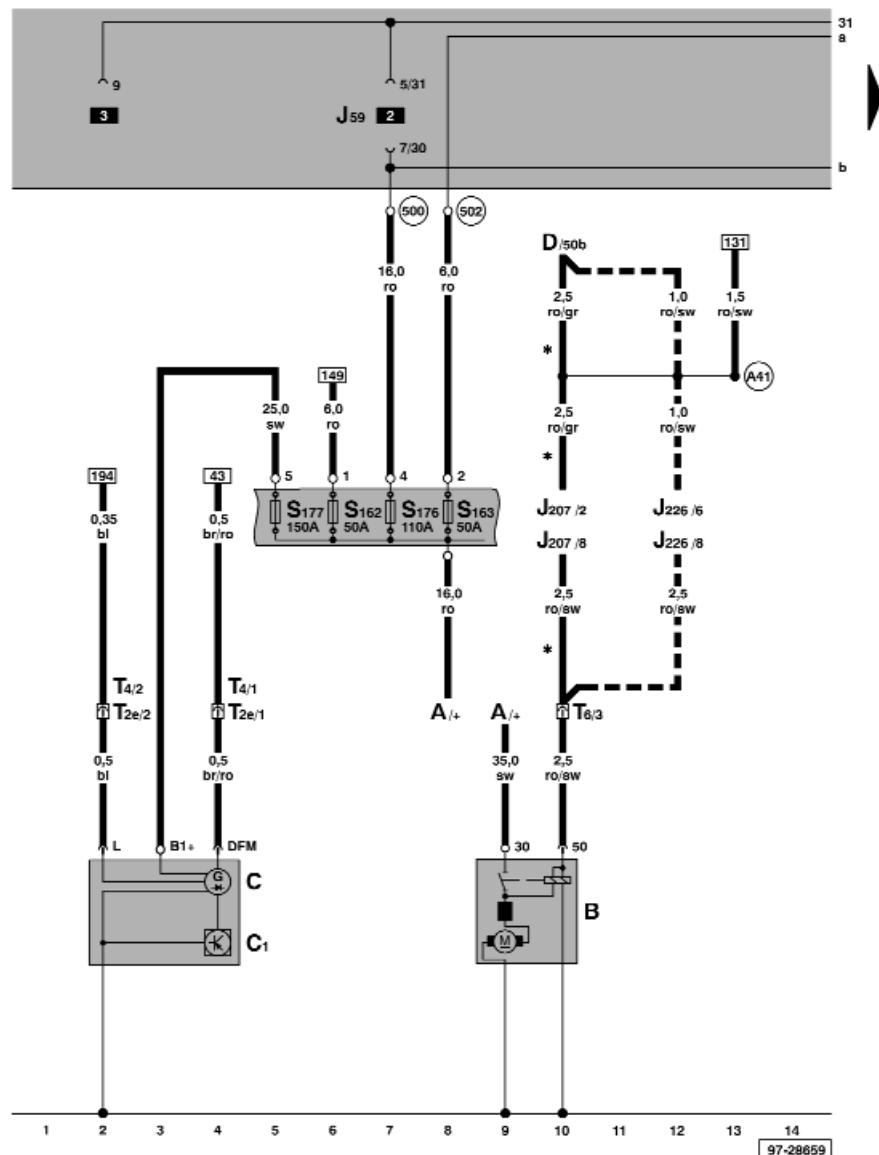
Note: Number in parentheses indicates production control number stamped on relay housing.

Fuse colors

- 30 A - green
- 25 A - white
- 20 A - yellow
- 15 A - blue
- 10 A - red
- 7.5 A - brown
- 5 A - beige
- 3 A - violet

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~A to the correctness of the

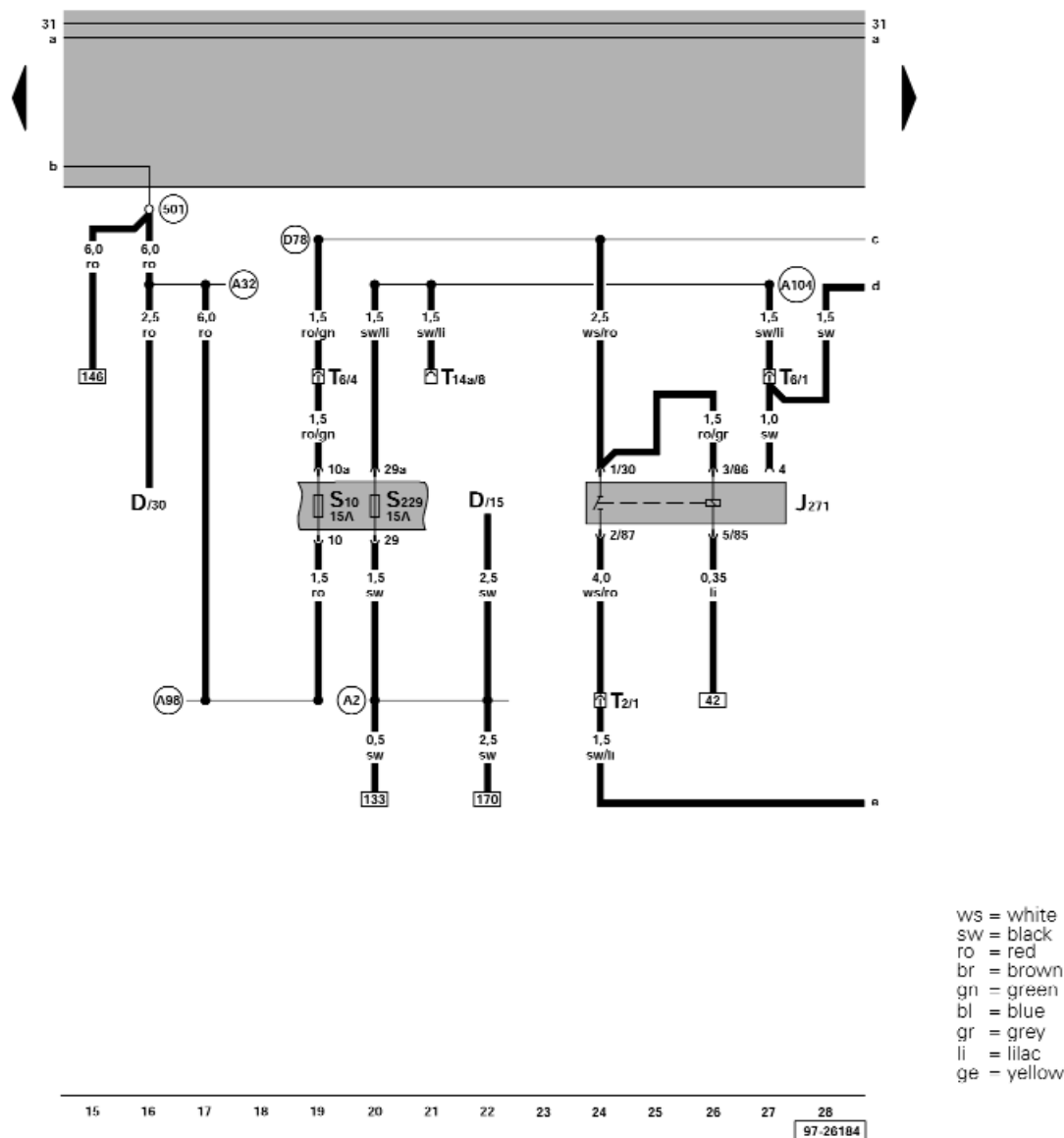


- A - Battery
- B - Starter
- C - Generator (GEN)
- C1 - Voltage Regulator (VR)
- D - Ignition/Starter Switch
- J59 - Load Reduction Relay
- J207 - Starting Interlock Relay
- J226 - Park/Neutral Position (PNP) Relay
- S162 - Fuse -1- (30) in fuse bracket/battery
- S163 - Fuse -2- (30) in fuse bracket/battery
- S176 - Fuse -4- (30) in fuse bracket/battery
- S177 - Fuse -5- (30) in fuse bracket/battery
- T2e - Double Connector, near starter (vehicles without air conditioning)
- T4 - 4-Pin Connector, near starter (vehicles with air conditioning only)

T6 - 6 Pin Connector, brown, in protective housing for connectors, in plenum chamber, left

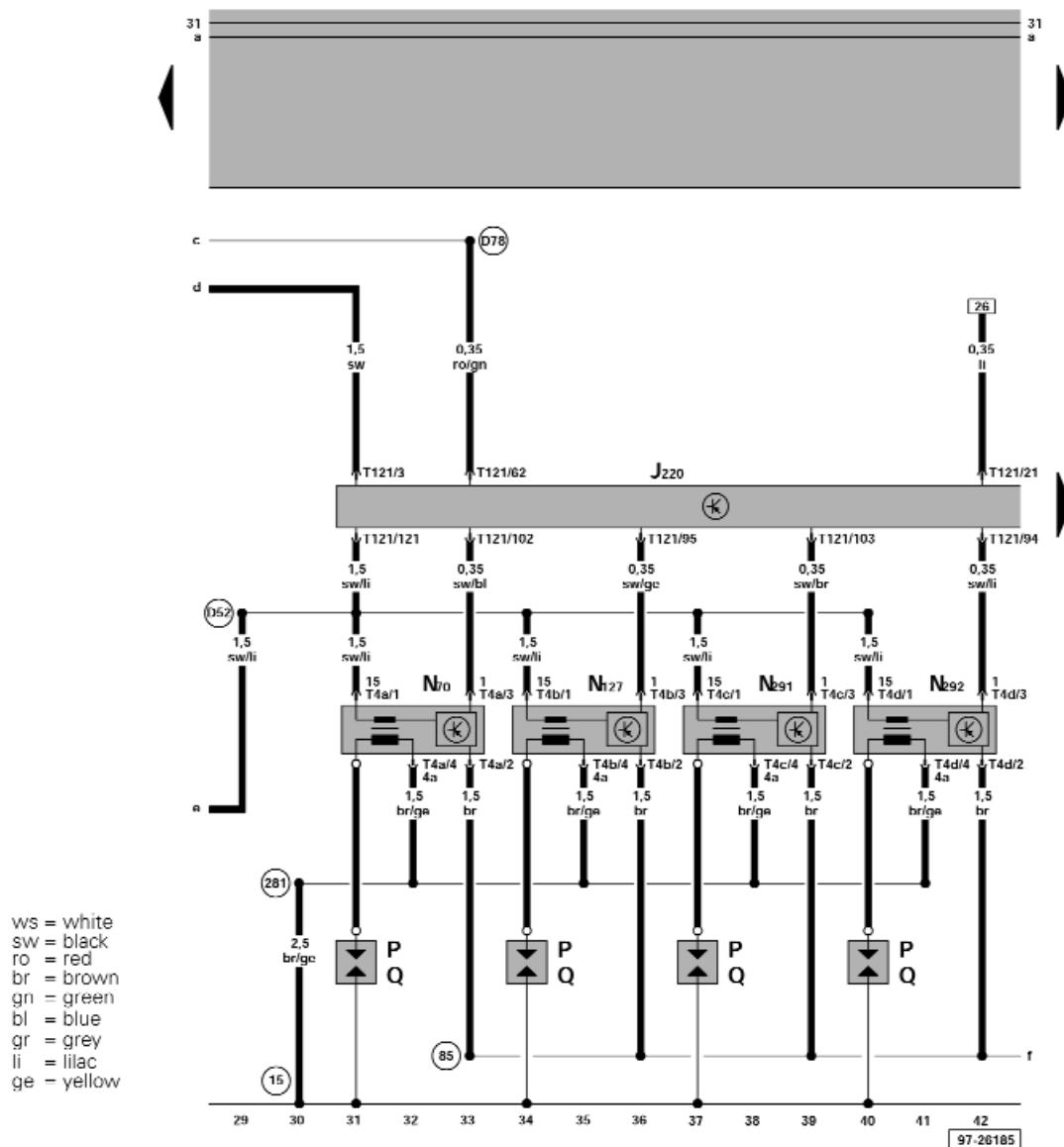
- (500) - Threaded connection -1- (30) on the relay plate
- (502) - Threaded connection -1- (30a) on the relay plate
- (A41) - Plus connection (50), in instrument panel wiring harness

- * - Manual transmission only
- Automatic transmission only



- D - Ignition/Starter Switch
J271 - Motronic engine control module (ECM) power supply relay, in protective housing, in engine compartment, left, production control number (428)
S10 - Fuse 10 in fuse holder
S229 - Fuse 29 in fuse holder
T2 - 2-Pin Connector, in engine compartment, left
T6 - 6-Pin Connector, brown, in protective housing for connectors, in plenum chamber, left
T14a - 14-Pin Connector, in engine compartment, in wiring duct, left
(501) - Threaded connection -2- (30) on the relay plate

- (A2) - Plus connection (15), in instrument panel wiring harness
(A32) - Plus connection (30), in instrument panel wiring harness
(A98) - Plus connector -4- (30), in instrument panel wiring harness
(A104) - Plus connector -2- (15), in instrument panel wiring harness
(D78) - Plus connector -1- (30a), in engine compartment wiring harness



J220 - Motronic Engine Control Module (ECM), in plenum chamber, center

N70 - Ignition Coil 1 with Power Output Stage

N127 - Ignition Coil 2 with Power Output Stage

N291 - Ignition Coil 3 with Power Output Stage

N292 - Ignition Coil 4 with Power Output Stage

P - Spark Plug Connectors

Q - Spark Plugs

T4a - 4-Pin Connector

T4b - 4-Pin Connector

T4c - 4-Pin Connector

T4d - 4-Pin Connector

T121 - Connector, 121 point

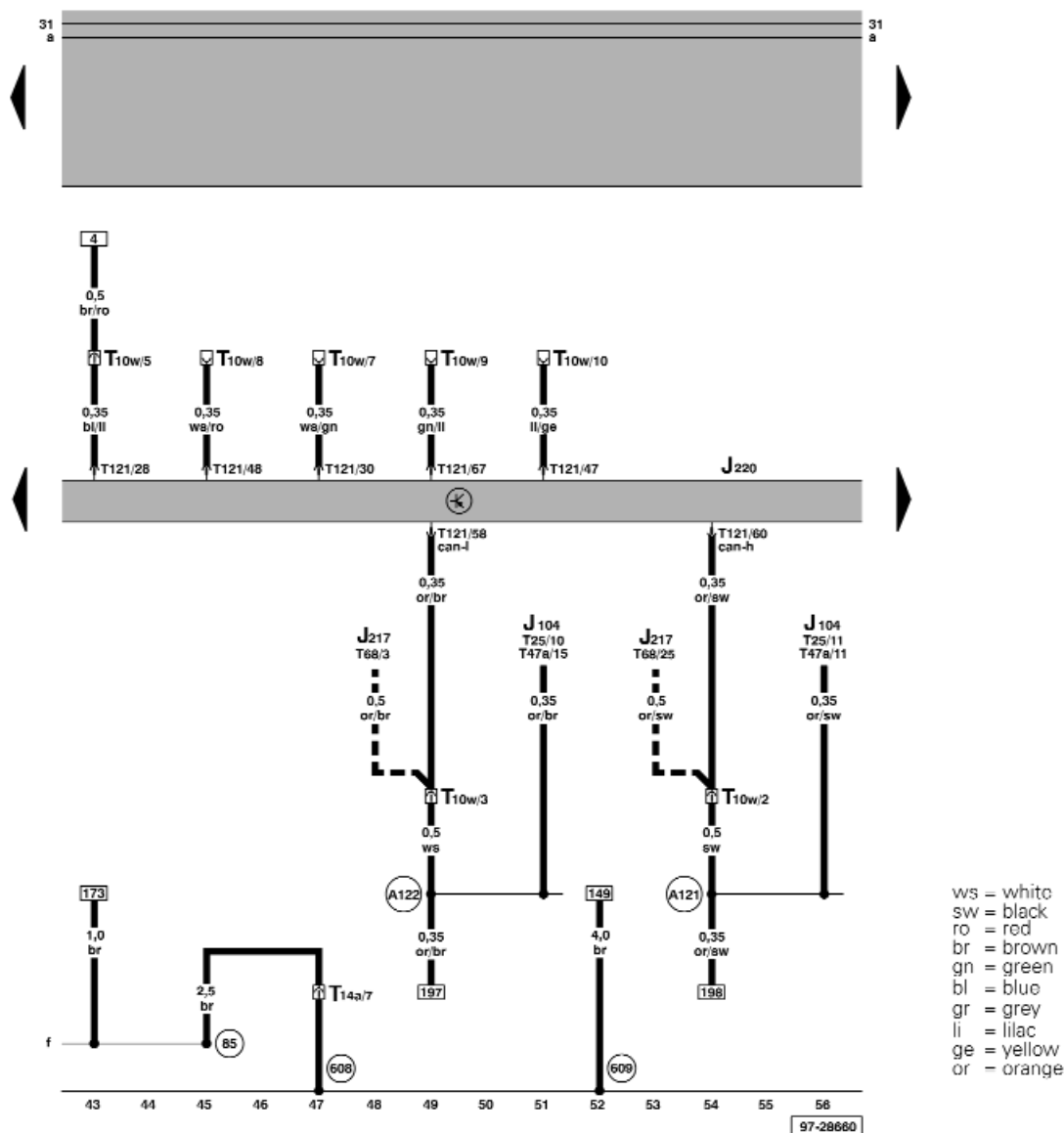
(15) - Ground connection, on cylinder head

(85) - Ground connection -1-, in engine compartment wiring harness

(281) - Ground connector -1-, in wiring harness engine pre-wiring

(D57) - Plus connector (15a), in engine compartment wiring harness

(D78) - Plus connector -1- (30a), in engine compartment wiring harness



- J104 - ABS Control Module (w/EDL), in engine compartment, left
J217 - Transmission Control Module (TCM), in plenum chamber, center
J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
T10w- 10-Pin Connector, white, in protective housing for connectors, in plenum chamber, left
T14a - 14-Pin Connector, in engine compartment, in wiring duct, left
T25 - 25-Pin Connector, on ABS Control Module (through July 2000)
T47a - 47-Pin Connector, on ABS Control Module (beginning August 2000)

T68 - 68-Pin Connector, on Transmission Control Module (TCM)

T121 - Connector, 121 point

85 - Ground connection -1-, in engine compartment wiring harness

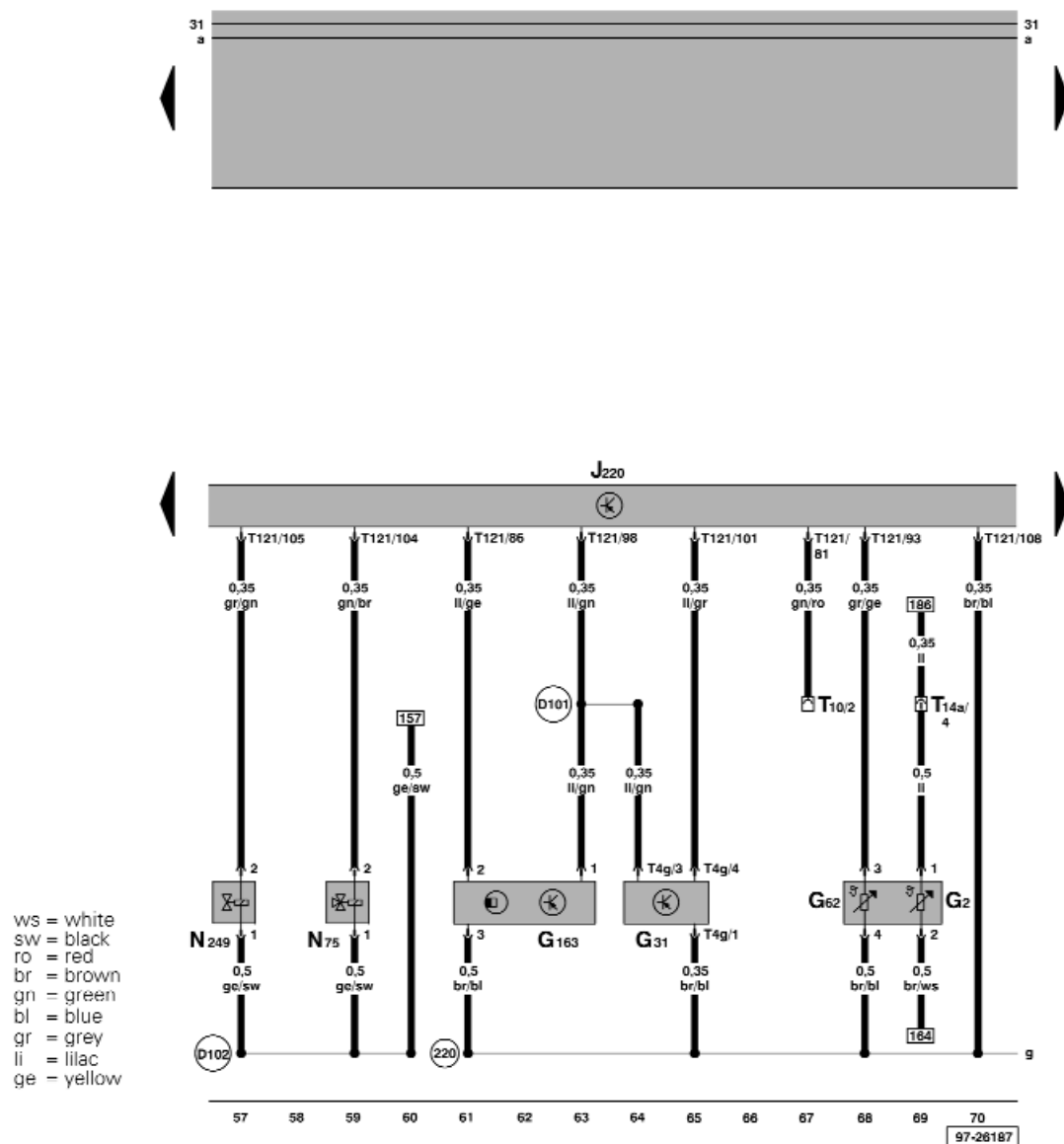
608 - Ground Connection (in center plenum chamber)

609 - Ground Connection (in right plenum chamber)

A121 - Connection (high bus), in instrument panel wiring harness

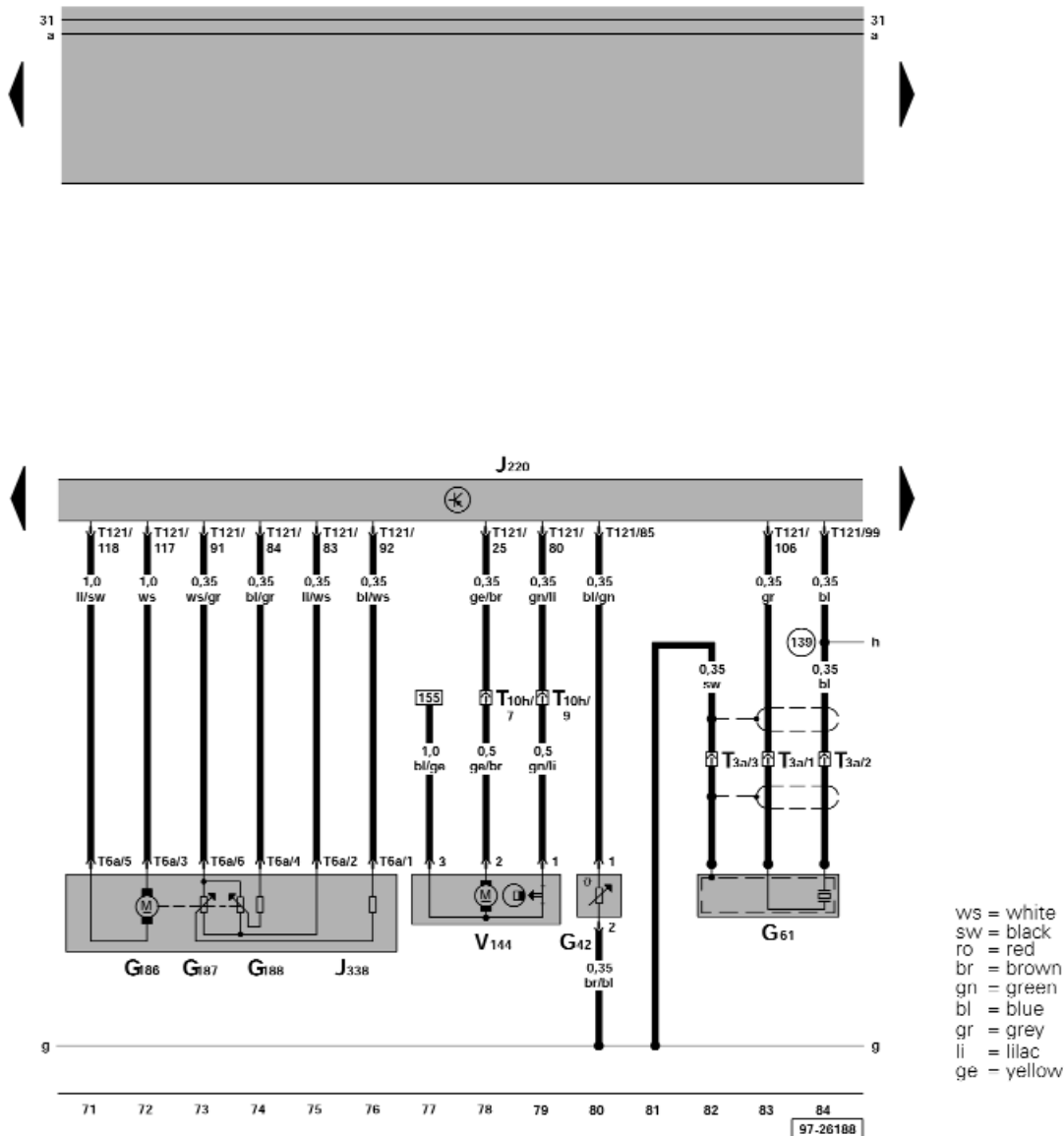
A122 - Connection (low bus), in instrument panel wiring harness

--- Automatic transmission only



- G2 - Engine Coolant Temperature (ECT) Sensor
- G31 - Charge Air Pressure Sensor
- G62 - Engine Coolant Temperature (ECT) Sensor
- G163 - Camshaft Position (CMP) Sensor 2
- J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
- N75 - Wastegate Bypass Regulator Valve
- N249 - Recirculating valve for turbocharger
- T4g - 4-Pin Connector
- T10 - 10-Pin Connector, orange, in protective housing for connectors, in plenum chamber, left
- T14a - 14-Pin Connector, in engine compartment, in wiring duct, left
- T121 - Connector, 121 point

- (220) - Ground connection (sensor ground), in engine compartment wiring harness
- (D101) - Wire connection -1-, in engine compartment wiring harness
- (D102) - Wire connection -2-, in engine compartment wiring harness



- G42 - Intake Air Temperature (IAT) Sensor
G61 - Knock Sensor (KS) 1
G186 - Throttle drive (power accelerator actuation)
G187 - Angle sensor -1- for throttle drive (power accelerator actuation)
G188 - Angle sensor -2- for throttle drive (power accelerator actuation)
J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
J338 - Throttle Valve Control Module
T3a - 3-Pin Connector, in engine compartment, front
T6a - 6-Pin Connector
T10h - 10-Pin Connector, blue, in protective housing for connectors, in plenum chamber, left
T121 - Connector, 121 point

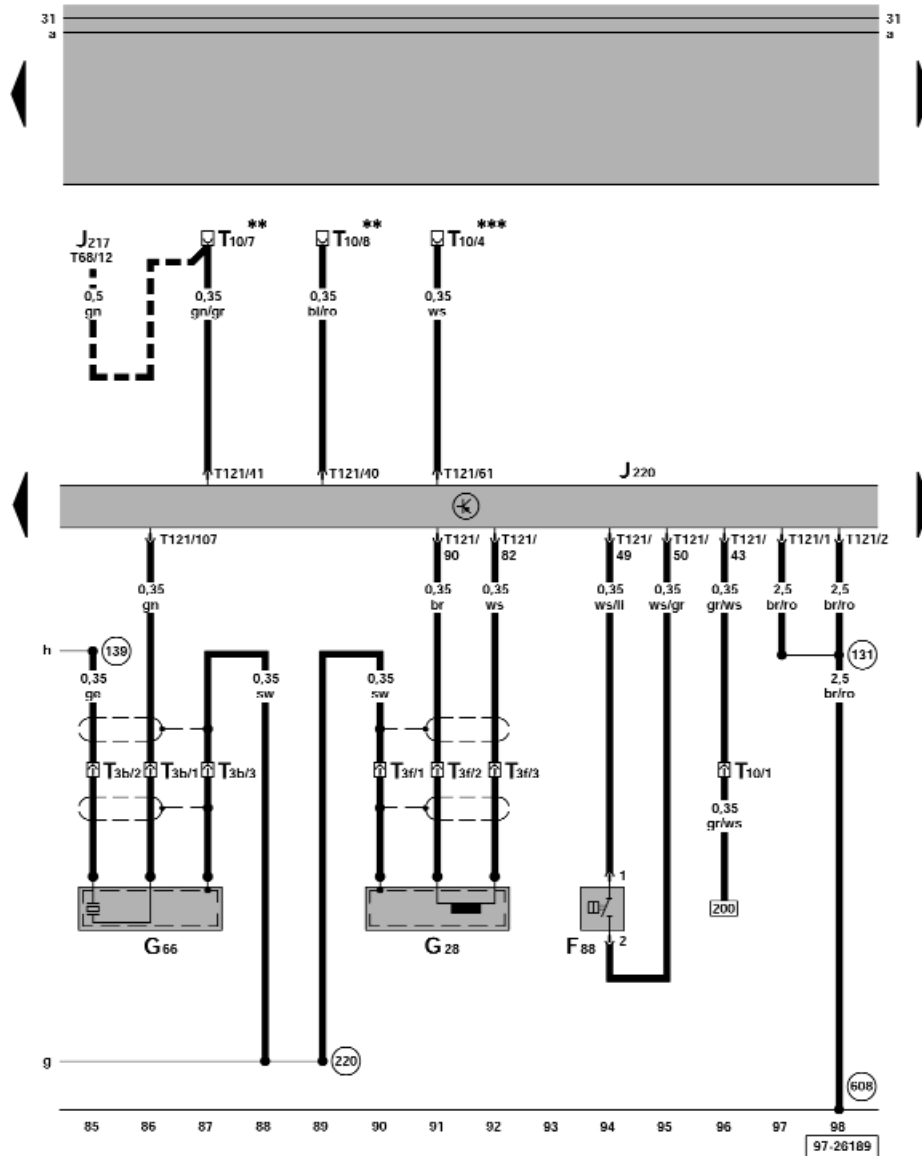
V144 - Leak detection pump (LDP)

(139) - Ground connection (sensor Ground), in Motronic Multiport Fuel Injection (MFI) wiring harness

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Wiring diagram

Golf/Jetta



- F88 - Pressure switch/power steering
G28 - Engine Speed (RPM) Sensor
G66 - Knock Sensor (KS) 2
J217 - Transmission Control Module (TCM), in plenum chamber, center
J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
T3b - 3-Pin Connector, in engine compartment, front
T3f - 3-Pin Connector, in engine compartment, front
T10 - 10-Pin Connector, orange, in protective housing for connectors, in plenum chamber, left
T68 - Connector, 68 point
T121 - Connector, 121 point
- (139) - Ground connection -2-, in engine compartment wiring harness

- (139) - Ground connection (sensor Ground), in Motronic Multiport Fuel Injection (MPI) wiring harness
(220) - Ground connection (sensor ground), in engine compartment wiring harness
(608) - Ground Connection (in center plenum chamber)
- ** - A/C connection
*** - Coolant Fan Control Module connection
--- - Automatic transmission only

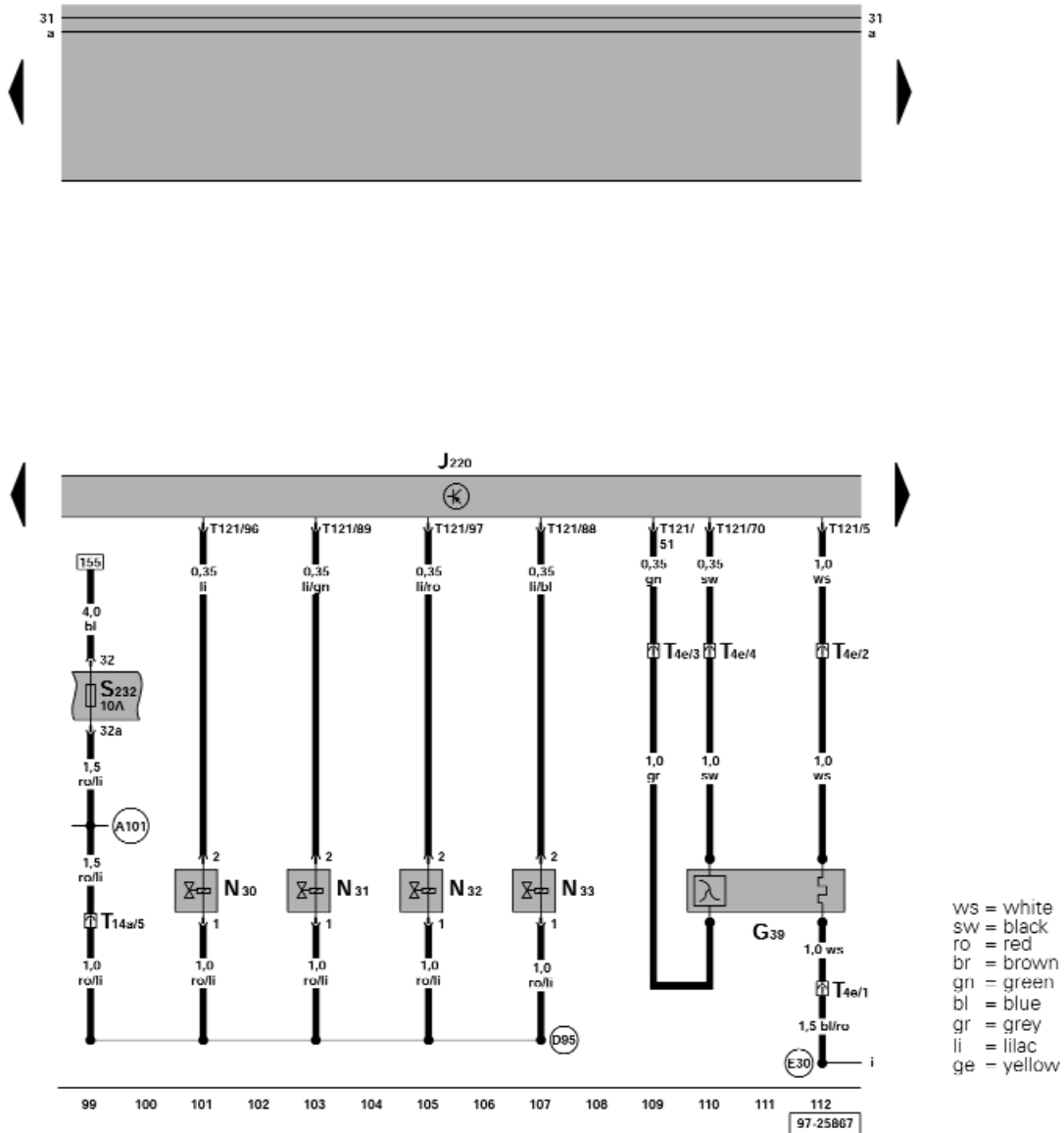
Motronic engine control module (ECM), pressure switch/power steering, engine speed (RPM) sensor, knock sensor (KS) 2

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Golf/Jetta

Wiring diagram

No. 41/9

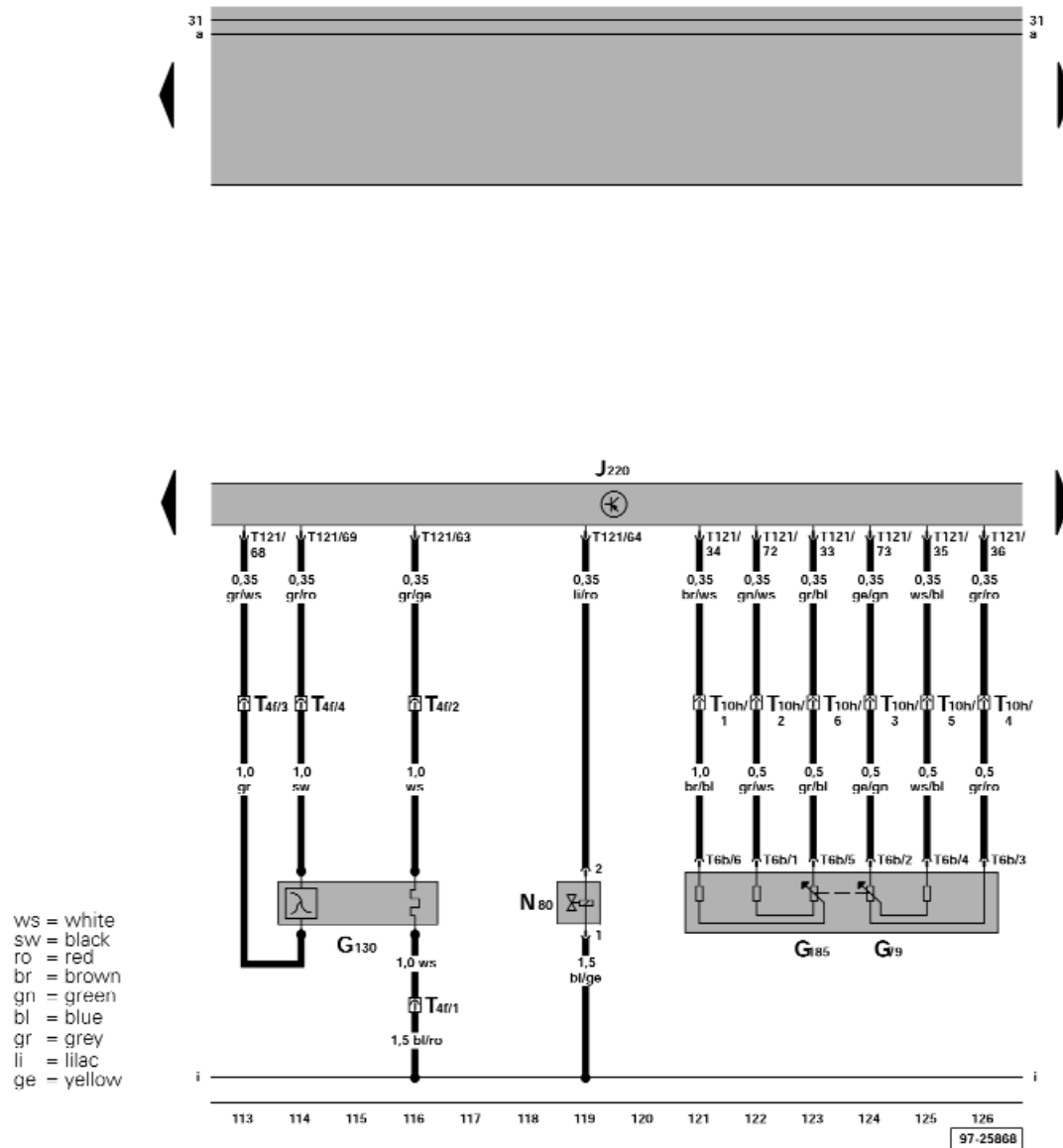


- G39 - Heated Oxygen Sensor (HO2S)
- J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
- N30 - Cylinder 1 Fuel Injector
- N31 - Cylinder 2 Fuel Injector
- N32 - Cylinder 3 Fuel Injector
- N33 - Cylinder 4 Fuel Injector
- S232 - Fuse 32 in fuse holder
- T4e - 4-Pin Connector, in protective housing for connectors under right floor
- T14a - 14-Pin Connector, in engine compartment, in wiring duct, left
- T121 - Connector, 121 point
- A101 - Connector -3- (87a), in instrument panel wiring harness

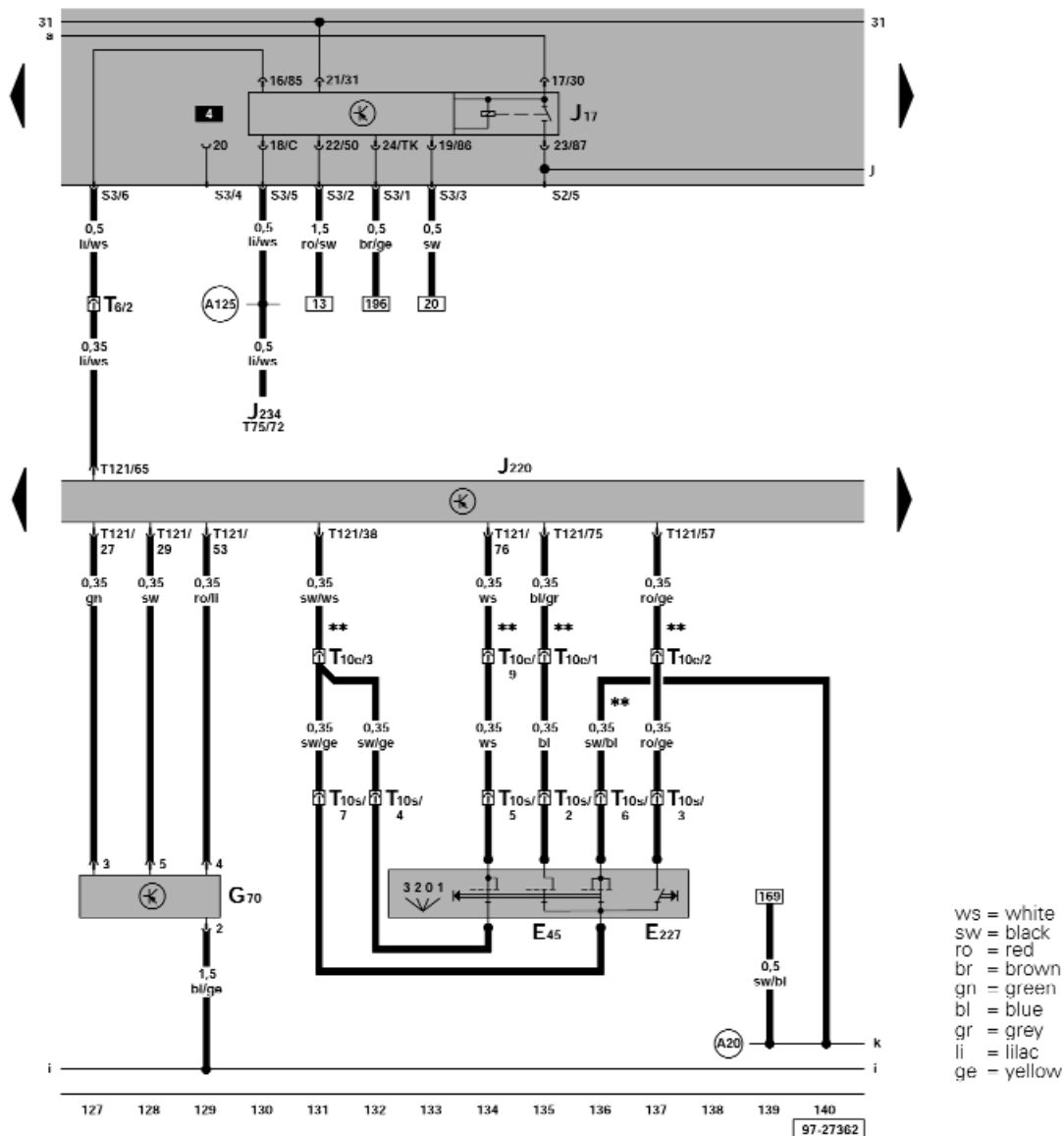
- D95 - Wire connection (injectors), in engine compartment wiring harness
- E30 - Connector (87a), in wiring harness engine

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Motronic engine control module (ECM), heated oxygen sensor (HO2S), injectors



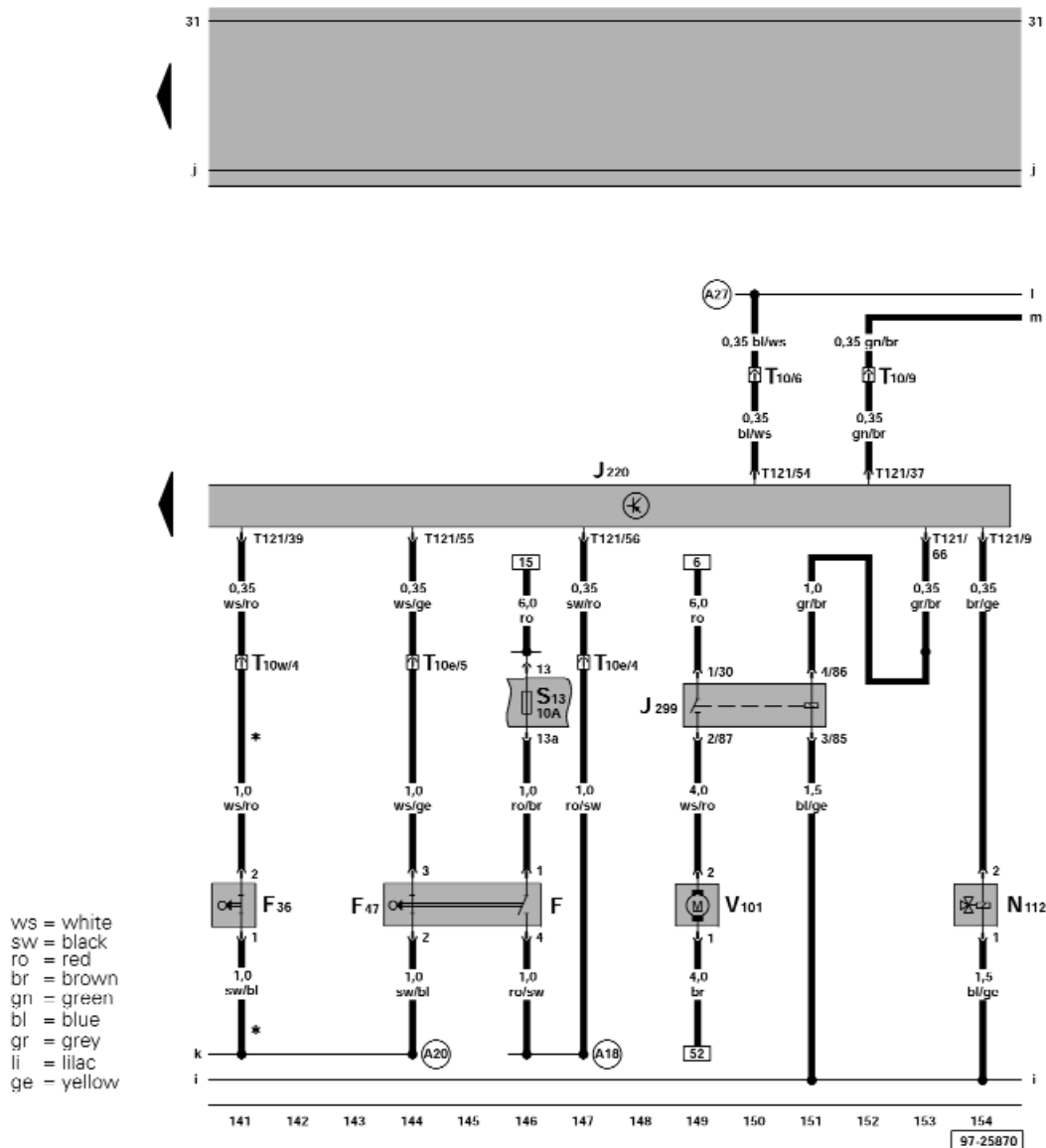
- G79 - Throttle Position (TP) Sensor
- G130 - Oxygen Sensor (O2S) behind Throce Way Catalytic Converter (TWC)
- G185 - Sender -2- for accelerator pedal position
- J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
- N80 - Evaporative Emission (EVAP) Canister Purge Regulator Valve
- T4f - 4-Pin Connector, in protective housing for connectors under right floor
- T6b - 6-Pin Connector
- T10h - 10-Pin Connector, blue, in protective housing for connectors, in plenum chamber, left
- T121 - Connector, 121 point



- E45 - Cruise Control Switch**
E227 - Cruise Control Push Button (SET)**
G70 - Mass Air Flow (MAF) Sensor
J17 - Fuel Pump (FP) Relay
J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
J234 - Airbag Control Module
T6 - 6 Pin Connector, brown, in protective housing for connectors, in plenum chamber, left
T10e - 10-Pin Connector, black, in protective housing for connectors, in plenum chamber, left
T10s - 10-Pin Connector, near steering column
T75 - 75-Pin Connector
T121 - Connector, 121 point

- (A20) - Wire connection (15a), in instrument panel wiring harness
(A125) - Connection (crash signal) in instrument panel wiring harness

** - Vehicles with cruise control only



- F - Brake Light Switch
F36 - Clutch Vacuum Vent Valve Switch
F47 - Brake Vacuum Vent Valve Switch for cruise control
J220 - Motronic Engine Control Module (ECM), in plenum chamber, center
J299 - Secondary Air Injection (AIR) Pump Relay, in protective housing, in engine compartment, left, production control number (100)
N112 - Secondary Air Injection (AIR) Solenoid Valve
S13 - Fuse 13 in fuse holder
T10 - 10-Pin Connector, orange, in protective housing for connectors, in plenum chamber, left
T10e - 10-Pin Connector, black, in protective housing for connectors, in plenum chamber, left

- T10w - 10-Pin Connector, white, in protective housing for connectors, in plenum chamber, left
T121 - Connector, 121 point
V101 - Secondary Air Injection (AIR) Pump Motor

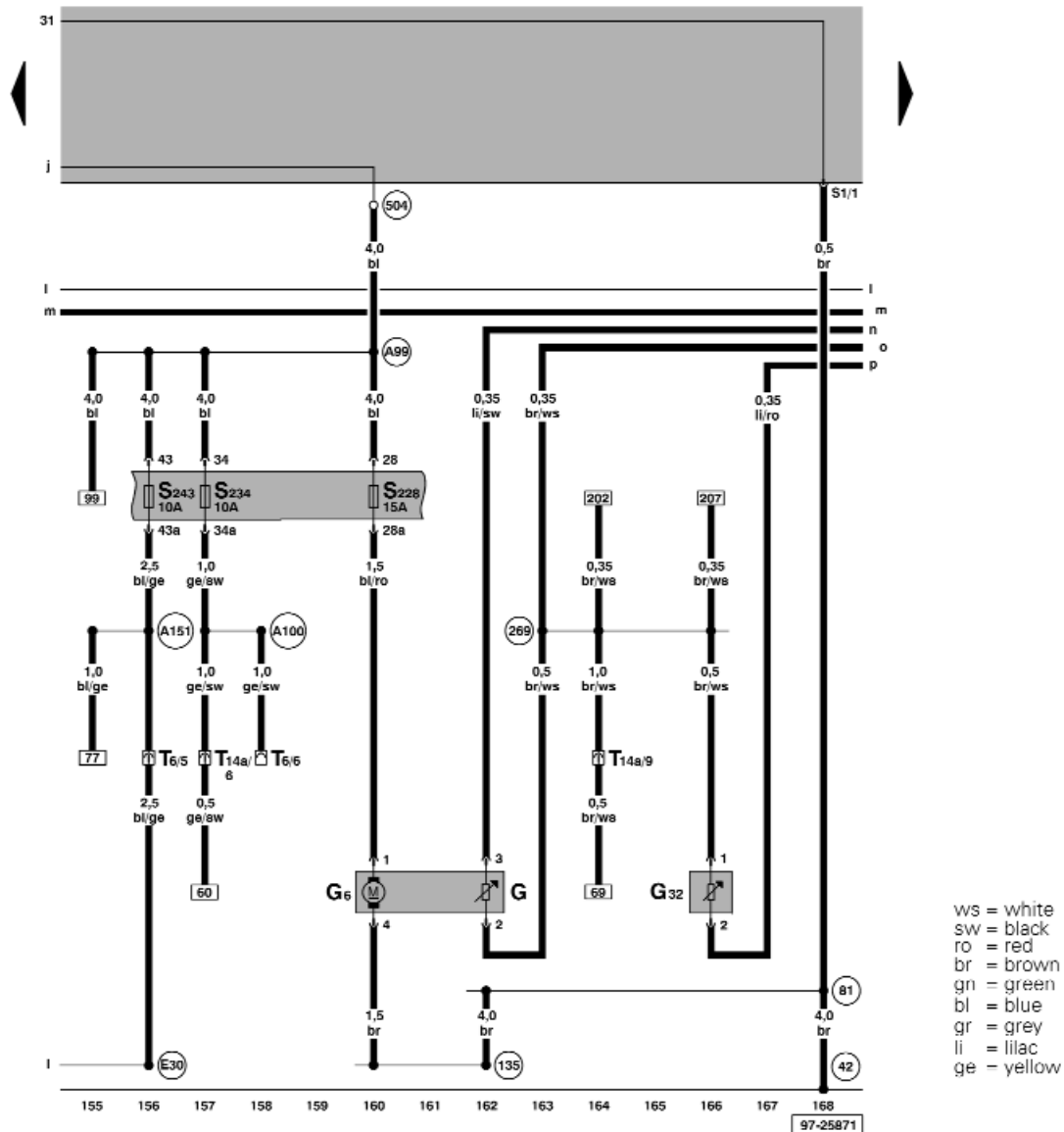
- (A18) - Wire connection (54), in instrument panel
(A20) - Wire connection (15a), in instrument panel wiring harness
(A27) - Wire Connection (vehicle speed signal), in instrument panel wiring harness

* - Manual transmission only

Golf/Jetta

Wiring diagram

No. 41/13

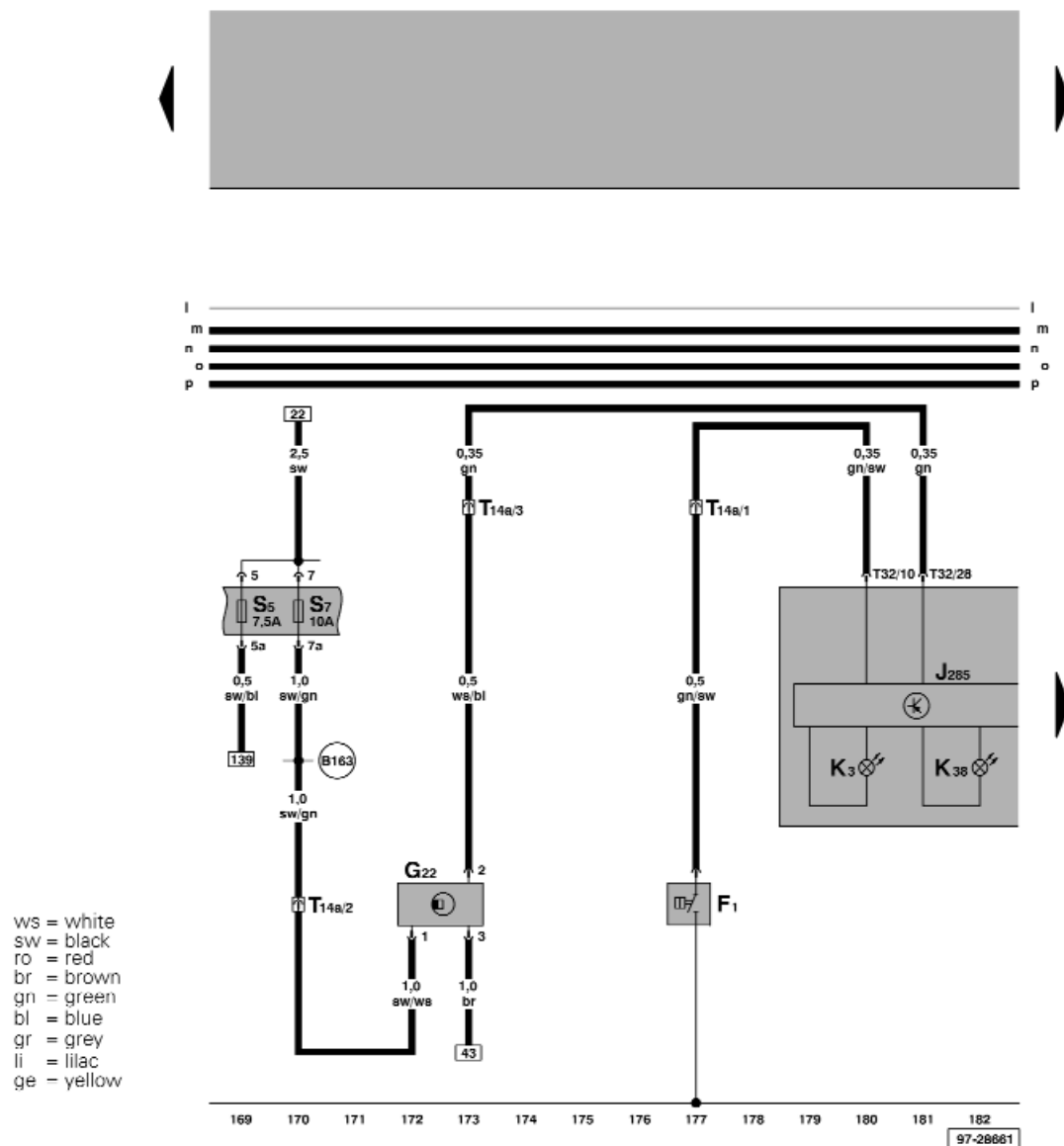


- G - Fuel Level Sensor
- G6 - Fuel Pump (FP)
- G32 - Engine Coolant Level (ECL) Sensor
- S228 - Fuse 28 in fuse holder
- S234 - Fuse 34 in fuse holder
- S243 - Fuse 43 in fuse holder
- T6 - 6-Pin Connector, brown, in protective housing for connectors, in plenum chamber, left
- T14a - 14-Pin Connector, in engine compartment, in wiring duct, left
- (42) - Ground connection, beside steering column
- (81) - Ground connection -1-, in instrument panel wiring harness

- (135) - Ground connection -2-, in instrument panel wiring harness
- (269) - Ground connection (sensor ground) -1-, in instrument panel wiring harness
- (504) - Threaded connection -1- (87) on the relay plate
- A99 - Connector -1- (87), in instrument panel wiring harness
- A100 - Connector -2- (87), in instrument panel wiring harness
- A151 - Connector -4- (87), in instrument panel wiring harness
- E30 - Connector (87a), in wiring harness engine

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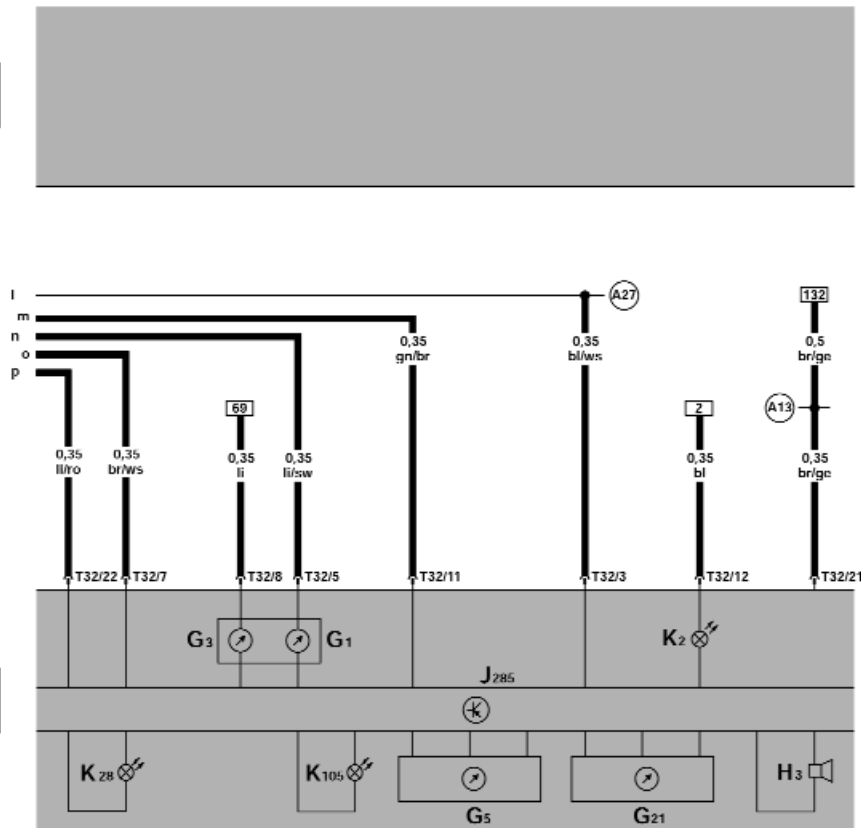
Fuel pump (FP), fuel level sensor, engine coolant level (ECL) sensor



- F1 - Oil Pressure Switch
- G22 - Speedometer Vehicle Speed Sensor (VSS)
- J285 - Control module with indicator unit in instrument panel insert
- K3 - Oil Pressure Warning Light
- K38 - Engine Oil Level Indicator Light
- S5 - Fuse 5 in fuse holder
- S7 - Fuse 7 in fuse holder
- T14a - 14-Pin Connector, in engine compartment, in wiring duct, left
- T32 - 32-Pin Connector, blue
- (F163)** - Plus connector -1- (15) in wiring harness interior

Instrument cluster, oil pressure switch, speedometer vehicle speed sensor (VSS), oil pressure warning light

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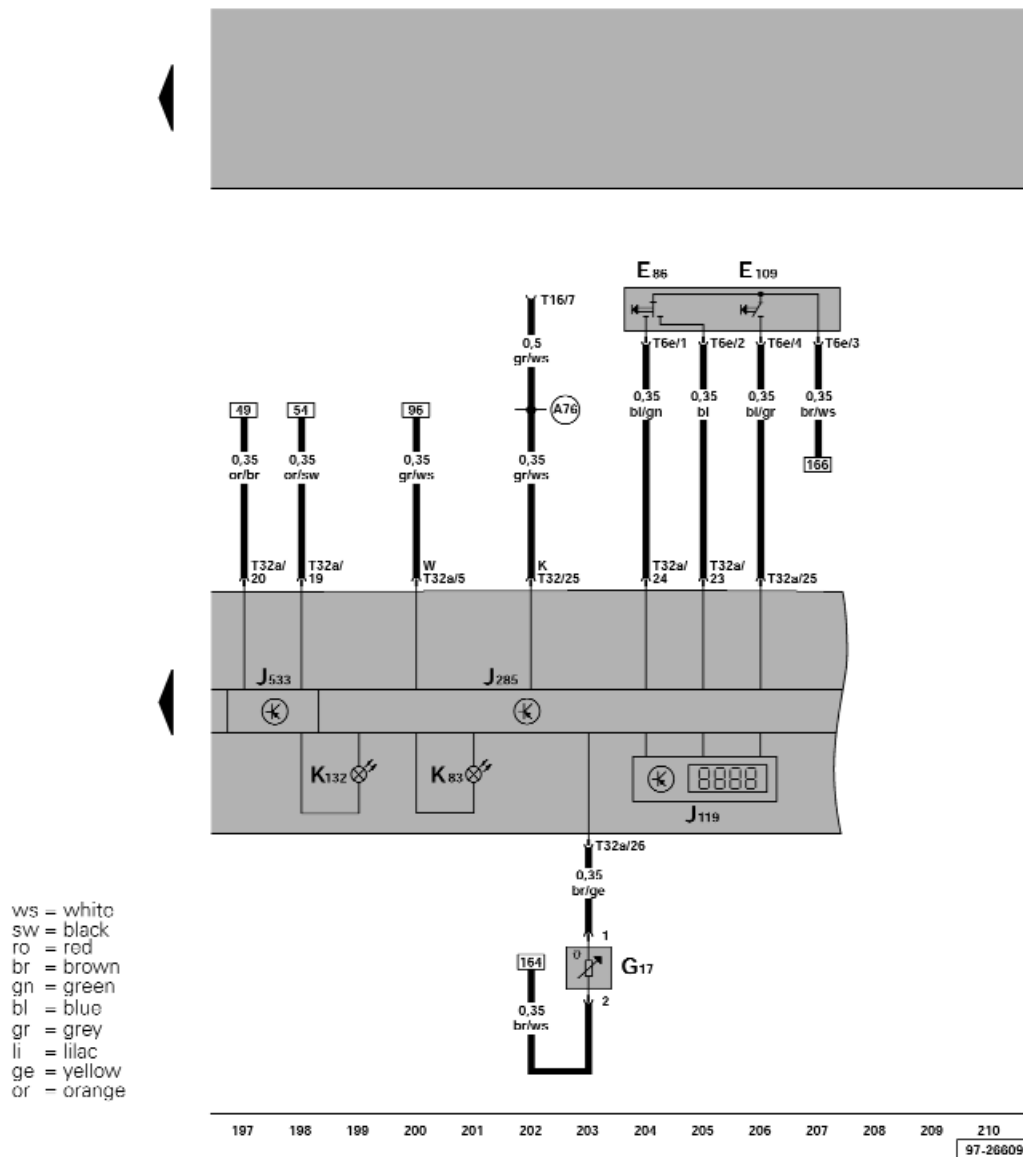
ws = white
 sw = black
 ro = red
 br = brown
 gn = green
 bl = blue
 gr = grey
 li = lilac
 ge = yellow

- 183 184 185 186 187 188 189 190 191 192 193 194 195 196
 97-26192
- G1 - Fuel gauge
 - G3 - Engine Coolant Temperature (ECT) Gauge
 - G5 - Tachometer
 - G21 - Speedometer
 - H3 - Warning Buzzer
 - J285 - Control module with indicator unit in instrument panel insert
 - K2 - Generator (GEN) Warning Light
 - K28 - Engine Coolant Level/Temperature (ECL/ECT) Warning Light
 - K105 - Low Fuel Level Warning Light
 - T32 - 32-Pin Connector, blue

- (A13) - Wire connection (door contact switch), in instrument panel wiring harness
- (A27) - Wire Connection (vehicle speed signal), in instrument panel wiring harness

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Instrument cluster, engine coolant temperature (ECT) gauge, fuel gauge, tachometer, speedometer, generator (GEN) warning light



E86 - Multi-Function Indicator Mode Select Switch
 E109 - Multi-Function Indicator Memory Switch
 G17 - Outside Air Temperature Sensor
 J119 - Multi-function Indicator (MFI)
 J285 - Control module with indicator unit in instrument panel insert
 J533 - Data Bus On Board Diagnostic Interface
 K83 - Malfunction Indicator Lamp (MIL)
 K132 - Electronic Power Control (EPC) Warning Lamp
 T6e - 6-Pin Connector
 T16 - Data Link Connector (DLC), below instrument panel, center
 T32 - 32-Pin Connector, blue
 T32a - 32-Pin Connector, green

(A76) - Connector (K diagnosis wire), in instrument panel wiring harness

Instrument cluster, multi-function indicator (MFI), outside air temperature sensor, electronic power control (EPC) warning lamp

Edition 06/00
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Glossary

American Wire Gauge (AWG): The American standard for wire size, expressed in units from 2 gauge to 22 gauge.

Central relay panel: A centralized location for relays and circuits in an automobile. Includes common circuits used to provide battery power, ignition "ON", and Ground to vehicle systems.

Control Module: A transistorized, rather than analog, switching circuit that relies on one or inputs to control the switching of a logic circuit.

Current track number: A method of following circuits within Volkswagen wiring diagrams. The wiring diagram is numbered along the bottom of the page to aid in navigating between disconnected pages.

DIN (Deutsche Institut fur Normung) standards: German institute for standards that establishes guidelines for manufacturing and nomenclature.

Electric motor: A motor driven by electromotive force.

Fuse: A component installed in series with a circuit, designed to disrupt the circuit when carrying more than its specified amperage. Fuses are placed in circuits to protect other components in the circuit from damage.

Ground: A wire connected to the vehicle's chassis (and therefore the Ground). This allows power to be supplied by one insulated wire, while the chassis acts as the return path.

Internal conductors: Connections inside of a component, often not designated in the wiring diagrams.

Load reduction circuit: A circuit designed to switch off high current electrical consumers during starting to insure optimal available voltage supply to the starter. The term comes from removing the load of the consumers from the Ignition switch.

Metric wire sizes: The metric system for wire size, expressed in mm². This designates the cross-sectional area of the wire. The wiring used in Volkswagen vehicles ranges from .35 to 35 mm².

Multi-point connectors: A connector installed in a wiring harness that allows the wiring harness to be detached from the component.

Potentiometer: see Rheostat

Glossary

Relay: Component that uses a low current to switch a high current circuit.

Resistor: A component that creates a voltage drop in a circuit. Resistors can be used for measurement of current flow, as well as to drop the applied voltage for certain consumers.

Rheostat: A resistor that varies its resistance based on a mechanical input.

Push-on connectors: See Multi-point connector.

Solenoid: Used to actuate other components, a solenoid consists of a coil of wire wrapped around an iron core. When a current is passed through a wire, a magnetic field is induced. This magnetic "pull" is used to operate other components.

Switch: A component used to disrupt or redirect current flow.

Symbols: Graphical representations of electrical components in Volkswagen wiring diagrams

Temperature dependent resistor: A resistor that varies its resistance with temperature.

Threaded Connectors: A connector consisting of a threaded stud to which other wires attach. Often used in high Amperage circuits.

Vehicle Chassis: See Ground.

Welded Connectors: A connection inside of a component created by compressing several wires together under high heat, partially melting the separate wires together.

X circuit: See Load reduction circuit.

How to Read Wiring Diagrams Teletest

The test accompanying this course, #873003, has been prepared and shipped as a separate document. Please refer to your copy of that document and follow the testing instructions to complete the Teletest.

Additional copies are available by contacting:

Certification Program Headquarters

Toll-free Hotline & Testing: 1-877-CU4-CERT (1-877-284-2378)

Fax: 1-877-FX4-CERT (1-877-394-2378)

Hotline assistance is available Monday-Friday

between 9:00 a.m. and 5:00 p.m., EST.

Cautions & Warnings

Please read these WARNINGS and CAUTIONS before proceeding with maintenance and repair work. You must answer that you have read and you understand these WARNINGS and CAUTIONS before you will be allowed to view this information.

- If you lack the skills, tools and equipment, or a suitable workshop for any procedure described in this manual, we suggest you leave such repairs to an authorized Volkswagen retailer or other qualified shop. We especially urge you to consult an authorized Volkswagen retailer before beginning repairs on any vehicle that may still be covered wholly or in part by any of the extensive warranties issued by Volkswagen.
- Disconnect the battery negative terminal (ground strap) whenever you work on the fuel system or the electrical system. Do not smoke or work near heaters or other fire hazards. Keep an approved fire extinguisher handy.
- Volkswagen is constantly improving its vehicles and sometimes these changes, both in parts and specifications, are made applicable to earlier models. Therefore, part numbers listed in this manual are for reference only. Always check with your authorized Volkswagen retailer parts department for the latest information.
- Any time the battery has been disconnected on an automatic transmission vehicle, it will be necessary to reestablish Transmission Control Module (TCM) basic settings using the VAG 1551 Scan Tool (ST).
- Never work under a lifted vehicle unless it is solidly supported on stands designed for the purpose. Do not support a vehicle on cinder blocks, hollow tiles or other props that may crumble under continuous load. Never work under a vehicle that is supported solely by a jack. Never work under the vehicle while the engine is running.
- For vehicles equipped with an anti-theft radio, be sure of the correct radio activation code before disconnecting the battery or removing the radio. If the wrong code is entered when the power is restored, the radio may lock up and become inoperable, even if the correct code is used in a later attempt.
- If you are going to work under a vehicle on the ground, make sure that the ground is level. Block the wheels to keep the vehicle from rolling. Disconnect the battery negative terminal (ground strap) to prevent others from starting the vehicle while you are under it.
- Do not attempt to work on your vehicle if you do not feel well. You increase the danger of injury to yourself and others if you are tired, upset or have taken medicine or any other substances that may impair you or keep you from being fully alert.
- Never run the engine unless the work area is well ventilated. Carbon monoxide (CO) kills.
- Always observe good workshop practices. Wear goggles when you operate machine tools or work with acid. Wear goggles, gloves and other protective clothing whenever the job requires working with harmful substances.
- Tie long hair behind your head. Do not wear a necktie, a scarf, loose clothing, or a necklace when you work near machine tools or running engines. If your hair, clothing, or jewelry were to get caught in the machinery, severe injury could result.
- Do not re-use any fasteners that are worn or deformed in normal use. Some fasteners are designed to be used only once and are unreliable and may fail if used a second time. This includes, but is not limited to, nuts, bolts, washers, circlips and cotter pins. Always follow the recommendations in this manual - replace these fasteners with new parts where indicated, and any other time it is deemed necessary by inspection.

Cautions & Warnings

- Illuminate the work area adequately but safely. Use a portable safety light for working inside or under the vehicle. Make sure the bulb is enclosed by a wire cage. The hot filament of an accidentally broken bulb can ignite spilled fuel or oil.
- Friction materials such as brake pads and clutch discs may contain asbestos fibers. Do not create dust by grinding, sanding, or by cleaning with compressed air. Avoid breathing asbestos fibers and asbestos dust. Breathing asbestos can cause serious diseases such as asbestosis or cancer, and may result in death.
- Finger rings should be removed so that they cannot cause electrical shorts, get caught in running machinery, or be crushed by heavy parts.
- Before starting a job, make certain that you have all the necessary tools and parts on hand. Read all the instructions thoroughly; do not attempt shortcuts. Use tools that are appropriate to the work and use only replacement parts meeting Volkswagen specifications. Makeshift tools, parts and procedures will not make good repairs.
- Catch draining fuel, oil or brake fluid in suitable containers. Do not use empty food or beverage containers that might mislead someone into drinking from them. Store flammable fluids away from fire hazards. Wipe up spills at once, but do not store the oily rags, which can ignite and burn spontaneously.
- Use pneumatic and electric tools only to loosen threaded parts and fasteners. Never use these tools to tighten fasteners, especially on light alloy parts. Always use a torque wrench to tighten fasteners to the tightening torque listed.
- Keep sparks, lighted matches, and open flame away from the top of the battery. If escaping hydrogen gas is ignited, it will ignite gas trapped in the cells and cause the battery to explode.
- Be mindful of the environment and ecology. Before you drain the crankcase, find out the proper way to dispose of the oil. Do not pour oil onto the ground, down a drain, or into a stream, pond, or lake. Consult local ordinances that govern the disposal of wastes.
- The air-conditioning (A/C) system is filled with a chemical refrigerant that is hazardous. The A/C system should be serviced only by trained automotive service technicians using approved refrigerant recovery/recycling equipment, trained in related safety precautions, and familiar with regulations governing the discharging and disposal of automotive chemical refrigerants.
- Before doing any electrical welding on vehicles equipped with anti-lock brakes (ABS), disconnect the battery negative terminal (ground strap) and the ABS control module connector.
- Do not expose any part of the A/C system to high temperatures such as open flame. Excessive heat will increase system pressure and may cause the system to burst.
- When boost-charging the battery, first remove the fuses for the Engine Control Module (ECM), the Transmission Control Module (TCM), the ABS control module, and the trip computer. In cases where one or more of these components is not separately fused, disconnect the control module connector(s).
- Some of the vehicles covered by this manual are equipped with a supplemental restraint system (SRS), that automatically deploys an airbag in the event of a frontal impact. The airbag is operated by an explosive device. Handled improperly or without adequate safeguards, it can be accidentally activated and cause serious personal injury. To guard against personal injury or airbag system failure, only trained Volkswagen Service technicians should test, disassemble or service the airbag system.

Cautions & Warnings

- Do not quick-charge the battery (for boost starting) for longer than one minute, and do not exceed 16.5 volts at the battery with the boosting cables attached. Wait at least one minute before boosting the battery a second time.
- Never use a test light to conduct electrical tests of the airbag system. The system must only be tested by trained Volkswagen Service technicians using the VAG 1551 Scan Tool (ST) or an approved equivalent. The airbag unit must never be electrically tested while it is not installed in the vehicle.
- Some aerosol tire inflators are highly flammable. Be extremely cautious when repairing a tire that may have been inflated using an aerosol tire inflator. Keep sparks, open flame or other sources of ignition away from the tire repair area. Inflate and deflate the tire at least four times before breaking the bead from the rim. Completely remove the tire from the rim before attempting any repair.
- When driving or riding in an airbag-equipped vehicle, never hold test equipment in your hands or lap while the vehicle is in motion. Objects between you and the airbag can increase the risk of injury in an accident.

I have read and I understand these Cautions and Warnings.